7 Embedded flash memory (FLASH)

7.1 Introduction

The embedded flash memory (FLASH) manages the accesses of any master to the up to 2 Mbytes of embedded nonvolatile memory. It implements read, program, and erase operations, error corrections, as well as various integrity and confidentiality protection mechanisms.

FLASH manages the automatic loading of nonvolatile user option bytes at power-on reset, and implements the dynamic update of these options. It also features a high-cycle data area, a one-time-programmable (OTP) area, a secure key storage area (OBKeys), and a read-only area configured by STMicroelectronics during manufacturing.

7.2 FLASH main features

- Up to 2 Mbytes of nonvolatile memory, divided into two 1-Mbyte banks
- Flash memory read operations supporting multiple lengths: 128 bits, 64 bits, 32 bits, 16 bits, or one byte
- Flash memory programming by 128 (user area, OBKeys) and by 16 bits (OTP and flash high-cycle data area)
- 8-Kbyte sector erase, bank erase and dual-bank mass erase
- Dual-bank organization supporting:
  - Simultaneous operations: read-while-write (program and erase) is supported, including flash high-cycle data area. The two banks share the same interface, hence write and erase cannot be performed in parallel (write-while-write is not supported).
  - Bank swapping: the address mapping of the user memory of each bank can be swapped, along with the corresponding registers. Security flags remain valid for the physical bank, so the data are not reveled by swapping to a bank with lower security configuration.
- Error code correction (ECC): one error detection/correction, or two errors detection per 128-bit flash word using nine ECC bits, on 16-bit words with six bits within configurable flash high-cycle data area.
- User configurable nonvolatile option bytes
- Flash memory enhanced protections, activated by option bytes
  - different product states for protecting memory content from debug access
  - sector group write-protection (WRPSG), protecting up to 32 groups of four sectors (32 Kbytes) per bank
  - two secure-only areas (one per user flash bank): when enabled, these areas are accessible only if the microcontroller operates in Secure access mode
  - HDP protection, providing temporal isolation for startup code
- 2-Kbyte one-time programmable (OTP) area
- Read-only area configured by STMicroelectronics
- Prefetch reads the next sequential instruction from flash memory

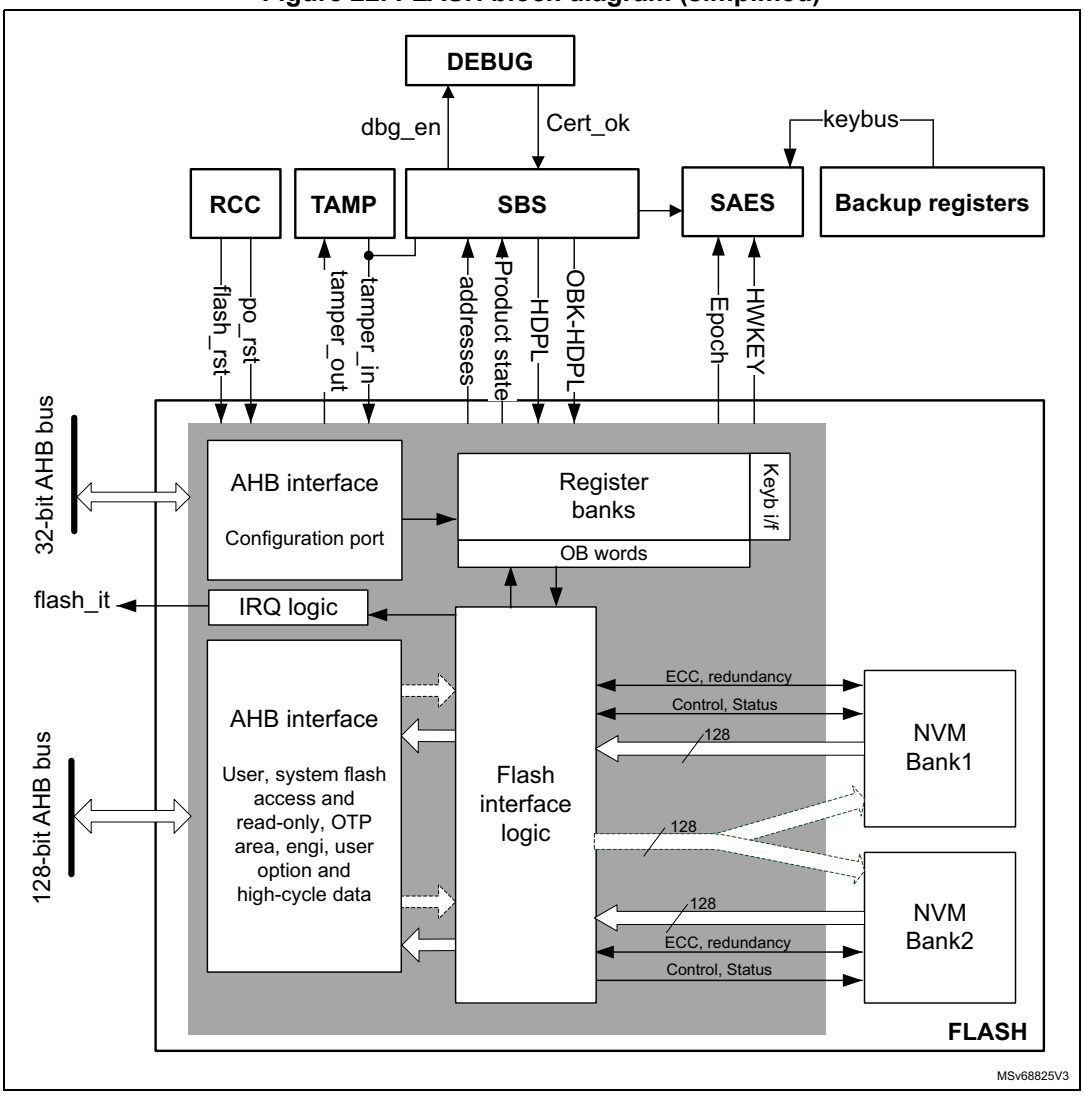
- Up to 48 Kbytes per bank supporting high-cycling capability (100 kcycles), to be used for data (EEPROM emulation)

7.3 FLASH functional description

7.3.1 FLASH block diagram

Figure 22 shows the embedded flash memory block diagram.

Figure 22. FLASH block diagram (simplified)



7.3.2 FLASH signals

The flash memory has two AHB connections, namely the flash AHB register interface and the main AHB interface.

Flash AHB register interface

- Data size is 32 bits
- Except for some registers (FLASH_NS/SECKEYR, FLASH_NS/SECOBKKEYR and FLASH_OPTKEYR, used to insert unlock sequences for control, and option registers that can be written by 32 bits), it is possible to read and write all registers by 8, 16 and 32 bits.
- When unlock sequence for control and option registers is wrong, a bus error is raised, otherwise no read or write errors are generated on the bus.

Main AHB interface

The AHB data bus size is 128 bits. This interface is used to handle three different targets:

- Code placed in user and system memory, protected by 9 bits of ECC.
- Secure keys placed in OBKey sectors, protected by 9 bits of ECC.
- OTP, read-only and flash memory high-cycle data area. protected by 6 bits of ECC.

The main AHB interface is implemented as follows:

- User and system memory, OBKeys storage:
  - Supports multiple length: 128-, 64-, 32-, 16- and 8-bit data width.
  - There is a read buffer of 128 bits for each bank where the last data read is stored. If data are available in the read buffer, no read access is given to the flash memory. Buffer is flushed when write access, OTP access, OBK swap, OBK alternate sector erase, high-cycle data area access, user option change request or erase operation occur.
  - There is a prefetch of the same size as the read buffer.
  - A 9-bit ECC is associated to each 128-bit data flash memory word.
- OTP, read-only and flash high-cycle data
  - Two dedicated data buffer of 137 bits are used to manage 16-bit data with 6-bit ECC
  - Reading two times the same address triggers two flash read accesses.
  - During a read access, two wait states are added in addition to the memory wait states. These wait states are necessary to parse the data buffer.
  - Each write access triggers a write in the flash memory.
  - 6-bit ECC is associated to each 16-bit data.

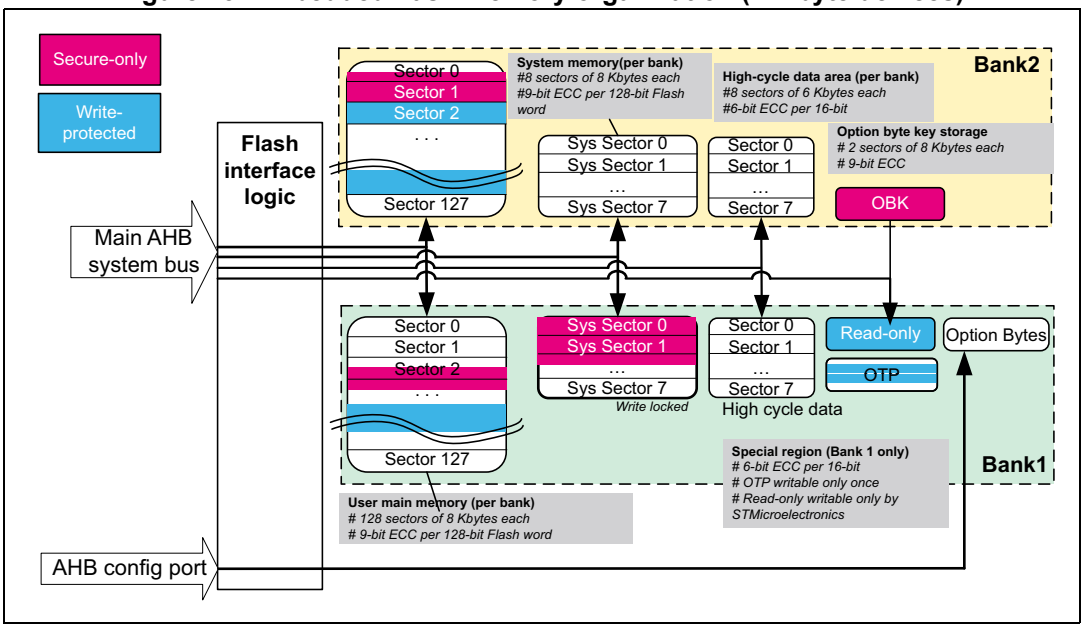
By default, all the AHB memory range is cacheable. For regions where caching is not practical (OTP, RO, data area), MPU must be used to disable local cacheability.

7.3.3 Flash memory architecture and usage

Flash memory architecture

Figure 23 shows the organization supported by the embedded flash memory. By default, all the AHB memory range is cacheable. For regions where caching is unpratical (OTP, read-only, data) use MPU to disable local cacheability.

Figure 23. Embedded flash memory organization (2-Mbyte devices)



The embedded flash nonvolatile memory is composed of:

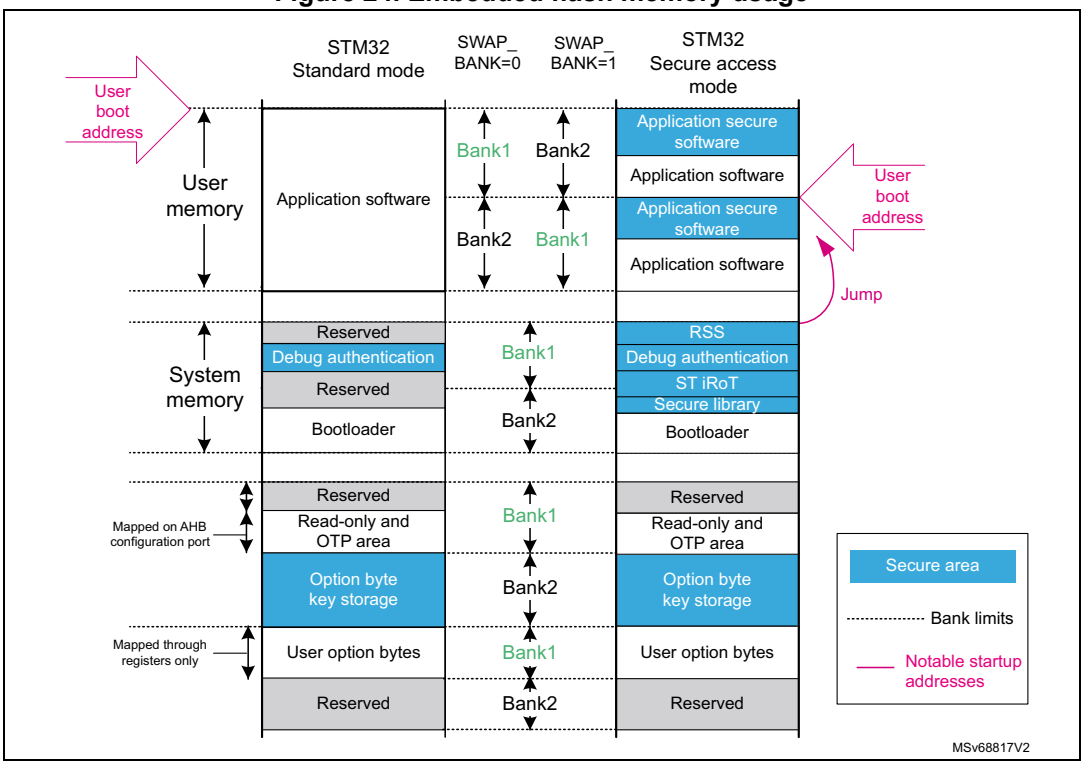
- A main memory block, organized in two banks. Each bank is divided in sectors of 8 Kbytes each, and features flash-word rows of 128 bits + 9 bits of ECC per word (see *Table 48*).
- A system memory block of 128 Kbytes, divided into two 64-Kbyte banks. Each bank is divided in eight 8-Kbyte sectors. The system flash memory is ECC protected (9-bit ECC per 128-bit word).
- A set of nonvolatile option bytes loaded at reset by the embedded flash memory and accessible by the application software only through the AHB configuration register interface.
- A 2-Kbyte one-time-programmable (OTP) area that can be written only once by the application software.
- A 2-Kbyte read-only area. It contains a unique device ID and product information.
- Two memory sectors (2 x 8 Kbytes) of secure key storage, OBKeys.
- Up to 16 sectors of user flash memory with high cycling capability (100 K cycles) for data, 8 sectors per bank.

The overall flash memory architecture and its corresponding access interface is summarized in *Table 48*.

Partition usage

Figure 24 shows how the embedded flash memory is used by STMicroelectronics and by the application software.

Figure 24. Embedded flash memory usage



User and system memories are used differently, according to product state and other option bytes settings:

- The user memory contains the application code and data, while the system memory is used with root secure services (RSS), the debug authentication code, and the STM32 bootloader. When a reset occurs, the core jumps to the boot address configured through the BOOT pin, the (SEC/NS)BOOTADD option bytes, and the product state.
- The unique boot entry (BOOT_UBE) is set to either ST iRoT located in the system flash memory, or iRoT in the user flash memory. This feature is available only on cryptography enabled devices.
- Secure service library in system flash memory is available in secure mode.
- If the debugger is attached to the product, the entry point is the debug authentication policy, used to unlock the device via the SBS when attached to debugger. A digital signature must be provided to perform a regression to product state, where debug is allowed.
- If (SEC/NS)BOOTADD is not yet configured, dedicated libraries can be used for secure boot. They are located in the system flash memory:
  - ST libraries in system flash memory assist the application software boot with special features, such as secure boot and secure firmware install (SFI-RSS)
  - ST iROT (immutable root of trust) secure software in user flash memory is used for secure firmware update and provisioning (SFU)

Note: For more information on option byte setup for boot, refer to Section 7.4.6.

7.3.4 FLASH read operations

Read operation overview

Read access to main and system flash memory operates as follows:

- There is a 128-bit read data buffer associated to each bank, which stores the last data read. If several consecutive read accesses request data belonging to the same flash data word (128 bits), data are read directly from the current data read buffer, without triggering additional flash read operations. This mechanism occurs each time a read access is granted. When a read access is rejected for security reasons, the corresponding read error response is issued by the embedded flash memory and no read operation to flash memory is triggered.
- The read data buffer is disabled when write access or OTP access, user option change request, OBK swap, OBK erase or other erase operation occur.

Read access to OTP, RO and flash high-cycle data operates as follows:

1. Flash data word of 137 bits is read and stored in a temporary buffer.
2. The interface parses the 137-bit data word, and selects the 16- or 32-bit data requested.
3. While parsing the 137-bit data word, (two) wait states are added, and the AHB bus is stalled.
4. If the application reads an OTP data or flash high-cycle data not previously written, **a double ECC error** is reported and **only a word full of set bits is returned** (see *Section 7.3.9* for details). The read data (in 16 bits) is stored in FLASH_ECCDR register, so that the user can identify if the double ECC error is due to a virgin data or a real ECC error.
5. Reading two times the same address triggers two reads in the flash memory.
6. For 8-bit accesses, an AHB bus error is generated.

Note: The embedded flash memory can perform single-error correction and double-error detection while read operations are being executed (see Section 7.3.8).

Instruction prefetch

The Cortex-M33 fetches instructions and literal pools (constants/data) over the C-Bus and through the instruction cache, if it is enabled. The prefetch block aims at increasing the efficiency of C-Bus accesses when the instruction cache is enabled, by reducing the cache refill latency.

Prefetch is efficient in case of sequential code; prefetch in the flash memory allows the next sequential instruction line to be read from the memory, while the current instruction line is being filled in instruction cache and executed by the CPU.

Prefetch is enabled by setting the PRFTEN bit in the FLASH access control register (FLASH_ACR). PRFTEN must be set only if at least one wait state is needed to access the flash memory.

Adjusting read timing constraints

The embedded clock must be enabled and running before reading data from a nonvolatile memory.

To correctly read data from the memory, the number of wait states (LATENCY) must be correctly programmed in the access control register (FLASH_ACR), according to the main AHB interface clock frequency, and the internal voltage range of the device (V_{core}).

Table 45 shows the correspondence between the number of wait states (LATENCY), the programming delay parameter (WRHIGHFREQ), the embedded flash memory clock frequency, and the supply voltage range.

Table 45. Recommended number of wait states and programming delay

Number of wait states (LATENCY)	Programming delay (WRHIGHFREQ)	Interface clock frequency vs. V_{CORE} range(1)			
		VOS3 range 0.95 to 1.05 V	VOS2 range 1.05 to 1.15 V	VOS1 range 1.15 to 1.26 V	VOS0 range 1.30 to 1.40 V
0 WS (1 FLASH clock cycle)	00	0 to 20 MHz	0 to 30 MHz	0 to 34 MHz	0 to 42 MHz
1 WS (2 FLASH clock cycles)		20 to 40 MHz	30 to 60 MHz	34 to 68 MHz	42 to 84 MHz
2 WS (3 FLASH clock cycles)	01	40 to 60 MHz	60 to 90 MHz	68 to 102 MHz	84 to 126 MHz
3 WS (4 FLASH clock cycles)		60 to 80 MHz	90 to 120 MHz	102 to 136 MHz	126 to 168 MHz
4 WS (5 FLASH clock cycles)	10	80 to 100 MHz	120 to 150 MHz	136 to 170 MHz	168 to 210 MHz
5 WS (6 FLASH clock cycles)		N/A	N/A	170 to 200 MHz	210 to 250 MHz

1. Voltage range from 1.26 to 1.30 V is not supported.

Adjusting system frequency

After power-on, the embedded flash memory is clocked by the 64 MHz high-speed internal oscillator (HSI), with a voltage range set at a scaled value of VOS3: a conservative 3 wait-state latency is specified in FLASH_ACR register (see *Table 45*).

When changing the bus frequency, the application software must follow the sequence described below, to tune the number of wait states required to access the memory.

To increase the CPU frequency:

1. If necessary, program the LATENCY and WRHIGHFREQ bits to the right value in the FLASH_ACR register, as described in *Table 45.*
2. Check that the new number of wait states is taken into account by reading back the FLASH_ACR register.
3. Modify the embedded flash memory clock source and/or the clock prescaler in the RCC_CFGR register of the reset and clock controller (RCC).
4. Check that the new embedded flash memory clock source and/or the new AHB clock prescaler value are taken in account by reading back the embedded flash memory clock source status and/or the prescaler value in the RCC_CFGR register of the reset and clock controller (RCC).

To decrease the CPU frequency:

1. Modify the embedded flash memory clock source and/or the clock prescaler in the RCC_CFGR register of reset and clock controller (RCC).
2. Check that the embedded flash memory new clock source and/or the new clock prescaler value are taken into account by reading back the embedded flash memory clock source status and/or the AHB interface prescaler value in the RCC_CFGR register of reset and clock controller (RCC).
3. If necessary, program the LATENCY and WRHIGHFREQ bits to the right value in FLASH_ACR register, as described in *Table 45.*
4. Check that the new number of wait states has been taken into account by reading back the FLASH_ACR register.

Error code correction (ECC)

The memory embeds an error correction mechanism. Single-error correction and double-error detection are performed for each read operation. For more details, refer to *Section 7.3.8*.

Read errors

When the ECC mechanism is unable to correct the read operation, the memory reports read errors, as described in *Section 7.9.10*.

Read interrupts

See *Section 7.10* for details.

7.3.5 FLASH program operations

Program operation overview

Program operation consists in issuing write commands. The memory supports the execution of only one write-command at a time. Write-while-write is not supported. Nothing prevents overwriting a non-virgin flash word, but this is not recommended. The result may lead to invalid data and inconsistent ECC code.

User flash, OBK storage and system flash memories sectors

For the user and system flash memories, 9-bit ECC is associated to each 128-bit data flash word. In this case, the embedded flash memory must always perform write operations to nonvolatile memory with a 128-bit word granularity. Once the write buffer is full (128 bits), the Busy flag is set, and a programming operation is triggered.

There is a write buffer common to Bank1 and 2, which supports multiple write-access types (128, 64, 32, 16 or 8 bits). The application can decide to write from 8 bits to 128 words. In this case, a force-write mechanism to the 128 bits + ECC is used (see FW bit of FLASH_NS/SECCR register).

When the write request is issued to the memory, any new write request stalls the main AHB bus. Moreover, while a write operation is ongoing, any new read request to the same bank stalls the main AHB bus.

OTP, RO, flash high-cycle data

When the target memory is OTP, RO and flash high-cycle data sectors, 6-bits ECC code is associated to each 16-bit data flash word. The embedded flash memory supports 16- or 32-bit write operations (8-bit write operations are not supported).For 8-bit accesses, write accesses are ignored. There is no write data buffer. Each write access triggers a write in the flash memory.

Note: The OTP area is typically write-protected on the final product, as described in Section 7.3.9.

The write protection check is performed at the reception of the write request (during address phase).Write protection is not checked anymore at the output of the write buffer. If a write protection violation is detected, the write operation is canceled, and write protection error (WRPERR) is raised in FLASH_NS/SECSR register.

Note: For write protections of main flash, ICP, and OTP see Section 7.6.

Monitoring ongoing write operations

The application software can use a status flag located in FLASH_NS/SECSR to monitor ongoing write operations. Since only one operation is possible at a time, this flag indicates if any operation (write, erase, option change) is ongoing, whatever the bank.

- **BSY**: indicates that an effective write, erase, option byte change, OBK swap, OBK alt sector erase is ongoing in the nonvolatile memory. This flag is not dedicated to a specific bank. It is set when an operation is starting on the memory, whatever the bank. An operation is triggered by:
  - An erase (FLASH_NS/SECCR.STRT)
  - A write (FLASH_NS/SECCR.PG + AHB write)
  - An option modification (FLASH_OPTCR.OPTSTRT)
  - OBKeys sector swap and OBKeys sector erase (FLASH_NS/SECOBKCFGR.SWAP_SECT_REQ and FLASH_NS/SECOBKCFGR.ALT_SECT_ERASE)

They are cleared when the current operation ends, or in case of error.

- **WBNE**: this bit indicates that the embedded flash memory is waiting for new data to complete the 128-bit write buffer. In this state the write buffer is not empty. It is reset as soon as the application software fills the write buffer, or forces the writes by using FW bit in FLASH_NS/SECCR, or an error is detected. When WBNE is high, it is not possible to launch an erase, an option modification, an OBK swap or OBK alternate sector erase operation on flash memory.
- **DBNE**: this bit indicates that the data buffer for parsing 16-bits data is not empty:
  - 16-bit data write access is received, and the data buffer is being filled. It is set at the receipt of a valid write access, and reset as soon as the write request preparation has been processed.

Note: If the memory is busy at the receipt of the AHB write request, the CPU execution is stalled.

Enabling write operations

Before programming the user flash memory in Bank1 or in Bank2, the application software must ensure that the PG bit is set to 1 in FLASH_NS/SECCR. If not, an unlock sequence must be used (see *Section 7.6.7*), and the PG bit must be set.

When an option byte or an option byte key must be modified, or a mass erase must be started, the application software must ensure that FLASH_OPTCR is unlocked. If this is not the case, an unlock sequence must be used (see *Section 7.6.7*).

A separate mechanism with similar use exists for FLASH_NS/SECOBKCFGR. The control register must be unlocked prior to the start of any OBKeys storage modification. Writing the correct sequence to the FLASH_NSOBKKEYR unlocks FLASH_NSOBKCFGR. FLASH_SECOBKKEYR is linked to FLASH_SECOBKCFGR, as described in *Section 7.6.7*.

Note: The application software must not unlock an already unlocked register, otherwise this register remains locked until the next system reset.

If needed, the application software can update the programming delay, as described in *Adjusting programming timing constraints*.

Writing to the FLASH control register FLASH_NS/SECCR and FLASH_OPTCR

The FLASH_NS/SECCR, FLASH_OPTCR and FLASH_NS/SECOBKCFGR registers are not accessible in write mode when the BSY bit is set. Any attempt to write these registers while the BSY bits is set causes the AHB bus to stall until SEC/NSBSY bit is cleared.

Single-write sequence

The recommended single-write sequence is the following:

1. Make sure protection mechanism does not prevent programming
2. Check that no memory operations are ongoing by checking the BSY bit in the FLASH_NS/SECSR register and CDBNE bits in the FLASH_NS/SECSR register. Check that the write buffer is empty by checking the WBNE bit in the FLASH_NS/SECSR register
3. Check and clear all the error flags due to previous programming/erase operation
4. Unlock the FLASH_NS/SECCR register, as described in *Section 7.6.7* (only if this register is not already unlocked)
5. Enable write operations by setting PG bit in the FLASH_NS/SECCR register
6. Write one flash-word at aligned address

Note: NS/SECWBNE flag indicates if the 128-bit write buffer is waiting for new data.

Note: No erase request, options change request, OBK operation is allowed between the first write and the completion of the write operation.

7. Wait for the BSY bit to be cleared in the corresponding FLASH_NS/SECSR register
8. Clear PG bit in FLASH_NS/SECCR register if there are not any more programming requests

If step *6* is executed incrementally (for example byte per byte), the write buffer can become partially filled. In this case the application software can decide to force-write what is stored in the write buffer by using FW bit in FLASH_NS/SECCR register. In this particular case, the unwritten bits are automatically set to 1. If no bit in the write buffer is cleared to 0, the FW bit has no effect.

Note: The usage of a force-write operation prevents the application from updating, in a later stage, the missing bits with a value different from 1. This can lead to unexpected or inconsistent data, or ECC.

Adjusting programming timing constraints

Program operation timing constraints depend of the memory clock frequency, which directly impacts the performance. If timing constraints are too tight, the nonvolatile memory does not operate correctly, if they are too lax, the programming speed is not optimal.

The user must therefore trim the optimal programming delay through the WRHIGHFREQ parameter in the FLASH_ACR register. Refer to *Table 45* for the recommended programming delay, depending upon the memory clock frequency.

FLASH_ACR configuration register is common to both banks.

The application software must check that no program/erase operation is ongoing before modifying WRHIGHFREQ.

Caution: Modifying WRHIGHFREQ while programming/erasing the memory can corrupt its content.

Programming errors

When a program operation fails, an error is reported, as described in *Section 7.9*.

Programming interrupts

See *Section 7.10: FLASH interrupts* for details.

7.3.6 FLASH erase operations

Erase operation overview

The memory can perform erase operations on 8-Kbyte user sectors, on one user flash memory bank, or on two user flash memory banks (for example mass erase). For more details in user flash memory, ICP, user options and OTP erase protection, see *Section 7.6*.

Erase commands are issued through the AHB configuration interface. The memory supports one operation at a time, if it receives simultaneously a write and an erase request an error flag is raised, and both operations are canceled. See *Section 7.9* for details.

After successful completion of erase, all the bytes in the erased flash sectors are set to the flash default 0xFF value.

Erase and WRP

If the application software attempts to erase a write-protected user sector, the sector erase operation is aborted, and the WRPERR flag is raised in the FLASH_NS/SECSR register, as described in *Section 7.9.2*.

Flash busy

Busy signals is described in *Monitoring ongoing write operations*.

Writing to the FLASH control register FLASH_NS/SECCR and FLASH_OPTCR

Refer to *Writing to the FLASH control register FLASH_NS/SECCR and FLASH_OPTCR*.

Enabling erase operations

Before erasing a sector, the application software must make sure that FLASH_NS/SECCR is unlocked. If this is not the case, an unlock sequence must be used (see *Section 7.6.7*).

Note: The application software must not unlock a register that is already unlocked, otherwise this register remains locked until next system reset. This can be used to deliberately lock-out a register from further accesses.

Similar constraints apply to bank erase requests.

Standard flash sector erase sequence

To erase an 8- or 6-Kbyte data user sector without security protections, proceed as follows:

1. Make sure protection mechanism does not prevent sector erase (WRP, secure flag, HDP).
2. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_NSSR register, and that the write buffer is empty by checking the WBNE bit in the FLASH_NSSR register.
3. Check and clear all the non-secure error flags due to previous programming/erase operation. Refer to *Section 7.9* for details.
4. Unlock the FLASH_NSCR register, as described in *Section 7.6.7* (only if this register is not already unlocked).
5. Set the BKSEL bit, the SER bit and SNB bitfield in the FLASH_NSCR register. BKSEL indicates in which physical bank sector must be erased, then SER indicates a sector erase operation, while SNB contains the target sector number. In case of data sector, use the number of the corresponding regular sector.
6. Set the STRT bit in the FLASH_NSCR register.
7. Wait for the BSY bit to be cleared in the FLASH_NSSR register.
8. STRT bit is automatically cleared at the end of the sector erase, or in case of error.
9. Clear SER in FLASH_NSCR register if there are not anymore sector erase request to be issued.

Note: If another erase flag is requested simultaneously to the sector erase, a PGSERR error is generated.

Secure flash memory sector erase sequence

To erase an 8- or 6-Kbyte data secure configured user sector, proceed as follows:

1. Make sure write protection or HDP mechanism does not prevent sector erase.
2. Ensure that no memory operations are ongoing (by checking the BSY and DBNE bits in the FLASH_SECSR register), and that the write buffer is empty (by checking the WBNE bit in the FLASH_SECSR register).
3. Check and clear all the secure error flags due to previous programming/erase operations. Refer to *Section 7.9* for details.
4. Unlock the FLASH_SECCR register, as described in *Section 7.6.7* (only if this register is not already unlocked).
5. Set the BKSEL bit, the SER bit and SNB bitfield in the FLASH_SECCR register. BKSEL indicates in which physical bank sector must be erased, then SER indicates a sector erase operation, while SNB contains the target secure sector number. In case of data sector, use the number of the corresponding regular sector.

6. Set the STRT bit in the FLASH_SECCR register.
7. Wait for the BSY bit to be cleared in the FLASH_SECSR register.
8. STRT bit is automatically cleared at the end of the sector erase or in case of error.
9. Clear SER in FLASH_SECCR register if there are not any secure sector erase request to be issued.

Note: If another erase flag is requested simultaneously to the sector erase, a PGSERR error is generated.

Standard flash memory bank erase sequence

To erase bank where no sector is configured as secure:

1. Make sure protection mechanism does not prevent sector erase.
2. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_NSSR register and that the write buffer is empty by checking the WBNE bit in the same register.
3. Check and clear all the non-secure error flags due to previous programming/erase operation. Refer to *Section 7.9* for details.
4. Unlock the FLASH_NSCR register, as described in *Section 7.6.7* (only if this register is not already unlocked).
5. Set the BKSEL bit and the BER bit in the FLASH_NSCR register to the targeted physical bank (swap setting is ignored).
6. Set the STRT bit in the FLASH_NSCR register to start the bank erase operation. Then wait until the BSY bit is cleared in the FLASH_NSSR register.
7. STRT bit is automatically cleared at the end of erase sequence or in case of error.
8. Clear BER in FLASH_NSCR register if there is not other bank erase request to be issued.

Secure flash memory bank erase sequence

To erase bank where all sectors are configured with secure flag:

1. Make sure protection mechanism does not prevent sector erase (for mass erase the HDP is also considered, it cannot be fully executed from HDPL= 2, 3 if HDP is defined).
2. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_SECSR register and that the write buffer is empty by checking the WBNE bit in the same register.
3. Check and clear all the secure error flags due to previous programming/erase operation. Refer to *Section 7.9* for details.
4. Unlock the FLASH_SECCR register, as described in *Section 7.6.7* (only if this register is not already unlocked).
5. Set the BKSEL bit and the BER bit in the FLASH_SECCR register to the targeted physical bank (swap setting is ignored).
6. Set the STRT bit in the FLASH_SECCR register to start the bank erase operation. Then wait until the BSY bit is cleared in the FLASH_SECSR register.
7. STRT bit is automatically cleared at the end of erase sequence or in case of error.
8. Clear BER in FLASH_SECCR register if there is not other bank erase request to be issued.

Flash mass erase sequence

To erase all sectors of both banks, using non-secure access, all sectors must be configured as non-secure. The application software can set the MER bit to 1 in FLASH_NSCR register, as described below:

1. Make sure protection mechanisms do not prevent mass erase (*Section 7.6*).
2. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_NSSR register and that the write buffer is empty by checking the WBNE bit in the FLASH_NSSR register.
3. Check and clear all the non-secure error flags due to previous programming/erase operation. Refer to *Section 7.9* for details.
4. Unlock the FLASH_NSCR register as described in *Section 7.6.7* (only if the registers are not already unlocked).
5. Set the MER bit to 1 in FLASH_NSCR register.
6. Set the STRT bit in the FLASH_NSCR register. Then wait until BSY bit is cleared in the FLASH_NSSR register.
7. STRT bit is cleared automatically at the end of the erase sequence, or in case of error.
8. Clear MER in FLASH_NSCR register.

Secure flash memory mass erase sequence

To erase all sectors of both banks simultaneously, using secure access, all sectors must be configured as secure. The application software can set the MER bit to 1 in FLASH_SECCR register, as described below:

1. Make sure protection mechanisms do not prevent mass erase (*Section 7.6*).
2. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_SECSR register and that the write buffer is empty by checking the WBNE bit in the FLASH_SECSR register.
3. Check and clear all the secure error flags due to previous programming/erase operation. Refer to *Section 7.9* for details.
4. Unlock the FLASH_SECCR register as described in *Section 7.6.7* (only if the registers are not already unlocked).
5. Set the MER bit to 1 in FLASH_SECCR register.
6. Set the STRT bit in the FLASH_SECCR register. Then wait until BSY bit is cleared in the FLASH_SECSR register.
7. STRT bit is cleared automatically at the end of the erase sequence, or in case of error.
8. Clear MER in FLASH_SECCR register.

Note: Mass and bank erase also erase high-cycle data sectors aliased from the erased bank.

7.3.7 FLASH parallel operations

As the memory is divided into two independent banks, the embedded flash memory interface supports a read in one bank while a write (RWW. read while write) or an erase is executed in the other bank. It does not support write-while-write, nor read-while-read. Same is valid for the high-cycle data area, system flash libraries, or the OBK (located on Bank2).

In all cases, the sequences described in *Section 7.3.4*, *Section 7.3.5* and *Section 7.3.6* apply.

7.3.8 FLASH error protections

Error correction codes (ECC)

The embedded flash memory supports an error correction code (ECC) mechanism, based on the SECDED algorithm, to correct single errors and detect double errors.

This mechanism uses nine ECC bits per 128-bit flash word, and applies to user and system memory.For read-only, OTP, flash high-cycle data, a stronger six ECC bits per 16-bit word is used. A double ECC error is generated for an OTP or flash high-cycle data virgin word (for example a word with 22 bits at 1). When this OTP or flash high-cycle data word is no more virgin, the ECC error disappears.

More specifically, during each read operation from a 128-bit flash word, the embedded flash memory retrieves the 9-bit ECC information, computes the ECC of the flash word, and compares the result with the reference value. If they do not match, the corresponding ECC error is raised, as described in *Section 7.9.10*.

During each program operation, a 9-bit ECC code is associated to each 128-bit data flash word, and the resulting 137-bit flash word information is written in nonvolatile memory.

A similar mechanism applies to read-only and OTP areas, but with 6-bit ECC for 16-bit data.

7.3.9 OTP and RO memory access

OTP and RO memory are accessed through main AHB interface. The OTP is accessible at addresses 0x08FF_F000 to 0x08FF F7FF, and the read-only section is accessible from 0x08FF F800 to 0x08FF_FFFF.

FLASH one-time programmable area

The embedded flash memory offers a 2048-byte memory area dedicated to application non-confidential, one-time programmable data (OTP). This area is composed by 1024 words of 16 bits (plus 6 bits of ECC). It cannot be erased, and can be written only once. The OTP area can be accessed through the main AHB interface from address 0x08FF F000 to 0x08FF F7FE.

OTP data can be programmed by the application software by 16-bit chunks. Overwriting an already programmed16-bit half-word can lead to data and ECC errors, and is therefore not supported.

Note: The OTP area is virgin when the device is delivered by STMicroelectronics.

When reading OTP data with a single error corrected or a double error detected, the embedded flash memory reports read errors, as described in *Section 7.9.10*.

When reading OTP data not written by the application software (such as virgin OTP), the ECC correction reports a double-error detection (ECCD), and the data are to be found in the FLASH_ECCDR register. ECCD implies an NMI raised.

OTP write protection

OTP data are organized as 32 blocks of 32 OTP words, as shown in *Table 46*. An entire OTP block can be protected (locked) from write accesses by setting the LOCKBLi bit (i = 0 to 31) corresponding to each OTP block in the FLASH_OTPBLR register. A block can be write-protected, if it has been programmed (even partially) or not.

The OTP block locking operation is irreversible, and independent from the product life state.

Note: The OTP area can be accessed only in read mode.

Table 46. Flash memory OTP organization

OTP block	AHB address	AHB word		Lock bit
		[31:16]	[15:0]	
Block 0	0x08FF F000	OTP001	OTP000	LOCKBL0
	0x08FF F004	OTP003	OTP002	
	...			
	0x08FF F03C	OTP031	OTP030	
Block 1	0x08FF F040	OTP033	OTP032	LOCKBL1
	0x08FF F044	OTP035	OTP034	
	...			
	0x08FF F07C	OTP063	OTP062	
Block 2	0x08FF F080	OTP065	OTP064	LOCKBL2
	0x08FF F084	OTP067	OTP066	
	...			
	0x08FF F0BC	OTP95	OTP94	
...				
Block 31	0x08FF F7C0	OTP993	OTP992	LOCKBL31
	0x08FF F7C4	OTP995	OTP994	
	...			
	0x08FF F7FC	OTP1023	OTP1022	

OTP write sequence

Follow the sequence below to write an OTP word:

1. Check that no memory operations are ongoing by checking the BSY bit in the FLASH_NSSR register and that the data buffer is empty by checking the DBNE bit in the FLASH_NSSR register.
2. Check and clear all the error flags due to previous programming/erase operation.
3. Set PG bit in the FLASH_NSCR register.
4. Check the protection status of the target OTP word (see *Table 46*). The corresponding LOCKBLi bit must not be set to 1.
5. Write two OTP words (32 bits) corresponding to the 4-byte aligned address shown in *Table 46*. Alternatively, the application software can program separately the 16-bit MSB or 16-bit LSB. In this case the first 16-bit write operation starts immediately, without waiting for the second one.
6. Wait for the BSY bit to be cleared in the FLASH_NSSR register.
7. Clear PG bit in FLASH_NSCR register if there is not any programming request anymore in the bank.
8. Optionally, lock the OTP block using LOCKBLi to prevent further data changes.

Note: Do not write twice an OTP 16-bit word, otherwise an ECC error is generated.

Writing OTP data at byte level is not supported, and generates a bus error.

To avoid data corruption, it is important to complete the OTP write process (for example by reading back the OTP value), before starting an option change.

Flash read-only area

The embedded flash memory offers a 2-Kbyte area to store read-only data. This area can be accessed through the AHB main port, and is protected by a robust ECC scheme, as detailed in *Section 7.3.8*.

The read-only information (programmed by STMicroelectronics) that can be used by the application software is detailed in *Table 47*.

Table 47. Read-only public data organization

Read-only data name	Address	Comment
Unique device ID	0x08FF F800	U_ID[31:0]
	0x08FF F804	U_ID[63:32]
	0x08FF F808	U_ID[96:64]
Flash memory size/ package	0x08FF F80C	Flash memory size[15:0] / Package code[15:0]
Reserved	0x08FF F810 to 0x08FF FFFF	Reserved information

7.3.10 Flash high-cycle data

The embedded flash memory offers up to 96 Kbytes (maximum) memory area with high-cycling capability (100 kcycles) to store data and emulate EEPROM. It can be accessed through the AHB system port from address 0x0900_0000 to 0x0901_7FFF (see *Figure 25*). It is mapped in the 8 (or 4) last sectors of Bank 1 and 2. This area is protected by a robust 6-bit ECC, enabling a 16-bit read and write granularity, at the expense of having sector size shrunk to 6 Kbytes.

A threshold per bank (EDATA(1/2)_STRT) is programmable to determine the beginning of the data flash area. By default, the whole memory is used for code.

For example, if 48 Kbytes of data are needed in Bank 1, set EDATA1_EN to 1, and EDATA1_STRT to 7. If no data are needed in Bank2, set EDATA2_EN to 0. In this case the data are accessible from address 0x0900_0000 to 0x0900_BFFF for Bank 1.

For greater efficiency, it is recommended to use sector on the other bank for flash high-cycle data, so that the application benefits from RWW capability of the dual-bank arrangement.

When SWAP_BANK feature is enabled, the banks are swapped: the flash high-cycle data in Bank 2 are accessible from 0x0900_000 to 0x0900_BFFF, and the data in Bank 1 are accessible from 0x0900_C000 to 0x0901_7FFF.

A bus error is generated on:

- Attempt to access an address between 0x0900_0000 to 0x0901_7FFF and this address is not valid (EDATA(1/2)_EN not enabled or EDATA(1/2)_STRT not correct).
- Attempt to fetch instructions from flash high-cycle data area.

Erasing the data area sector is possible by normal erase request for the corresponding user flash sector (120-127 for the STM32H562/563/573xx devices, 24-31 for the STM32H523/33xx devices).

Protections and security of high-cycle area are detailed in *Section 7.6.9*.

Figure 25. Flash high-cycle data memory map on 2-Mbyte devices

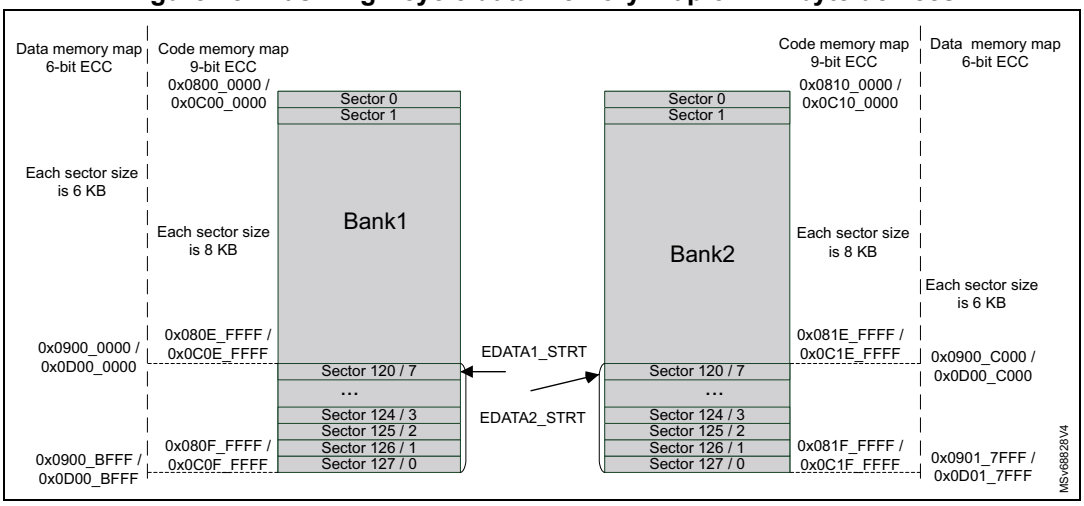


Figure 26. Flash high-cycle data memory map on 1-Mbyte devices

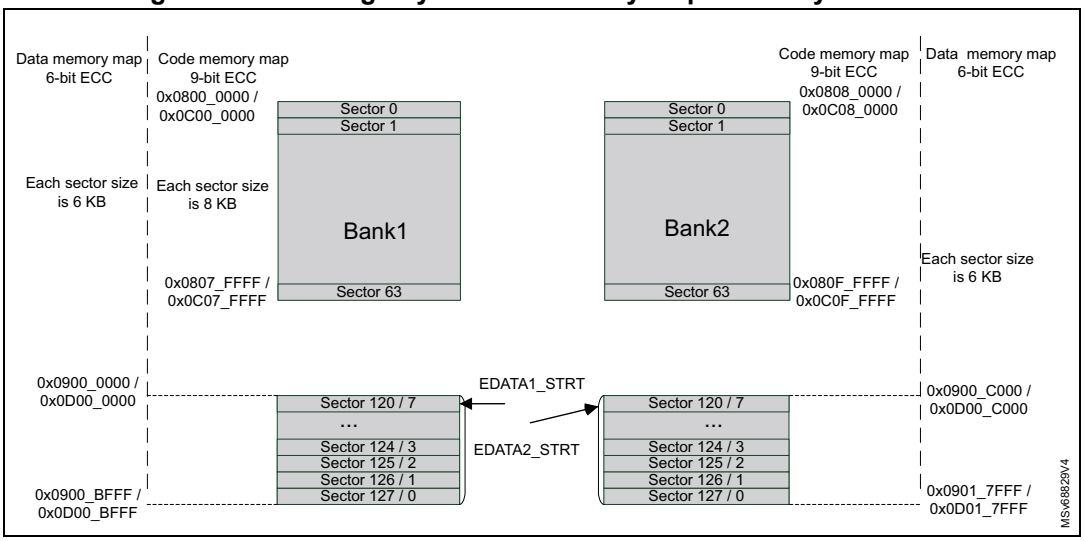


Figure 27. Flash high-cycle data memory map on 512-Kbyte devices

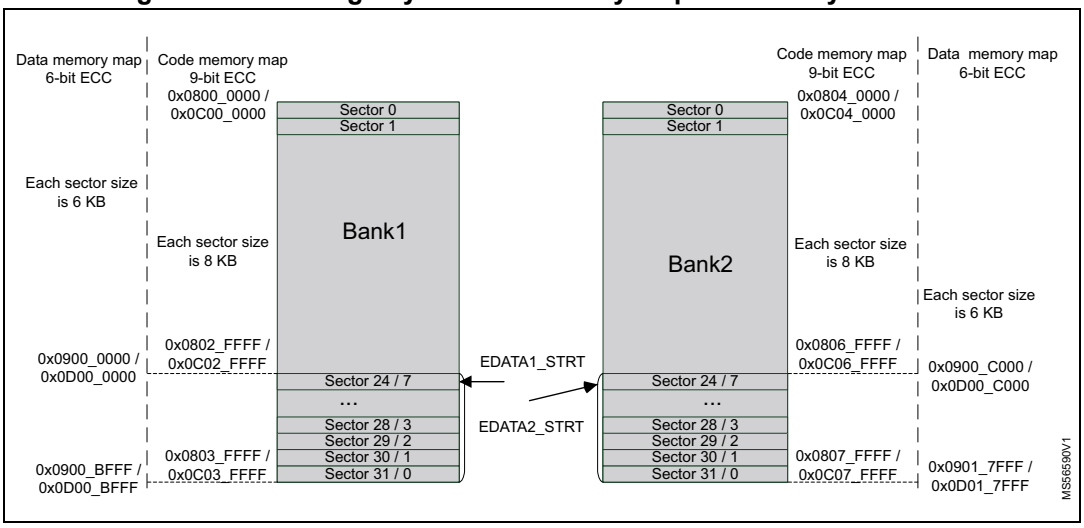
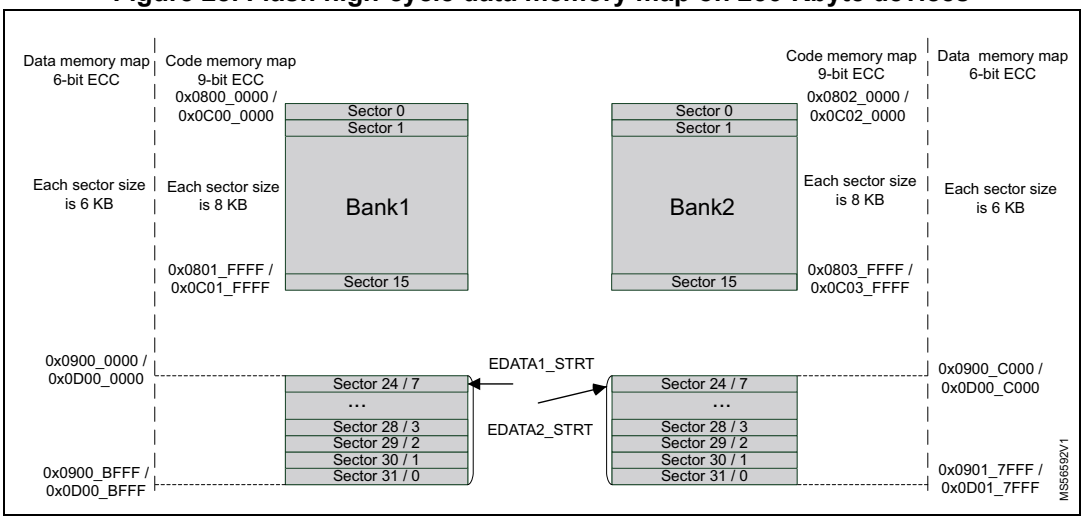


Figure 28. Flash high-cycle data memory map on 256-Kbyte devices



Note: When flash high-cycle data area on Bank1 is enabled, the code memory map is not continuous from Bank1 to Bank2 on 2M- and 512K-byte devices.

7.3.11 Flash bank swapping

Bank1 and Bank2 can be swapped for the user flash. This feature can be used after a firmware upgrade to restart the device on the new firmware. Bank swapping is an user option byte flag controlled by the SWAP_BANK bit of the FLASH_OPTCR register.

Bank specific settings for data area and security attributes follow the original bank and its contents. Control bit BKSEL always refers to physical bank, not the SWAP_BANK setting.

Table 48 shows the accessible memory, depending upon the SWAP_BANK bit value.

Table 48. Memory map and swapping options (STM32H562/563/573xx devices)

Area	Corresponding bank		Start address	End address	Size (bytes)	Region name
	SWAP_BANK = 0	SWAP_BANK = 1				
User main memory	Bank1	Bank2	0x0800 0000	0x0800 1FFF	8 K	Sector 0
			0x0800 2000	0x0800 3FFF	8 K	Sector 1
			...	...	...	...
			0x080F E000	0x080F FFFF	8 K	Sector 127
	Bank2	Bank1	0x0810 0000	0x0810 1FFF	8 K	Sector 0
			0x0810 2000	0x0810 3FFF	8 K	Sector 1
			...	...	...	...
			0x081F E000	0x081F FFFF	8 K	Sector 127
System memory	Bank1		0x0BF8 0000	0x0BF8 1FFF	8 K	System 1 Sector 0
			0x0BF8 2000	0x0BF8 3FFF	8 K	System 1 Sector 1
			...	...	...	...
			0x0BF8 E000	0x0BF8 FFFF	8 K	System 1 Sector 7
	Bank2		0x0BF9 0000	0x0BF9 1FFF	8 K	System 2 Sector 0
			0x0BF9 0000	0x0BF9 3FFF	8 K	System 2 Sector 1
			...	...	...	...
			0x0BF9 E000	0x0BF9 FFFF	8 K	System 2 Sector 7

Table 49. Memory map and swapping options (STM32H523/533xx devices)

Area	Corresponding bank		Start address	End address	Size (bytes)	Region name
	SWAP_BANK = 0	SWAP_BANK = 1				
User main memory	Bank1	Bank2	0x0800 0000	0x0800 1FFF	8 K	Sector 0
			0x0800 2000	0x0800 3FFF	8 K	Sector 1
			⋮	⋮	⋮	⋮
			0x0803 E000	0x0803 FFFF	8 K	Sector 31
	Bank2	Bank1	0x0804 0000	0x0804 1FFF	8 K	Sector 0
			0x0804 2000	0x0804 3FFF	8 K	Sector 1
			⋮	⋮	⋮	⋮
			0x0807 E000	0x0807 FFFF	8 K	Sector 31

Table 49. Memory map and swapping options (STM32H523/533xx devices) (continued)

Area	Corresponding bank		Start address	End address	Size (bytes)	Region name
	SWAP_BANK = 0	SWAP_BANK = 1				
System memory	Bank1		0x0BF8 0000	0x0BF8 1FFF	8 K	System 1 Sector 0
			0x0BF8 2000	0x0BF8 3FFF	8 K	System 1 Sector 1
			...	...	...	...
			0x0BF8 E000	0x0BF8 FFFF	8 K	System 1 Sector 7
	Bank2		0x0BF9 0000	0x0BF9 1FFF	8 K	System 2 Sector 0
			0x0BF9 0000	0x0BF9 3FFF	8 K	System 2 Sector 1
			...	...	...	...
			0x0BF9 E000	0x0BF9 FFFF	8 K	System 2 Sector 7

The SWAP_BANK bit in FLASH_OPTCR register is loaded from the SWAP_BANK option bit only after system reset or POR.

To change the SWAP_BANK bit (for example to apply a new firmware update), follow the sequence below:

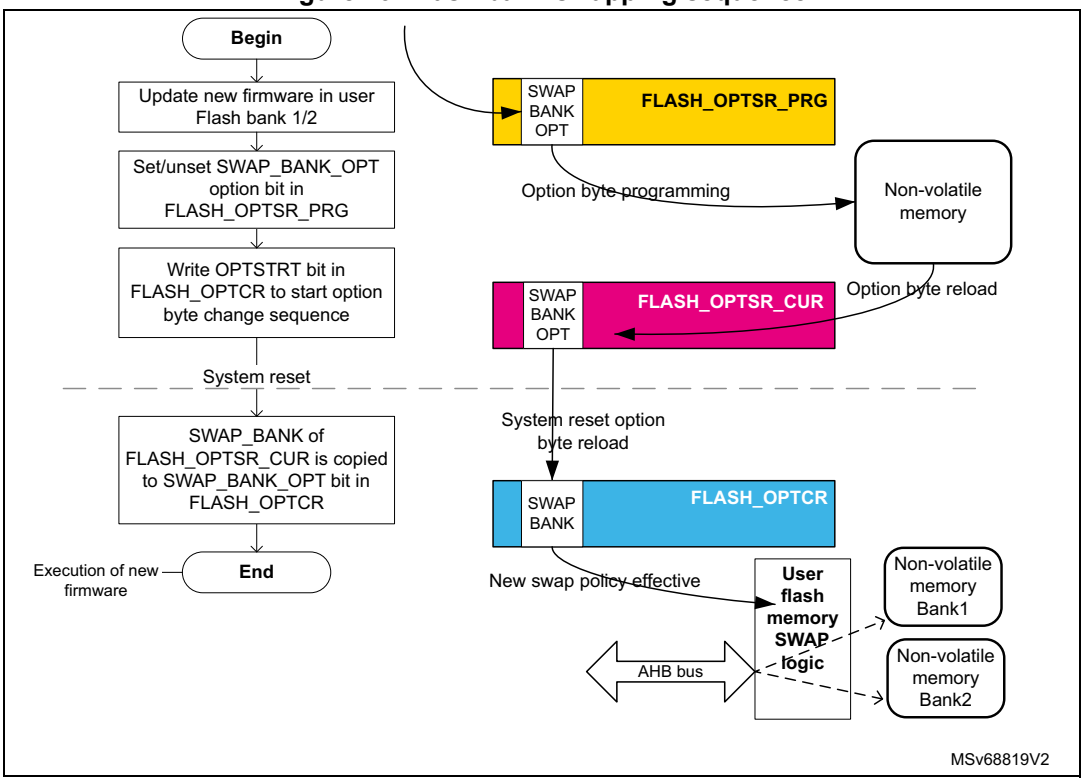
1. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_NS/SECSR register and that the write buffer is empty by checking the WBNE bit in the FLASH_NS/SECSR register.
2. Clear all error flags due to a previous operation.
3. Unlock OPTLOCK bit, if not already unlocked.
4. Set the new desired SWAP_BANK value in the FLASH_OPTSR_PRG register.
5. Start the option byte change sequence by setting the OPTSTRT bit in the FLASH_OPTCR register.
6. Once the option byte change has completed, FLASH_OPTSR_CUR contains the expected SWAP_BANK value, but SWAP_BANK bit in FLASH_OPTCR has not yet been modified and the bank swapping is not yet effective.
7. Force a system reset or a POR. When the reset rises up, the bank swapping is effective (SWAP_BANK value updated in FLASH_OPTCR) and the new firmware shall be executed.

Note: The SWAP_BANK bit in FLASH_OPTCR is read-only, and cannot be modified by the application software.

The SWAP_BANK option bit in FLASH_OPTSR_PRG can be modified whatever the product state. Instead of being locked by PRODUCT_STATE, it is locked by (NS/SEC)BOOT_LOCK User OB.

Figure 29 gives an overview of the bank swapping sequence.

Figure 29. Flash bank swapping sequence



7.3.12 FLASH reset and clocks

Reset management

The embedded flash memory can be reset by a core domain reset, driven by the reset and clock control (RCC). The main effects of this reset are the following:

- All registers, except for option byte registers, are cleared, including read and write latencies. If the bank swapping option is changed, it is applied.
- Most control registers are automatically protected against write operations. To unprotect them, new unlock sequences must be used as described in *Section 7.6.7*.

The memory can be reset by a power-on core domain reset, driven by the reset and clock control (RCC). When the reset falls, all option byte registers are reset. When the reset rises up, the option bytes are loaded, potentially applying new features. During this loading sequence, the device remains under reset, and the memory is not accessible.

The reset signal can have a critical impact on the memory: the content is not guaranteed if a device reset occurs during a write or erase operation.

Reset occurring during flash operation

If a reset occurs during a flash operation (programing, erase or option change), the content of the memory is not guaranteed. It is mandatory for integrity to restart the operation. The status register FLASH_OPSR gives information about operations interrupted by a reset.

FLASH_OPSR.CODE_OP gives opcode of operation. *Table 50* indicates how to use FLASH_OPSR, and which operation is required.

Table 50. Recommended reactions to FLASH_OPSR contents

CODE_OP	Operation interrupted	OTP_OP	SYSF_OP	BK_OP	DATA_OP	Flash area	ADDR_OP min	ADDR_OP max	Recommended action
0x000	No operation ongoing while reset	0	0	0	0	-(1)	-	-	No extra action
0x001	Write operation	0	0	0/1	0	User flash	0x0000	0xFFFF	Erase sector and rewrite
		0	1	0/1	0	System flash	0x0000	0x0FFF	
		1	0	0	0	OTP	0x0600	0x07FF	
		1	0	1	0	OBKeys(2)	0x0000	0x03FF	
		0	0	0/1	1	Data area(3)	0xF000	0xFFFF	
0x010	OBK alternate sector erase	0	0	1	0	OBKeys	0	0	Relaunch the alternate sector erase
0x011	Sector erase(4)	0	0	0/1	0	User flash	0x0000	0xFFFF	Relaunch sector erase
		0	1	0/1	0	System flash	0x0000	0x0FFF	
0x100	Bank erase	0	0	0/1	0	User flash	-	-	Relaunch bank erase
0x101	Mass erase	0	0	0	0		-	-	Relaunch mass erase
0x110	Option change	0	0	0	0	User configuration	-	-	New attempt on option change
0x111	OBK swap sector	0	0	1	0	OBKeys	-	-	Erase alternate sector, reprogram new OBKeys and relaunch swap

1. Dash represents “does not matter”.
2. Depends on current OBK sector.
3. Depends on EDATA setting in the OB.
4. Addresses indicated are aligned to sector start, data area sectors are erased by erase request to corresponding user flash memory sector.

Clock management

The memory uses the microcontroller system clock (sys_ck), here the AHB interface clock.

7.4 FLASH option bytes

7.4.1 About option bytes

The memory includes a set of nonvolatile option bytes. They are loaded at power-on reset and can be read and modified only through configuration registers.This section details:

- when option bytes are loaded
- how application software can modify them
- their detailed list, together with their initial values (before the first option byte change, user default configuration).

7.4.2 Option bytes loading

There are multiple ways of loading the option bytes:

- **Power-on wake-up**

  When the device is first powered, the embedded flash memory automatically loads all the option bytes. During the option byte loading sequence, the device remains under reset and the embedded flash memory cannot be accessed.
- **Wake-up from system Standby**

  When the core power domain, which contains the embedded flash memory, is switched from Standby mode to Run mode, the embedded flash memory behaves as during a power-on sequence. **During loading time the device is not under reset, unlike power-on sequence.**
- **Dedicated option byte reloading by the application**

  When the user application successfully modifies the option byte content through the memory registers, the nonvolatile option bytes are programmed and the memory automatically reloads all option bytes to update the option registers.

Note: The option byte read sequence is protected by error correction code.In case of error, the option bytes are loaded with default values (see Section 7.4.3), different from the initial values (user default configuration), and more restrictive.

7.4.3 Option bytes modification

Changing user option bytes

A user option byte change operation can be used to modify the configuration and the protection settings saved in the nonvolatile option byte area.

There are several rules enforced when attempting to change a user option byte, they are summarized in *Section 7.4.7*. Failing to stick to those rules usually results in errors, described in *Section 7.9.12*.

The embedded flash memory features two sets of option byte registers:

- The first register set contains the current values of the option bytes. Their names have the _CUR extension. All these registers are read-only. Their values are automatically loaded from the nonvolatile memory after power-on reset, wake-up from system standby or after an option byte change operation.
- The second register set allows the modification of the option bytes. Their names contain the _PRG extension. All “_PRG” registers can be accessed in read/write mode.

When the OPTLOCK bit in FLASH_OPTCR register is set, it is not possible to modify the FLASH_XXX_PRG registers.

When OPTSTRT bit is set to 1, the memory checks the programming sequence (PGSERR) and the conditions described in *Section 7.4.7* (OPTCHANGEERR). If no error has been detected (PGSERR/OPTCHANGEERR), the flash interface launches the option byte modification, and updates the option byte registers with _CUR extension.

If one of the condition described in *Section 7.4.7*, *Section 7.9.12* or *Section 7.9.5*, alternatively *Section 7.9.4* is not respected, the memory aborts the option byte change operation. In this case, the FLASH_XXX_PRG registers are not overwritten by the current option value. The user application can check what was wrong in their configuration.

Unlocking the option byte modification

After reset, the OPTLOCK bit is set to 1 and the FLASH_OPTCR is locked. As a result, the application software must unlock the option configuration register before attempting to change the option bytes. The FLASH_OPTCR unlock sequence is described in *Section 7.6.7*.

Option bytes modification sequence

To modify user option bytes, follow the sequence below:

1. Check that no memory operations are ongoing by checking the BSY bit in the FLASH_NS/SECSR register and that the write buffer is empty by checking the WBNE bit in the FLASH_NS/SECSR register.
2. Check the data buffer is empty (DBNE = 0) in FLASH_NS/SECSR register.
3. Clear all error flags due to a previous operation.
4. Unlock FLASH_OPTCR register as described in *Section 7.6.7*, unless the register is already unlocked.
5. Write the desired new option byte values in the corresponding option registers (FLASH_XXX_PRG).
6. Set the option byte start change OPTSTRT bit to 1 in the FLASH_OPTCR register.
7. Wait until BSY bit is cleared in FLASH_NS/SECSR register.
8. OPTSTRT bit is cleared automatically at the end of the sequence (or in case of error).
9. Reset the device. This step is always recommended, becomes mandatory when one of the following option bytes is impacted:
   a) SECBOOTADDR
   b) NSBOOTADDR
   c) TZEN
   d) BOOT_UBE

Note: If a reset or a power-down occurs while the option byte modification is ongoing, the original option byte value is kept. A new option byte modification sequence is required to program it.

Option bytes overview

Table 51 lists all the user option bytes managed through the memory registers, as well as the initial values before the first option byte change (user default configuration).

Table 51. Option bytes organization

Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLASH_ OPTSR	SWAP_BANK	Res.	BOOT_UBE								IWDG_STDBY	IWDG_STOP	Res.	Res.	IO_VDDIO2_HSLV	IO_VDD_HSLV	PRODUCT_STATE								NRST_STDBY	NRST_STOP	Res.	WWDG_SW	IWDG_SW	BORH_EN	BOR_LEV	
	0	0	1	0	1	1	0	1	0	0	1	1	0	0	0	0	1	1	1	0	1	1	0	1	1	1	0	1	1	0	0	0
FLASH_ OPTSR2	TZEN								Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	USBPD_DIS	Res.	SRAM2_ECC	SRAM3_ECC	BKPRAM_ECC	SRAM2_RST	SRAM13_RST	Res.	Res.
	1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	0	0
FLASH_ BOOTR	SECBOOTADD[23:8]																SECBOOTADD[7:0]								SECBOOT_LOCK							
	0x0C00																0x00								0xC3							
FLASH_ NSBOOTR	NSBOOTADD[23:8]																NSBOOTADD[7:0]								NSBOOT_LOCK							
	0x0800																0x00								0xC3							
FLASH_ SECWM1R	Res	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_END							Res	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_STRT						
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
FLASH_ SECWM2R	Res	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_END							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_STRT						
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
FLASH_ WRPSGN1R	WRPSG[127:124]	WRPSG[123:120]	...												WRPSG[71:68]	WRPSG[67:64]	WRPSG[63:60]	WRPSG[59:56]	...												WRPSG[7:4]	WRPSG[3:0]	
	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
FLASH_ WRPSGN2R	WRPSG[127:124]	WRPSG[123:120]	...												WRPSG[71:68]	WRPSG[67:64]	WRPSG[63:60]	WRPSG[59:56]	...												WRPSG[7:4]	WRPSG[3:0]	
	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 51. Option bytes organization (continued)

Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLASH_ OTPBLR	LOCKBL																															
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
FLASH_ EDATA1R	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DATA_EN1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECTOR_START_1		
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
FLASH_ EDATA2R	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DATA_EN1	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECTOR_START_2		
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
FLASH_ NSEPOCHR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	NS_EPOCH																							
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
FLASH_ SECEPOCHR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SEC_EPOCH																							
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
FLASH_ HDP1R	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_END								HDP1_STRT							
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
FLASH_ HDP2R	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_END								HDP2_STRT							
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

7.4.4 Description of user and system option bytes

The general-purpose option bytes that can be used by the application are listed below:

- Watchdog management
  - IWDG_STOP: independent watchdog IWDG (also known as WDGLS_CD) counter active in Stop mode if 1 (stop counting or freeze if 0)
  - IWDG_STDBY: independent watchdog IWDG (also known as WDGLS_CD) counter active in Standby mode if 1 (stop counting or freeze if 0)
  - IWDG_SW: hardware (0) or software (1) IWDG (also known as WDGLS_CD) watchdog control selection

Note: If the hardware watchdog "control selection" feature is enabled (set to 0), the watchdog is automatically enabled at power-on, thus generating a reset unless the watchdog key register is written to or the down-counter is reloaded before the end-of-count is reached.

Depending on the configuration of IWDG_STOP and IWDG_STDBY options, the IWDG can continue counting (1) or not (0) when the device is, respectively, in Stop or Standby mode.

When the IWDG is kept running during Stop or Standby mode, it can wake up the device from these modes.

- Reset management
  - BOR_LEV: Brownout level option, indicating the supply level threshold that activates/releases the reset
  - BORH_EN: enabling a high BOR level
  - NRST_STDBY: generates a reset when entering Standby mode if cleared to 0
  - NRST_STOP: generates a reset when entering Stop mode if cleared to 0.

Note: Whenever a Standby (Stop) mode entry sequence is successfully executed, the device is reset instead of entering Standby (Stop) mode if NRST_STDBY (NRST_STOP) is cleared to 0.

- Device options
  - IO_VDDIO2_HSLV: enables the configuration of pads below 2.7 V for VDDIO2 power rail if set to 1
  - IO_VDD_HSLV: enables the configuration of pads below 2.7 V for VDD power rail if set to 1
  - USBPD_DIS: bit to disable USB PD

At delivery, the values programmed in the user option bytes are the following:

- Watchdog management
  - IWDG (also known as WDGLS_CD) active in Standby and Stop modes: 0x1
  - IWDG (also known as WDGLS_CD) not automatically enabled at power-on: 0x1
- Reset management
  - BOR: brownout level option (reset level = 2.1 V): 0x0. A reset is not generated when the device enters Standby, Stop low-power mode (value = 0x1)
- Device working in the full voltage range with I/O speed optimization at low-voltage disabled (IO_VDDIO2_HSLV = 0 and IO_VDD_HSLV = 0)

Refer to *Section 7.11* for details.

7.4.5 Description of data protection option bytes

The option bytes that can be used to enhance data protection are listed below:

- PRODUCT_STATE[7:0]: A product life cycle state (see *Section 7.6.11* for details).
- WRPSGn1/2: write protection option of the corresponding group of four consecutive sectors in Bank1 (respectively Bank2). It is active low. Refer to *Section 7.6.8* for details.
  - Bit N: Group embedding sectors 4 x N to 4 x N + 3
- SECWMx: TrustZone® secure only watermark area definition (refer to *Section 7.6.1* for details).
  - SECWM1_STRT (respectively SECWM1_END) contains the first (respectively last) sector of the secure access only zone in Bank1
  - SECWM2_STRT (respectively SECWM2_END) contains the first (respectively last) sector of the secure access only zone in Bank2.
- TZEN: this nonvolatile option can be used by the application to activate the secure access mode, as described in *Section 7.6*. For security reasons the TZEN is stored with redundancy on 8 bits. While the TZEN OB value is set immediately after

programming, to actually use TZ features, reset is required to set the TZ_STATE in SBS.

When TZEN is activated, secure watermark settings and secure boot address can be reset to default to prevent a deadlock in case the previous values were not correct.

- HDPx: Secure hide protection - HDPL exclusive area control in the user flash memory.

When factory programmed values of the data protection option bytes are the following:

- Product state depends on sales type
- Flash bank erase operations do not impact watermarked secure data areas
- Secure watermark areas protections disabled (start addresses higher than end addresses)
- Write protection disabled (all option byte bits set to 1)
- TrustZone Secure access mode disabled (TZEN option byte value = 0xC3)

Refer to *Section 7.11* for details.

7.4.6 Description of boot address option bytes

Below the list of option bytes that can be used to configure the appropriate boot address for an application:

- PRODUCT_STATE
- BOOT_UBE: Selects either ST-iRoT or User flash as default boot address. Also used by RSS(SFI) to choose either Bootloader or ST-iRoT as next stage. Available only on products embedding cryptographic acceleration (STM32H533/573xx).
- NSBOOTADD: Selects default boot address when TZ is disabled.
- SECBOOTADD: Selects default boot address when TZ is enabled.
- (NS/SEC)BOOT_LOCK: Protects the boot configuration from further modification attempts.
- SWAP_BANK: bank swapping option, set to 1 to swap user flash banks after boot (see *Section 7.3.11*). If BOOT_LOCK corresponding to TZEN state (SEC/NS) is active, the value in SWAP_BANK is fixed, read-only.

When STMicroelectronics delivers the device, the PRODUCT_STATE is Open, BOOT_LOCK is not set (0xC3) and the BOOT_UBE is set to user flash (0xB4). Addresses are SECBOOTADD = 0x0C00 0000, NSBOOTADD = 0x0800 0000.

Refer to *Section 7.11* for details.

7.4.7 Specific rules for modifying option bytes

On top of OPTLOCK bit and register access rules, there are other protections for selected security-sensitive option byte fields.

Different option bytes can be modified simultaneously, but if they rely on each other for protection, both states are checked. For example, when trying to modify PRODUCT_STATE and TZEN simultaneously, resulting state must be coherent with the rules. With few exceptions, listed in this section, failing to uphold the rules results in raising OPTCHANGEERR flag (*Section 7.9.12* for additional details).

Table 52. Specific modifying rules

Option byte	HDPL	TZ secure	Value	Product state
PRODUCT_STATE	1(1)	Some values only when TZ is enabled	Set of possible transitions	Set of possible transitions (*Table 45*)
HDP	1(2)	(3)	-	Refer to *Table 53*
TZEN	-	-	-	Regression, Open or Provisioning
EPOCH	-	-	PRG > CUR	Regression
EPOCH_NS	-	-	PRG > CUR	NS-Regression, Regression
SECWM	-	Secure mode only(2)	-	Refer to *Table 53*
BOOT_UBE	-	-	-	Open or Provisioning, SECBOOT_LOCK disabled
SECBOOTADD	-	Secure mode only(2)	-	
NSBOOTADD	-	-	-	Open or Provisioning, NSBOOT_LOCK disabled
LOCKBL	-	-	One way switch(2)	-
SWAP_BANK	-	Fixed if BOOT_LOCK corresponding to TZEN is set	-	-
SECBOOT_LOCK	-	Secure mode only(2)	-	Open, Regression or Provisioning to unlock
NSBOOT_LOCK	-	-	-	

1. Most transitions are possible regardless of HDPL, only a few require specific HDPL, see text below.
2. No OPTCHANGEERR raised in violation of this.
3. Dash means there is no limitation from this side.

Even in the closed PRODUCT_STATE progression, some OBs can still be modified, if all the other constraints are satisfied. An overview is presented in *Table 53*.

Table 53. OB modifiable in closed product

PRODUCT_STATE	OBs that can be modified
TZ-Closed	SWAP_BANK, LOCKBL, PRODUCT_STATE, SECWM, HDP
Closed	SWAP_BANK, LOCKBL, PRODUCT_STATE
Locked	SWAP_BANK, LOCKBL

Specific rules must be respected to update the following OB:

- **PRODUCT_STATE**

  PRODUCT_STATE transitions follow a state machine with two types of transitions. Locking down the product is allowed without restriction. Opening the product is possible only using a debug interface and following a digital signature verification. The regression to a less secure state results in the erase of the protected content. More details are given in *Section 7.6.11*.

  Summary: Selected changes are possible only in HDPL1. Some states are accessible only when TZEN is enabled.

- **HDP**

  Can be modified only in HDPL1. Do not set it to overlap with flash high-cycle data area.
- **TrustZone access mode (TZEN)**

  Can be changed only in Open, Regression and Provisioning.
- **SEC_EPOCH and NS_EPOCH**

  The value programmed must be greater than current value.The increment is done in specific product states: Regression and NS-Regression.
- **Secure watermark area (SECWM1/2_STRT and SECWM1/2_END)**

  Can be changed only in secure mode (TZ_state = 0xB4). Automatically reset to default value when TZEN is enabled. Do not set it to to overlap with flash high-cycle data area.
- **BOOT_UBE**

  Available only on cryptography enabled devices. Can be changed only in product states open for debug like Open and also in Provisioning. SECBOOT_LOCK must be disabled to change.
- **SECBOOTADD**

  Can be changed only in Open and Provisioning. Locked by SECBOOT_LOCK. Automatically reset to default value when TZEN is enabled.
- **NSBOOTADD**

  Can be changed only in Open and Provisioning. Locked by NSBOOT_LOCK.
- **LOCKBL**

  Can be changed freely only in one direction. A permanent irreversible switch.
- **SWAP_BANK**

  Not modifiable when both TZEN and SECBOOT_LOCK are 0xB4 (set) or when TZEN = 0xC3 (disabled) and NSBOOT_LOCK is active (0xB4).
- **SECBOOT_LOCK**

  Can be changed only freely in locking direction. Unlock is possible in Open, Provisioning and Regression.
- **NSBOOT_LOCK**

  Can be changed only freely in locking direction. Unlock is possible in Open, Provisioning and Regression.

Note: For all user option bytes above: default values are loaded and Tamper is signaled when a double ECC error occurs during OBL.

7.5 Option bytes key (OBK) management

This section describes the key storage tied with SAES and OBK-HDPL value set by the SBS (refer to *Section 14.3.7: SBS hardware secure storage control*). The OBKs are stored in 128 bits and protected by 9-bit ECC (SEC/DED). They are written alternatively in two sectors.

7.5.1 OBK loading

OBKs are memory mapped, and accessible through the main AHB bus (C-Bus), starting from address 0x0FFD 0000. A 9-bit ECC is associated to each 128-bit data flash word.

Note: Read access is allowed in the current and alternate sector, except for the last address of the current and alternate sector. In case of read access, no error is reported but the read data is always 0x0 (see Section 7.6.12).

7.5.2 OBK access per HDPL level

Table 54 describes the option byte key areas. The key storage is not dedicated to a particular key, and the usage is defined by the application.

An option byte key can be accessed only if the OBK-HDPL (set in SBS) matches the HDPL associated to the storage offset, indicated below (refer to *Section 14.3.7: SBS hardware secure storage control*). If this is not the case, an OBKERR error is raised.

Table 54. Option bytes key area

OBK-HDPL	Address offset start	Address offset end	Comment
-	0x0000	0x00FF	Reserved
1	0x0100	0x08FF	OEM iRoT keys
2	0x0900	0x0BFF	uRoT, OS or secure application
3	0x0C00	0x17FF	HDPL 3 secure keys
3	0x1800	0x1FEF	HDPL 3 non-secure keys

7.5.3 OBK programming sequence

The sequence to write OBK is the same as a write to main flash except that the “alternate sector” (ALT_SECT) bit must be set to write in the alternate OBK sector. The ALT_SECT bit is checked each time a write access is received. If it does not match the one stored in the write buffer, an error is raised.

Note: Before writing in the alternate OBK sector the user must ensure that the whole key space (128, 256, or 512 bits) is free, by reading the alternate OBK sector.

If the alternate sector is not free, erase the sector using the command ALT_SECT_ERASE.

Secure programming method

Using the secure register if TrustZone is active (TZ_STATE = 0xB4):

1. Program the ALT_SECT bit.
2. Make sure protection mechanism does not prevent programming (TZ, PRIV and OBK-HDPL).
3. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_SECSR register and that the write buffer is empty by checking the WBNE bit in the FLASH_SECSR register.
4. Check and clear all the secure error flags due to previous operation.
5. Unlock the FLASH_SECCR register only if this register is not already unlocked, as described in *Section 7.6.7*.
6. Enable write operations by setting the SECPG bit in the FLASH_SECCR register.
7. Write one flash-word corresponding to 16-byte data starting at 16-byte aligned address.

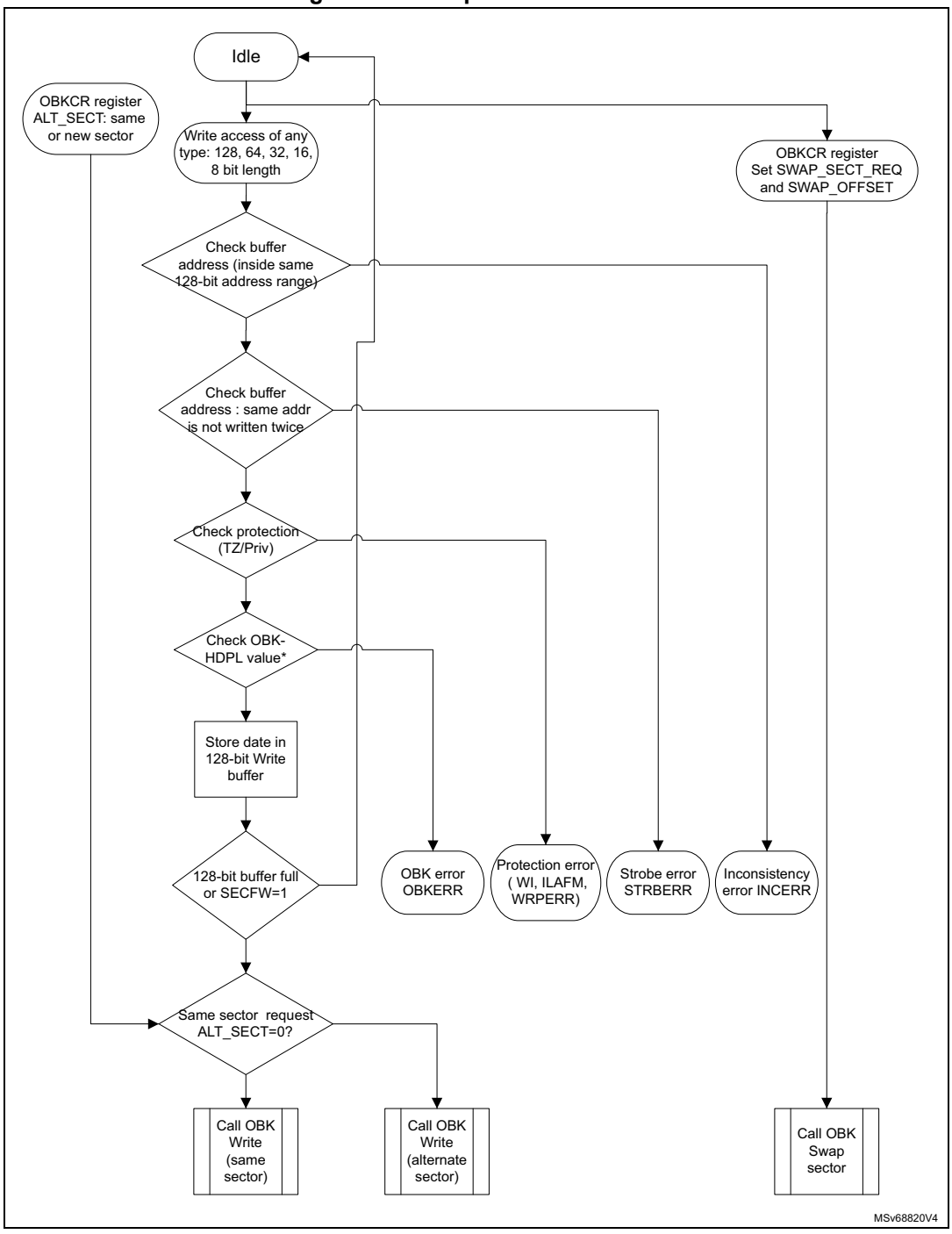
Note: WBNE flag indicates if the 128-bit write buffer is waiting for new data.

Note: No erase request, options change request, OBK swap sector or OBK alternate sector erase is allowed between first write and the completion of flash write operation.

8. Wait for the BSY bit to be cleared in the corresponding FLASH_SECSR register.
9. Clear PG bit in FLASH_SECCR register if there is not any programming request anymore in the bank.

If step 7 is executed incrementally (byte per byte), the write buffer can become partially filled. In this case the application software can decide to force-write what is stored in the write buffer by using FW bit in FLASH_SECCR register. In this particular case, the unwritten bits are automatically set to 1. If no bit in the write buffer is cleared to 0, the FW bit has no effect.

Figure 30. OBK protection checks

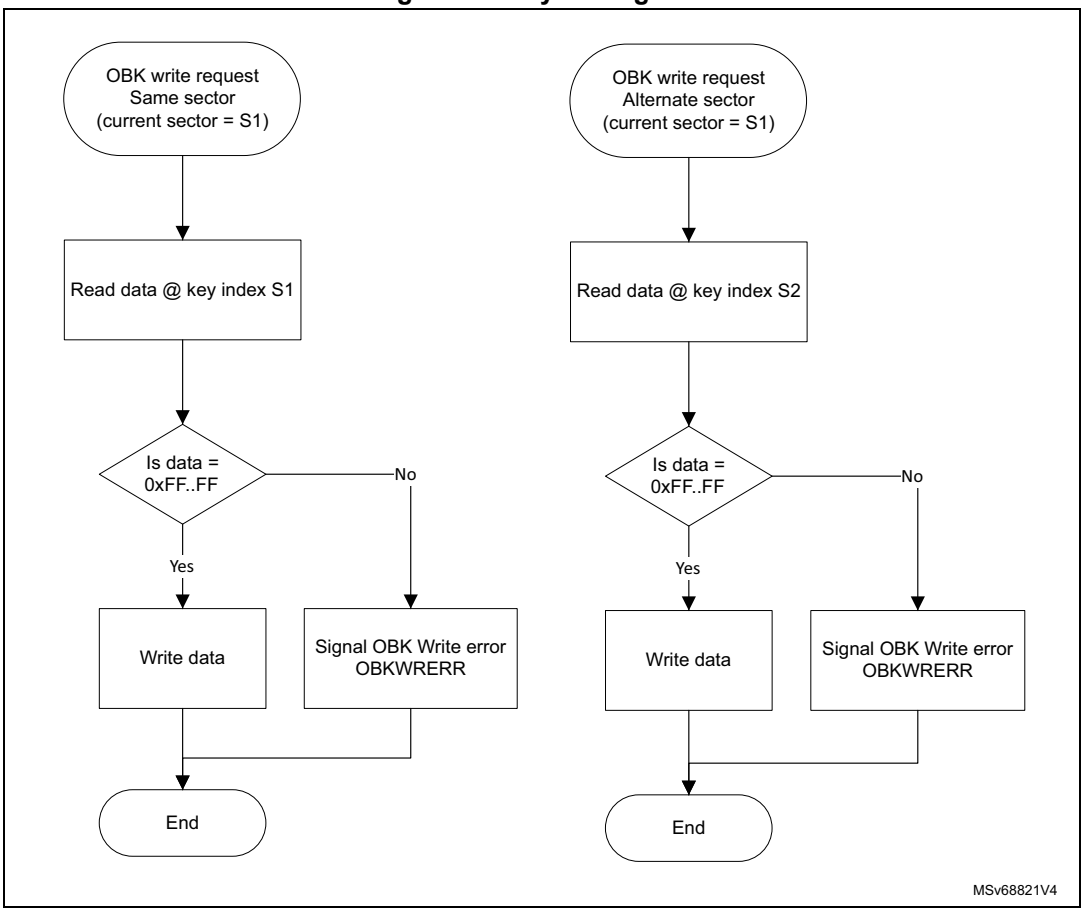


7.5.4 OBK programming finite state machine

When the OBK write buffer is filled, the OBK programing is launched automatically in the current OBK sector or in the alternate OBK sector, depending upon the ALT_SECT bit.

If the address is not virgin, the OBKWRERR is raised.

Figure 31. Key writing flow



7.5.5 OBK swap sector

The user can request a swap of the option byte keys by setting the bit SWAP_SECT_REQ in the NS/SECOBKCFGR register. The number of keys swapped is defined by the SWAP_OFFSET in NS/SECOBKCFGR register.

Swap is limited by OBK-HDPL value, the following SWAP_OFFSET values are permitted:

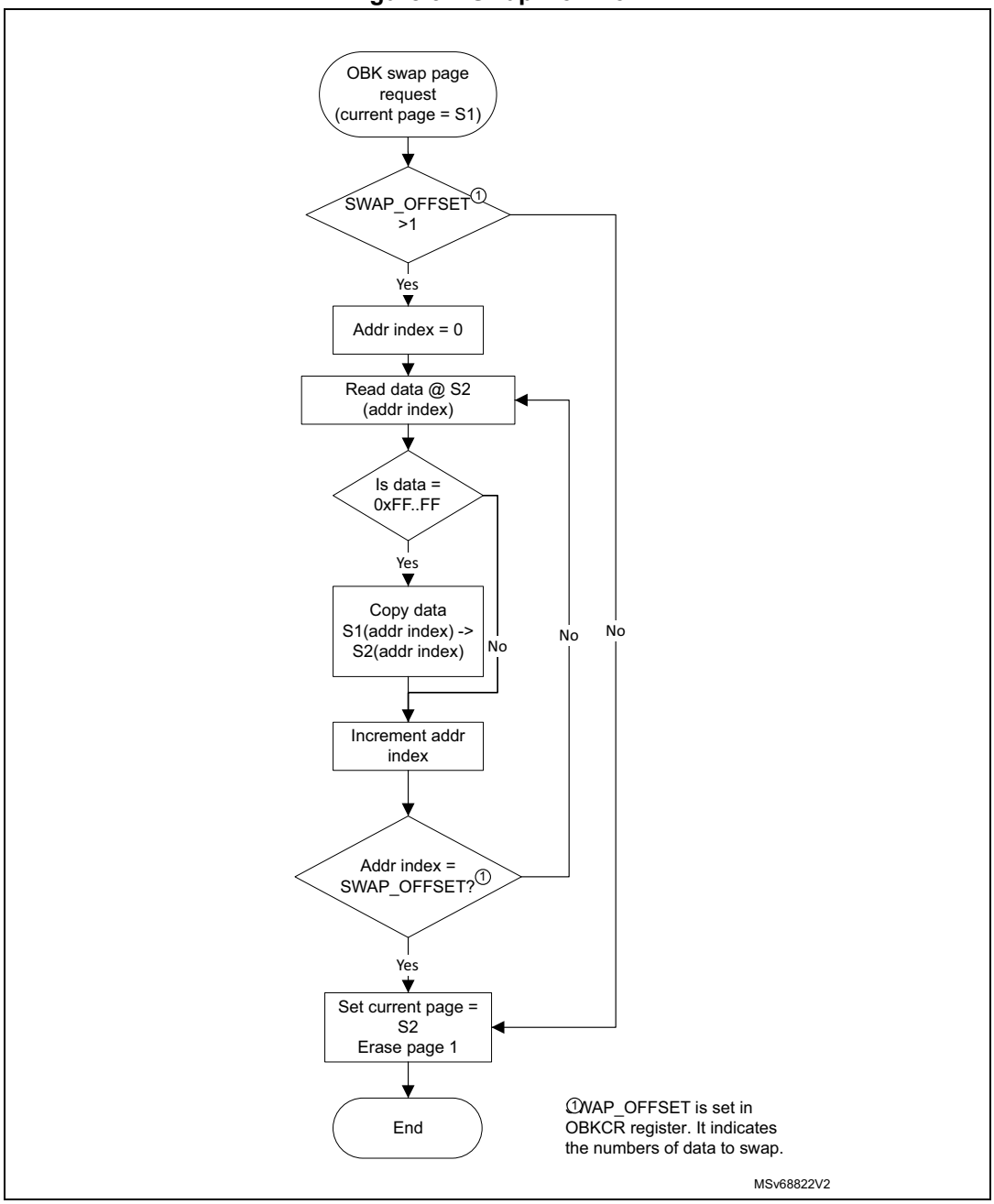
- OBK_HDPL1_OFFSET = 144
- OBK_HDPL2_OFFSET = 192
- OBK_HDPL3SEC_OFFSET = 384
- OBK_HDPL3NS_OFFSET = 511

The programming sequence is the following:

1. Check that no memory operations are ongoing by checking the BSY and DBNE bit in the FLASH_SEC/NSSR register and that the write buffer is empty by checking the WBNE bit in the FLASH_SEC/NSSR register.
2. Check and clear all the secure/non-secure error flags due to previous operation.
3. Unlock NS/SECOBKCFGR register if not unlocked already thanks to FLASH_NS/SECOBKKEYR.

4. Define the numbers of keys to be swapped thanks to SWAP_OFFSET field in NS/SECOBKCFGR. SWAP_OFFSET must be equal to or greater than OBK_HDPL<N-1>_OFFSET.
5. Set bit SWAP_SECT_REQ in NS/SECOBKCFGR.
6. Wait for BSY bit to be cleared. In case of error OBKERR flag is set.
7. SWAP_SECT_REQ is cleared automatically at the end of the operation or in case of error.

Figure 32. Swap workflow



While the OBK SWAP operation is ongoing, the BSY flag is set and no other operation (write, erase, user opt change) can be launched in parallel.

7.5.6 OBK alternate sector erase

The user can request the erase of the alternate OBK sector by setting the bit ALT_SECT_ERASE in the FLASH_NS/SECOBKCFGR register.

The programming sequence must be:

1. Check that no memory operations are ongoing by checking the BSY and DBNE bits in the FLASH_SEC/NSSR register and that the write buffer is empty by checking the WBNE bit in the FLASH_SEC/NSSR register.
2. Check and clear all the secure/non-secure error flags due to previous operation.
3. Set bit ALT_SECT_ERASE in NS/SECOBKCFGR.
4. Wait for BSY bit to be cleared.
5. ALT_SECT_ERASE is cleared automatically at the end of the operation or in case of error.

7.6 FLASH security and protections

To protect sensitive information against unwanted operations (such as reading confidential areas, illegal programming of immutable sectors, or malicious flash memory erasing), the FLASH implements the following protection mechanisms:

- TrustZone backed watermark and block security protection
- Temporal isolation protection (HDP)
- Configuration protection
- User flash write protection
- Device nonvolatile security life cycle and application boot state management
- OTP locking

The microcontroller can have TrustZone active (TZ_STATE = 0xB4) or not (TZ_STATE = 0xC3).

While TrustZone is active the flash memory can be accessed in four basic modes:

- non-secure and unprivileged
- non-secure and privileged
- secure and privileged
- secure and unprivileged

While TrustZone is disabled, only two modes are possible:

- non-secure and unprivileged
- non-secure and privileged

The flash interface evaluates the access restrictions in the following order:

1. TrustZone security
2. Write protection (for write access)
3. HDP
4. Privilege

7.6.1 TrustZone security protection

The global TrustZone system security is activated by setting the TZEN option in the FLASH_OPTSR2_PRG register. The actual state of TZ is however determined by the SBS and the setting of TZ_STATE (refer to *Section 14.3.5: SBS boot control*).

When TrustZone is active (TZ_STATE = 0xB4), additional security features are available:

- Secure watermark-based user options bytes defining secure areas
- Secure or non-secure block-based areas can be configured on-the-fly after reset. This is a volatile secure area
- Additional product states associated with **closed_secure** provide protection of secure domain against external access (debug or bootloader)
- Erase or program operation can be performed in secure or non-secure mode with associated configuration bit. When the TrustZone is disabled (TZ_STATE = 0xC3), the above features are deactivated and all secure registers are RAZ/WI.

Activating TrustZone security

When the TrustZone is activated (TZ_STATE = 0xB4), the SECBOOTADD and the secure watermark areas are set to a secure default value. Default settings are also used in case of corrupted configuration. In this default state, user flash memory is secure (see *Table 55*).

Table 55. Default secure watermark

Secure watermark in case of configuration error	Security attribute
SECWMx_STRT = 0 SECWMx_END = 0x7F(1)	All flash memory is secure

1. Default and maximum values depend upon the total number of sectors in the user flash memory.

Illegal access generation

A non-secure access to a secure flash memory area is RAZ/WI, it generates an illegal access event. An illegal access interrupt is generated if the FLASHIE illegal access interrupt is enabled in the TZIC_IER2 register.

A non-secure access to a secure flash register generates an illegal access event. An illegal access interrupt is generated if the FLASH_REGIE illegal access interrupt is enabled in the TZIC_IER2 register.

Table 56. Flash memory TZ protection summary

TZ protection (TZ_STATE = 0xB4)		Main flash	
		Non-secure sector	Secure sector
Secure	Fetch	BUS ERROR	OK
	Read	RAZ, ILAFM	
	Write	WI, WRPERR, ILAFM	If not WRP: OK else: WI, WRPERR
	Erase		

Table 56. Flash memory TZ protection summary (continued)

TZ protection (TZ_STATE = 0xB4)		Main flash	
		Non-secure sector	Secure sector
Non-secure	Fetch	OK	BUS ERROR
	Read		RAZ, ILAFM
	Write	If not WRP: OK else: WI, WRPERR	WI, WRPERR, ILAFM
	Erase		

Table 57. TZ protection and bank or mass erase summary

TZ protection (TZ_STATE = 0xB4)	Main flash		
	Non-secure flash	Secure flash	Mix of secure and non-secure
MES	WI, WRPERR, ILAFM	OK	WI, WRPERR, ILAFM
MENS	OK	WI, WRPERR, ILAFM	WI, WRPERR, ILAFM

Deactivate TrustZone security

Reversing of the TZ setting in OB (TZEN) is possible when the product state is ST-RoT-Ready or Provisioning. Deactivating TZEN from other states requires full regression (Regression state).

When the TrustZone is deactivated (TZ_STATE = 0xC3) after option bytes loading, the following security features are deactivated:

- Watermark-based secure area (refer to *Watermark-based secure flash area protection*)
- block-based secure area (refer to *Section 7.6.3*)
- Secure interrupts (refer to *Section 7.9.11*)
- All secure registers are RAZ/WI.

Watermark-based secure flash area protection

When TrustZone security is active (TZ_STATE = 0xB4), a part of the flash memory can be protected against non-secure read and write accesses. Up to two different nonvolatile secure areas can be defined by option bytes and can be read or written by a secure access only: one area per bank can be selected with a sector granularity.

The secure areas are defined by a start sector offset and end sector offset using the SECWMx_STRT and SECWMx_END option bytes. These offsets are defined in the secure watermark registers address registers *FLASH security watermark for Bank1 (FLASH_SECWM1R_CUR)*, *FLASH security watermark for Bank2 (FLASH_SECWM2R_CUR)* and modified by corresponding FLASH_XXX_PRG registers.

The SECWMx_STRT and SECWMx_END option bytes can be modified only by secure firmware.

Table 58. Secure watermark-based area

Secure watermark option bytes values (x = 1, 2)	Secure watermark protection area
SECWMx_STRT > SECWMx_END	Nonsecure area
SECWMx_STRT = SECWMx_END	One sector defined by SECWMx_STRT is secure watermark-based protected
SECWMx_STRT < SECWMx_END	The area between SECWMx_STRT and SECWMx_END is secure watermark-based protected

Caution: Switching a memory area from secure to nonsecure does not erase its content. The user secure software must perform the needed operation to erase the secure area before switching an area to non-secure attribute whenever is needed. It is also recommended to flush the instruction cache.

7.6.2 Hide protection (HDP)

The HDP area is independent of the flash watermark-based secure area. One nonvolatile secure hide protection (HDP) area per bank can be defined with a sector granularity. Access to the hide protection area can be denied by progressing the HDPL level in the SBS.

When the HDPL = 1, no user flash is protected by the HDP. With HDPL ≥ 2 the user OB defined HDP area in each bank is closed. No read, write, fetch or erase is allowed in the HDP area.

The HDP area can be extended, the extension setting is only a register value, not stored in OB. With HDPL = 3 the HDP area remains activated and the extension is added. If extension is added while the base user OB defined area is not active, the extension covers one more sector, because HDPx_END sector is also protected.

The HDPL level can be only cleared by a system reset, there is no means to deactivate the HDP area.

The protected HDP area is defined by setting its size using start and end sectors in a similar way as the secure watermark.

The size of the inaccessible area can be extended using register FLASH_HDPEXTR values HDPx_EXT, representing the number of sectors added to the HDP area (past the HDPx_END sector). The volatile HDPx_EXT value can be increased, but cannot decrement.

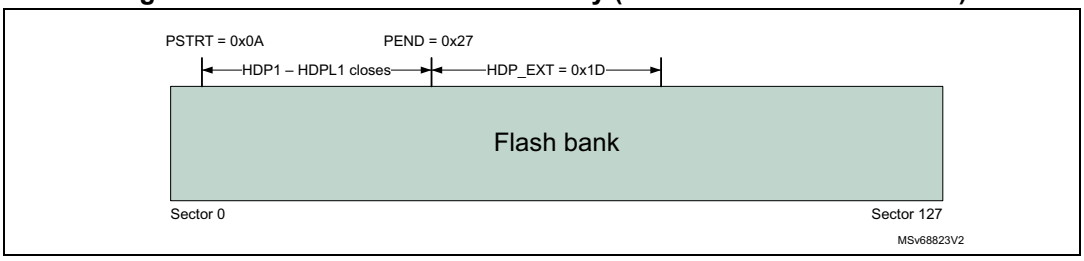
If HDPx_STRT > HDPx_END the extension covers (HDPx_EXT + 1) size area between sectors marked by HDPx_END and HDPx_END + HDPx_EXT.

By default HDPx_END is set to 0 and HDPx_STRT is set to 1. This means that the HDP area in user flash has a 0 size, and extends from the flash start address (first sector).

For example, to protect area by HDP from the address 0x0801 4000 (included) to the address 0x0804 FFFF (included):

- For physical Bank1, the option byte registers must be programmed with:
  - HDP1_STRT = 0x0A
  - HDP1_END = 0x27

Figure 33. HDP[a] in user flash memory (STM32H563/573xx devices)



- Once the HDPL is incremented to 2, the marked area becomes inaccessible. The HDPL = 2 code then sets the extension size to 29. Once the HDPL is incremented to 3, even the extension area becomes inaccessible.
- Alternatively, if:
  - HDP1_END = 0, HDP1_START = 1, HDPx_EXT = 1:
    Sector 0 and Sector 1 are HDP protected.
  - HDP1_END = 2, HDP1_START = 7, HDPx_EXT = 3 (a rather unusual example):
    Sectors 2 to 5 are HDP protected.
- If the two banks are swapped, the protection defined to physical Bank1 remains on the physical Bank1, unaffected by swapping. Separate protection applies to physical Bank2 and the option bytes registers must also be programmed with:
  - HDP2_STRT = 0x0A
  - HDP2_END = 0x27

Note: For more details on the bank swapping mechanism, refer to Section 7.6.6. The values have an upper limit, 127 for STM32H562/63/73xx devices, 31 for STM32H523/33xx devices.

Table 59. Secure hide protection

HDPx watermark option byte values (x = 1, 2)		Hide protection area
HDPL ≤ 1	-	No inaccessible HDP area in user flash memory
HDPL = 2	HDPx_END < HDPx_STRT	No inaccessible HDP area in user flash memory
	HDPx_END = HDPx_STRT	Exactly one sector is protected by HDP
	HDPx_END > HDPx_STRT	The area between HDPx_STRT and HDPx_END is HDP protected.
HDPL = 3	HDPx_END < HDPx_STRT and HDPx_EXT = 0	No inaccessible HDP area in user flash memory
	HDPx_END ≥ HDPx_STRT or HDPx_EXT > 0	The area between min (HDPx_STRT, HDPx_END) and (HDPx_END + HDPx_EXT) is HDP protected

a. The HDP works analogically, the number of sectors is the only differentiator.

Table 60. HDP protections summary

HDP protection	User flash			
	Access to OB HDP area		Access to EXT HDP area	
	HDPL 2 or 3	HDPL1	HDPL3	HDPL1 or HDPL 2
Fetch	BUS ERROR	OK	BUS ERROR	OK
Read	RAZ		RAZ	
Write	WI, WRPERR	If not WRP: OK Else: WI, WRPERR	WI, WRPERR	If not WRP: OK Else: WI, WRPERR
Erase (mass erase)				

7.6.3 Block-based secure flash memory area protection

Any sector can be programmed on-the-fly as secure or non-secure using the block-based configuration registers. FLASH_SECBB1x/FLASH_SECBB2x registers are used to configure the security attribute for, respectively, sectors in Bank1/Bank2.

When the sector security attribute bit SECBByx[i] is set, the security attribute is the same as the secure watermark-based area. The secure sector is accessible only by a secure access.

If the SECBBy[i] bit is set or reset for a sector already included in a secure watermark-based area, the sector keeps the watermark-based protection security attributes.

To modify a block-based sector security attribute, it is recommended to:

- Check that no flash operation is ongoing on the related sector.
- Add an ISB instruction after modifying the sector security attribute SECBByx[i].

Caution: Switching a sector from secure to non-secure does not erase the content of the associated sector. User secure software must perform the needed operations before switching to non-secure attribute:

- Erase sector content.
- Invalidate the instruction cache.

Note: For SECBByx[i] access control, refer to Table 61.

Execute protection for S-Bus (AHB register bus) is done in AHB decoder, not in flash memory interface itself.

Table 61. Secure configuration block-based registers access conditions

Access			Corresponding sector privilege status	Sector setting in SECBBxy
Fetch	Secure/ non-secure	Privilege/ unprivilege	-	BUS ERROR
Read			-	OK
Write	Secure	Privilege	-	OK
		Unprivilege	0	OK
			1	WI
	Non-secure	Privilege/ unprivilege	-	WI, ILAP

7.6.4 Block-based privileged flash memory area protection

Any sector can be programmed on-the-fly as privileged or unprivileged using the block-based configuration registers. FLASH_PRIVBB1x and FLASH_PRIVBB2x registers are used to configure the privilege attribute for, respectively, sectors in Bank1 and Bank2.

When the sector privilege attribute PRIVBByx[i] bit is set, the sector is accessible only by a privileged access. An unprivileged sector is accessible by a privileged or unprivileged access.

To modify a block-based privilege attribution, it is recommended to:

- Check that no flash operation is ongoing on the related sector.
- Add an ISB instruction after modifying the sector security attribute PRIVBByx[i].

Caution: Switching a sector from privileged to unprivileged does not erase the content of the associated sector.

Table 62. Privilege protection summary

Access (TZ not relevant)		Main flash	
		Unprivileged sector	Privileged sector
Privilege	Fetch	OK	
	Read		
	Write		
	Erase		
Unprivilege	Fetch	OK	RAZ
	Read		
	Write		WI, WRPERR
	Erase		

Table 63. Privilege and mass or bank erase

Access (TZ state not relevant)	Main flash		
	Unprivileged FLASH	Privileged FLASH	Mix unprivileged and privileged FLASH
MEP	OK		
MENP	OK	WI, WRPERR	

Note: For PRIVBByx[i] access control, refer to Table 64 and Table 65.

Table 64. Privilege configuration register access conditions (TZ enabled)

Access			Corresponding sector secure status(1)	Sector setting access in PRIVBBxy
Fetch	Privileged/ unprivileged	Secure/ non-secure	-	BUS ERROR
Read			-	OK (all sectors)

Table 64. Privilege configuration register access conditions (TZ enabled) (continued)

Access			Corresponding sector secure status(1)	Sector setting access in PRIVBBxy
Write	Privileged	Secure	-	OK (all sectors)
		Non-secure	NS	OK (only the sector corresponding to the bit)
			S	WI (only the sector corresponding to the bit)
	Unprivileged	Secure/ non-secure	-	WI

1. Either watermark or block-based.

Table 65. Privilege configuration register access conditions (TZ disabled)

Access		Sector setting access in PRIVBBRxy
Fetch	Privileged/ unprivileged	BUS ERROR
Read		OK
Write	Privileged	OK
	Unprivileged	WI

7.6.5 Flash memory register privileged and unprivileged modes

The flash registers can be read and written by privileged and unprivileged accesses depending on the SPRIV and NSPRIV bits in *FLASH privilege configuration register (FLASH_PRIVCFGR)*:

- When the SPRIV (respectively NSPRIV) bit is reset, all secure (respectively non-secure) flash registers can be read and written by both privileged or unprivileged access.
- When the SPRIV (respectively NSPRIV) bit is set, all secure (respectively non-secure) flash registers can be read and written by privileged access only. Unprivileged access to a privileged registers is RAZ/WI.

The registers related to key storage (FLASH_NSOBKCFGR, FLASH_SECOBKCFGR, FLASH_NSOBKKEYR, FLASH_SECOBKEYR) are exceptions to this rule and can be only accessed in privilege, regardless of other settings.

The next table summarizes flash registers access control.

Table 66. Flash register accesses(1)

Access			Non-secure register		Secure register	
			NSPRIV = 1	NSPRIV = 0	SPRIV = 1	SPRIV = 0
X	Secure/ non-secure	Privileged/ unprivileged	Bus error			
R/W	Secure	Privileged	OK			
R/W		Unprivileged	RAZ, WI	OK	RAZ, WI	OK

Table 66. Flash register accesses(1) (continued)

Access			Non-secure register		Secure register	
			NSPRIV = 1	NSPRIV = 0	SPRIV = 1	SPRIV = 0
R/W	Non-secure	Privileged	OK		RAZ, WI, ILAP	
R/W		Unprivileged	RAZ, WI	OK		

1. Some registers are privilege only, not configurable.

7.6.6 Flash memory banks attributes in case of bank swap

The SWAP_BANK option bit modifies the address of each bank in the memory map. When SWAP_BANK is reset, flash Bank1 is mapped at the lower address range. When SWAP_BANK is set, flash Bank1 is mapped at the higher address range. flash bank attributes follow their bank contents so there is no need to modify their setting registers when swapping banks:

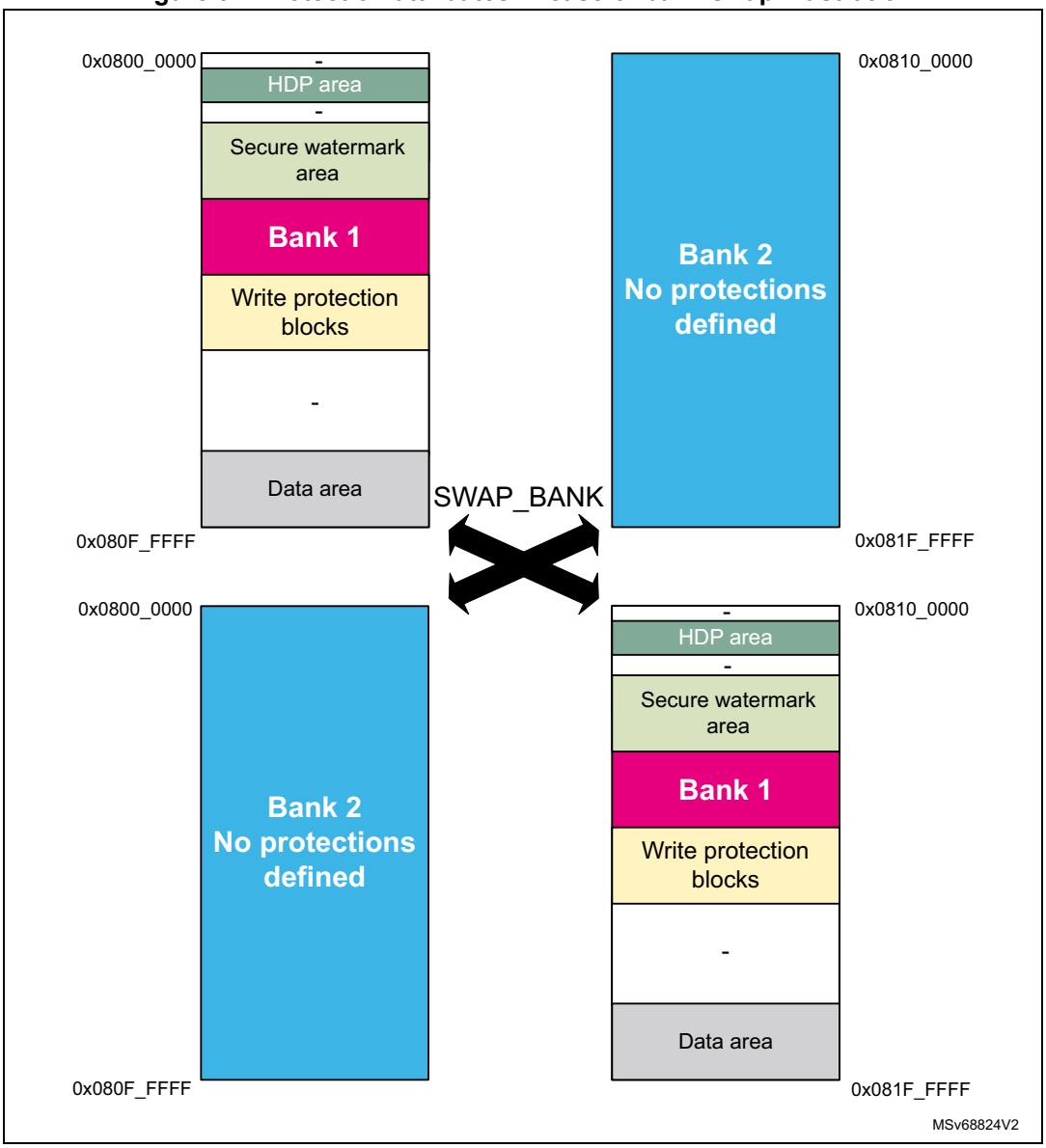
- Flash secure watermark x FLASH_SECWMx
- Flash write protection sector group FLASH_WRPSG (refer to *Section 7.6.8*)
- Hide protection in FLASH_HDPx registers
- Flash secure block based bank x register y FLASH_SECBBxy
- Flash privilege block based bank x register y FLASH_PRIVBBxy
- The data area configuration FLASH_EDATA

The SWAP_BANK is rendered immutable by setting BOOT_LOCK in the user OB. If TZEN is enabled, then SECBOOT_LOCK locks the option. If TZEN is disabled, then NSBOOT_LOCK locks the SWAP_BANK option.

Note: The BK_ECC bit in the FLASH ECC detection register (FLASH_ECCDETR) and FLASH ECC correction register (FLASH_ECCCORR), BKSEL bit in FLASH non-secure control register (FLASH_NSCR) and BKSEL bit in FLASH secure control register (FLASH_SECCR) always refers to Bank1 (respectively Bank2) when it is low (respectively high), regardless of the SWAP_BANK value.

Figure 34 shows how security attributes and protections behave in case of bank swap.

Figure 34. Protection attributes in case of bank swap illustration



7.6.7 Flash memory configuration protection

The memory uses hardware mechanisms to protect the following assets against unwanted or spurious modifications (such as software bugs):

- Option bytes change
- Write operations
- Erase commands
- Interrupt masking

The memory configuration registers protection is summarized in *Table 67*. Registers not present in this table are not protected by a key.

Table 67. Flash interface register protection summary

Register name	Unlocking register	Protected asset
FLASH_NSCR	FLASH_NSKEYR	Write/erase control
FLASH_SECCR	FLASH_SECKEYR	Secure write/erase control
FLASH_OPTCR + all _PRG registers	FLASH_OPTKEYR	Flash bank option byte words change
FLASH_NSOBKCFGR	FLASH_NSOBKKEYR	Flash OBKey storage manipulation
FLASH_SECOBKCFGR	FLASH_SECOBKKEYR	

7.6.8 Write protection

The purpose of the write protection is to prevent unwanted modifications to code and/or data.

Any group of four consecutive 8-Kbyte sectors can be independently write-protected or unprotected by clearing/setting the corresponding WRPSGn1/2 bit in the FLASH_WRPSG1/2 register (see *FLASH write sector group protection for Bank1 (FLASH_WRP1R_CUR)* and *FLASH write sector group protection for Bank2 (FLASH_WRP2R_CUR)*).

A write-protected group of sectors cannot be erased nor programmed. As a result, a bank erase cannot be performed if one group of sectors is write-protected, unless a NVSTATE transition to OPEN is triggered (erasing the whole user flash memory).

Note: Write protection errors are documented in Section 7.9.

7.6.9 Flash high-cycle data protections

The memory can be configured to have 96-Kbyte memory area with high-cycling capability (100 kcycles) to store data and emulate EEPROM, see *Section 7.3.10*.

Option byte controlled security features for user flash memory prevent sectors from being used as high-cycle data area. No OPTCHANGEERR is raised in case of setting conflict, it only makes the sectors unusable. Details are listed in *Table 68*.

Volatile settings using FLASH_SECBBxx and FLASH_PRIVBBxx can be applied. In this case the security setting of user flash sector 127 applies to flash high-cycle data sector 0 and so on, as indicated in *Figure 25*, *Figure 26*, and *Figure 29*.

Volatile settings of HDP extension that extends into sectors of data area cause the data area to be inaccessible.

Table 68. High-cycle area protection summary: access to data area address range

<table>
<tr><th colspan="2" rowspan="3">TZ protection (when TZ_STATE = 0xB4)(1)</th><th colspan="6">Data sector enabled</th><th rowspan="3">Data sector not enabled</th></tr>
<tr><th rowspan="2">HDP protected</th><th rowspan="2">OB secure area</th><th colspan="2">SecBB protected</th><th colspan="2">NS sector</th></tr>
<tr><th>16/32 bits access</th><th>Other</th><th>16/32 bits access</th><th>Other</th></tr>
<tr><td rowspan="4">Secure</td><td>Fetch</td><td colspan="2" rowspan="3">BUS ERROR</td><td colspan="4">BUS ERROR</td><td rowspan="5">BUS ERROR</td></tr>
<tr><td>Read</td><td>OK</td><td rowspan="2">BUS ERROR</td><td colspan="2">RAZ, ILAC</td></tr>
<tr><td>Write</td><td>If not WRP: OK else WI, WRPERR</td><td colspan="2">WI, WRPERR, ILAFM</td></tr>
<tr><td>Erase</td><td colspan="6">Same protection as in corresponding classic user flash sector</td></tr>
<tr><td rowspan="4">Non-secure</td><td>Fetch</td><td colspan="2" rowspan="3">BUS ERROR</td><td colspan="4">BUS ERROR</td></tr>
<tr><td>Read</td><td colspan="2">RAZ, ILAC</td><td>OK</td><td rowspan="2">BUS ERROR</td></tr>
<tr><td>Write</td><td colspan="2">WI, WRPERR, ILAFM</td><td>If not WRP: OK else WI, WRPERR</td></tr>
<tr><td>Erase</td><td colspan="6">Same protection as in corresponding classic user flash sector</td></tr>
</table>

1. If TZ_STATE = 0xC3, there is no secure state access, no SecBB, and no Secure Area. For HDP and NS access the rules remain the same.

Table 69. HDP protected definition

User flash	OB HDP area enabled	HDP extension enabled
HDPL1	HDP protected = 0	HDP protected = 0
HDPL2	HDP protected = 1	HDP protected = 0
HDPL3	HDP protected = 1	HDP protected = 1

Table 70. Privileged sectors and data area - Access to data area address range

<table>
<tr><th colspan="2" rowspan="2">Access</th><th colspan="2">Data area address range access</th></tr>
<tr><th>Unprivileged sector</th><th>Privileged sector</th></tr>
<tr><td rowspan="3">Privileged</td><td>Fetch</td><td colspan="2">BUS ERROR</td></tr>
<tr><td>Read</td><td>OK</td><td>OK</td></tr>
<tr><td>Write</td><td>OK</td><td>OK</td></tr>
<tr><td rowspan="3">Unprivileged</td><td>Fetch</td><td colspan="2">BUS ERROR</td></tr>
<tr><td>Read</td><td>OK</td><td>RAZ, ILAC</td></tr>
<tr><td>Write</td><td>OK</td><td>WI, WRPERR</td></tr>
</table>

7.6.10 Life cycle management

Non-volatile or debug states are determined by the product state set by user option bytes. Debug possibility and possibility to change selected security settings are related to it. While the state is stored in the OB, it is the SBS that controls the debug policy.

OBK storage access conditions are unaffected by the PRODUCT_STATE, but keys can be evicted in regression transitions.

HDP works regardless of the debug state. Even in debug the HDP area is hidden when HDPL increments.

The following states are defined:

Table 71. Product states, debug states and debug policy

PRODUCT_STATE	Code in OB	Description
Open	0xED	User flash open (TZ secure and non-secure open). External access (debug) N and NS enabled (~RDP0).
Provisioning	0x17	Provisioning - Immutable root of trust is being installed. Open if TZEN = 0xC3. The debug is opened in HDPL3-NS.
iROT-Provisioned	0x2E	The immutable root of trust is installed (~RDP 0.5).
TZ-Closed	0xC6	State for debugging the NS application. Debug is restrained to non-secure areas.
Closed	0x72	State for running a secure application. Debug disabled, regression is possible.
Locked	0x5C	Transition to other state, policy change or debug not permitted(1).
Regression	0x9A	The temporary state initiated by the debug authentication system in transition to Open.
NS-Regression	0xA3	The temporary state initiated by the debug authentication system in transition to TZ-Closed.

1. The same situation is Closed state with incorrect or not defined debug certificates.

1. No debug protection

Read, program and erase operations into the flash main memory area are possible. Least restriction on the option bytes.

2. Non-secure debug only

All read and write operations (if no write protection is set) from/to the non-secure flash memory are possible. The debug access to secure area is prohibited. Debug access to non-secure area remains possible.

The following rules are enforced:

- User mode: code executing in user mode (boot flash) can access flash main memory and option bytes with all operations (read, erase, program).
- Non-secure debug mode: The secure flash memory areas are inaccessible; the non-secure flash memory areas remain accessible for debug purpose.
- Boot RAM mode: boot from SRAM is not possible.
- Transition to Open state is possible with the consequence of flash mass erase and key slots revocation. Depends on debug unlock policy (see below).

3. Full product protection

All debug features are disabled and following rules enforced:

- User mode: code executing in user mode (boot flash) can access flash main memory and selected option bytes with all operations (read, erase, program).
- Boot RAM mode: boot from SRAM is not possible.
- Transition to TZ-Closed or Open state is possible with the consequence of flash mass erase and key slots revocation. Depends on debug unlock policy (see below).

7.6.11 Product state transitions

Progressing towards a more closed state is the normal product life cycle, that does not require security measures. Transition towards Open state is a regression, controlled by the debug authentication control. If debug unlock policy is set to “locked”, no regression is accepted. Locked state can be reinforced by invalidating the debug authentication certificates.

All transitions not listed in this chapter are invalid.

There is no restriction on reading the current product state.

Transitions in progress direction

Transition from Open to any of the closed states is matter of correct product configuration and provisioning. The transitions must be done in correct order.

The normal progression is:

- Open to Provisioning or iRoT-Provisioned
- Provisioning to iRoT-Provisioned
- Provisioning or iRoT-Provisioned to TZ-Closed, Locked or Closed
- TZ-Closed to Locked or Closed

Transition is managed either by a software running on the device, or directly, using a debug interface. Transitions from Closed are only possible by software running on the device. By software it is assumed that transitions are triggered by bootloader, or root-of-trust services, but generally any software running on the device can do a progress transition.

Transition to Open state

This transition is a full regression. The starting state is any except Open and Locked. The debug tools are used to authenticate debug regression access rights with the debug authentication library, running on the device in HDPL1 (refer to *Section 14.3.6: SBS debug control*). After verifying the credentials, it puts the device into intermediate state Regression. From this state the device regresses securely in HDPL0 to Open.

The transition has the following consequence:

- Epoch secure counter is incremented
- Key slots revocation
- Product state updated
- User options updated (security settings reset, TZEN reset)
- Flash memory mass erase
- Backup RAM erase is requested

Transition from closed to closed-secure debug state

This transition opens the device to authorize the debug of the non-secure application software without compromising the security of the ROT functions.

The starting state is Closed. The debug tools are used to authenticate debug regression access rights with the debug authentication library, running on the device in the HDPL1 (refer to *Section 14.3.6: SBS debug control*). After verifying the credentials, it puts the device into intermediate state NS-Regression. From this state, the device regresses securely in HDPL0 to Closed.

The transition has the following consequence:

- Epoch non-secure counter is incremented
- Non-secure slots revocation
- Product state updated
- Flash memory sectors not covered by the secure watermark are erased
- Backup RAM erase is requested

Table 72. PRODUCT_STATE transitions

From	To(1)							
	Open	Provisioning	iROT-Provisioned	TZ-Closed(2)	Closed	Locked	Regression	NS-Regression(2)
Open	-	OK	OK	X	X	X	X	X
Provisioning	X	-	OK	OK	OK	OK	OK (HDPL1)	X
iRoT-Provisioned	X	X	-	OK	OK	OK	OK (HDPL1)	X
TZ-Closed	X	X	X	-	OK	OK	OK (HDPL1)	X
Closed	X	X	X	X	-	X	OK (HDPL1)	OK (HDPL1)
Locked	X	X	X	X	X	-	X	X

Table 72. PRODUCT_STATE transitions (continued)

From	To[1]							
	Open	Provisioning	iROT-Provisioned	TZ-Closed[2]	Closed	Locked	Regression	NS-Regression[2]
Regression	OK (HDPL0)	X	X	X	X	X	-	X
NS-Regression	X	X	X	OK (HDPL0)	X	X	X	-

1. X = transitions not permitted (OPTCHANGEERR raised), - = no change, OK = valid transitions.
2. To transition into this state, both the current and programmed values of TZEN must be enabled (TZEN = 0xB4), otherwise the change is ignored with OPTCHANGEERR.

In *Table 72*, X indicates transitions that are not permitted, a dash indicates no change, OK indicates valid transitions. Some transitions are possible only in correct HDPL provided in parenthesis. Attempts to do an illegal transition result in OPTCHANGEERR.

Incorrect PRODUCT_STATE (no OBL) is interpreted as Locked.

7.6.12 OBK protection

At the receipt of an OBK access (read/write/execute), secure and privilege attributes are first checked, followed by OBK-HDPL. The OBK-HDPL value must exactly match the HDPL assigned to the key location, otherwise OBKERROR flag is raised. Flash interface response is described in *Table 73*.

Table 73. TZ OBK protection summary

TZ protection (when TZ_STATE = 0xB4)	OBK access		
	No tamper		Tamper
	HDPL0/1/2/3	OBK selector	
XS	BUS ERROR	BUS ERROR	RAZ, WI
RS	OK if privilege, otherwise RAZ	RAZ	
WS	OK if privilege, otherwise WI, WRPERR	WI, if not privilege also WRPERR	
XNS	BUS ERROR	BUS ERROR	
RNS	RAZ, ILAFM	RAZ, ILAFM	
WNS	WI, WRPERR, ILAFM	WI, WRPERR, ILAFM	
Erase	NA	NA	NA

Table 74. OBK protection summary with TZ disabled

TZ protection (when TZ_STATE = 0xC3)	OBK access		
	No tamper		Tamper
	HDPL1/2/3	OBK selector	
XNS	BUS ERROR	BUS ERROR	RAZ, WI
RNS	OK if privileged, otherwise RAZ	RAZ	
WNS	OK if privilege, otherwise WI, WRPERR	WI if privileged, otherwise WI, WRPERR	
Erase	NA	NA	NA

Note: Cacheable access is managed at system level and not in the flash interface. The MPU must be configured to disable cache where it is not desirable.

In case of tamper, all keys are read as 0x00s.

There are also rules for accessing the OBKFGCR and OBKKEYR registers.

Table 75. Access conditions to secure control register

-	S/NS	P/NP	TZ state	SECOBKCFGR and SECOBKKEYR access
X	-	-	-	Bus error
R/W	S	P	Active	OK
R/W	S	NP	Active	RAZ, WI
R/W	NS	P	Active	RAZ, WI, ILAP
R/W	NS	NP	Active	RAZ, WI, ILAP
R/W	-	-	Deactivated	RAZ, WI

Table 76. Access conditions to non-secure control register

	S/NS	P/NP	TZ state	NSOBKCFGR and NSOBKKEYR access
X	-	-	-	Bus error
R/W	-	-	Active	RAZ, WI
R/W	-	P	Deactivated	OK
R/W	-	NP	Deactivated	RAZ, WI

7.6.13 One-time-programmable and read-only memory protections

Sections of OTP/RO in the flash memory are described in details in *Section 7.3.9*. There is no protection provided by the flash memory interface other than the dedicated write protection.

No write is possible to the RO area, once it was established in manufacturing. The *OTP write protection* describes a method of locking out the OTP area.

The access conditions are summarized in *Table 77*.

Table 77. OTP/RO access constraints

-	8 Kbytes sector dedicated to RO/OTP			
	0x08FF_E000 - E7FF	**0x08FF_E800 - EFFF Reserved**	**0x08FF_F000 - F7FF OTP**	**0x08FF_F800 - FFFF RO**
XS	BUS ERROR	BUS ERROR		
RS		RAZ, ILAC		
WS		WI, WRPERR, ILAFM		
XNS	BUS ERROR	BUS ERROR		
RNS		If correct size(1) RAZ, else BUS ERROR	If correct size OK, else BUS ERROR	
WNS		If correct size WI, else BUS ERROR	BUS ERROR if incorrect size. WRPERR if block is protected by LOCKBL, else OK.	If correct size WI, else BUS ERROR
Erase	WI			

1. Word size is 16 bits. 32-bit access is possible. Other access attempt leads to bus error.

7.7 System memory

7.7.1 Introduction

System memory stores RSS (root secure services) firmware programmed by ST during production. The RSS provides runtime services to user firmware.

When TrustZone is enabled (user sets TZEN bitfield to 0xB4) then RSS provides secure services to secure user firmware only; when TrustZone is disabled (user sets TZEN bitfield to 0xC3), RSS provides services to user firmware.

7.7.2 RSS user functions

The RSS provides runtime services thanks to RSS library, whose functions are exposed to user within the CMSIS device header file provided by the STM32CubeH5 firmware package (see UM3065 "*Getting started with STM32CubeH5 for STM32H5 Series*" for more details).

RSS provides services through two different libraries, RSSLIB and NSSLIB.

Table 78. RSSLIB/NSSLIB accesses

TZEN configuration	User FW Non-Secure	User FW Secure	Bootloader/JTAG/SWD(1)
Enabled (0xB4)	N/A	RSSLIB	RSSLIB
Disabled (0xC3)	NSSLIB	N/A	

1. When the devices boot in bootloader, that is, in product state OPEN (with BOOT pin set to 1 and UBE set to 0xB4), or PROVISIONING.

Table 79. RSSLIB/NSSLIB entry point access

C defined macro	Location in flash memory
RSSLIB_PFUNC	0xBF9FB68
NSSLIB_PFUNC	0xBF9FB6C

RSSLIB

The secure user firmware calls RSSLIB functions using RSSLIB_PFUNC C defined macro, which points to a location within non-secure system memory.Before calling RSSLIB functions, the secure user firmware must define a non-secure region above this location within SAU of the Cortex®-M33, starting from RSSLIB_SYS_FLASH_NS_PFUNC_START (0xBF9FB78), up to RSSLIB_SYS_FLASH_NS_PFUNC_END (0xBF9FB84). These last addresses are provided within the CMSIS device header file.

The user can set this non-secure region either by using the CMSIS system partition header file, or by implementing its own code for SAU setup. The CMSIS system partition header file is part of the STM32CubeH5 firmware package.

Table 80. RSS lib interface functions

Library	Function	Attributes
RSSLIB_PFUNC	*JumpHDPLvl2*	Secure callable function
	JumpHDPLvl3	
	JumpHDPLvl3NS	
	DataProvisioning	Non-secure callable function

DataProvisioning

Secure attribute: Non-secure callable function.

Prototype:

```
uint32_t_ DataProvisioning (RSSLIB_DataProvisioningConf_t *pConfig)
```

User code function call example:

```
RSSLIB_PFUNC→NSC.DataProvisioning(&pConfig)
```

Arguments:

pConfig: input parameter. **RSSLIB_DataProvisioningConf_t**

C structure definition is described below:

```
typedef struct
{
  uint32_t *pSource;
  uint32_t *pDestination;
  uint32_t Size;
  uint32_t DoEncryption;
  uint32_t Crc;
} RSSLIB_DataProvisioningConf_t;
```

Structure elements

pSource Provides the address of data to be provisioned. Must be within SRAM3 address range (non-secure aliases).

pDestination Provides the address where to store data to be provisioned. Must be within OBKeys address range. To be aligned on 16 bytes.

Size Provides the size of data to be provisioned (the number of bytes, must be a multiple of 16).

DoEncryption Notifies RSSLIB_DataProvisioning function if it must encrypt or not data within OBKeys.

DoEncrypt can be either 0xF5F5A0AAU or 0xCACA0AA0U (the only one allowed on STM32H563xx devices)

– 0xF5F5A0AAU: notifies RSSLIB_DataProvisioning to encrypt data with relevant DHUK before programming it within OBKeys. DHUK is selected according to **pDestination** value.

– 0xCACA0AA0U: notifies RSSLIB_DataProvisioning to program in clear data within OBKeys.

Crc CRC over full source data buffer, **pConfig->pDestination** value, **pConfig->Size** value and finally **pConfig->DoEncryption** value.

CRC computation uses CRC-32 (Ethernet):

– CRC polynomial: 0x04C11DB7U

– Initial value: 0xFFFFFFFFU

Returned values

0xEAEAEAEAU	Success
0xF5F5E0E0U	Error: – **pConfig** or **pSource** are not in SRAM3 – **pDestination** is not aligned on 16 bytes – **pDestination + Size** does not fit within an OBKey section
0xF5F50E0EU	Error: **Size** is not a multiple of 16
0xF5F58080U	Error: computed CRC is not the expected one provided within **pConfig->CRC**
0xF5F58008U	Error: cannot program data within OBKeys destination section
0xF5F5E00EU	Error: wrong **DoEncryption** parameter value
0xF5F50880U	Error: encryption requested, butw platform does not support it
0xF5F50EE0U	Error: encryption error
0xF5F50808U.	Error: OBKeys programming error

RSSLIB_DataProvisioning receives in input a data buffer and programs it within OBKeys. A CRC prevents any data and parameters tampering issue.

The destination address (**pDestination**) added to the size (**Size**) of the data to be provisioned must not cross the OBKeys section boundaries, listed below.

OBKeys Level1	Start address: 0x0FFD0100UL
	End address: 0x0FFD08FFUL

(continued)

OBKeys Level2	Start address: 0x0FFD0900UL
	End address: 0x0FFD0BFFUL
OBKeys Level3 secure	Start address: 0x0FFD0C00UL
	End address: 0x0FFD17FFUL
OBKeys Level3 non-secure	Start address: 0x0FFD1800UL
	End address: 0x0FFD1FEFUL

When requested through **pConfig->DoEncryption** parameter, RSSLIB_DataProvisioning encrypts data before programming them within OBKeys.

RSSLIB_DataProvisioning uses AES CBC 128 bits with:

- IV: defined as (using C definition format)
  uint32_t IV = {0x8001D1CEU, 0xD1CED1CEU, 0xD1CE8001U, 0xCED1CED1U};
- Key: DHUK corresponding to the targeted OBKeys section defined by **pConfig->pDestination** parameter.

Note: Data encryption within OBkeys is supported only by STM32H533/573xx devices.

JumpHDPLvl2

Secure attribute: Secure callable function.

Prototype:

```
uint32_t JumpHDPLvl2(uint32_t VectorTableAddr, uint32_t MPUIndex)
```

User code function call example:

```
RSSLIB_PFUNC->S.JumpHDPLvl2((uint32_t)NextVectorTableAddr, 1U );
```

Arguments:

- VectorTableAddr:
  - Input parameter, address of the next vector table to apply.
  - The vector table format is the one used by the Cortex-M33 core.
- MPUIndex:
  - Input parameter, MPU region index. Caller function must define (but keep disabled) the corresponding MPU region before calling JumpHDPLvl2. The function enables the MPU region before jumping to the reset handler of the vector table. The vector table reset handler function belongs to the MPU region.

User calls JumpHDPLvl2 to close user Flash HDPL1 area by incrementing HDPL to 2, then jump to the reset handler embedded within the vector table, whose address is passed as input parameter.

After closing HDPL1, JumpHDPLvl2 enables the MPU region provided as input parameter. Once the MPU is enabled, the function sets the SP to the address provided by the passed vector table, and jumps to the reset handler function supported by it. JumpHDPLvl2 does not set the new vector table.

On successful execution, the function does not return and does not push LR onto the stack.

In case of failure (bad input parameter value), RSSLIB_Sec_JumpHDPLvl2 returns 0xF5F5F5F5.

JumpHDPLvl3

Secure attribute: Secure callable function.

Prototype:

```
uint32_t JumpHDPLvl3(uint32_t VectorTableAddr, uint32_t MPUIndex)
```

User code function call example:

```
RSSLIB_PFUNC->S.JumpHDPLvl3((uint32_t)NextVectorTableAddr, 1U );
```

Arguments:

- VectorTableAddr:
  - Input parameter, address of the next vector table to apply.
  - The vector table format is the one used by the Cortex-M33 core.
- MPUIndex:
  - Input parameter, MPU region index. Caller function must define but keep disabled the corresponding MPU region before calling JumpHDPLvl3. The function enables the MPU region before jumping to the reset handler of the vector table, whose function belongs to the MPU region.

User calls JumpHDPLvl3 to close user Flash HDPL1 and HDPL2 areas by incrementing HDPL up to 3, then jump to the reset handler embedded within the vector table, whose address is passed as input parameter.

After closing HDPL1/2, JumpHDPLvl3 enables the MPU region provided as input parameter. Once the MPU is enabled, the function sets the SP to the address provided by the passed vector table, and jumps to the reset handler function supported by the vector table. JumpHDPLvl3 does not set the new vector table.

On successful execution, the function does not return and does not push LR onto the stack.

In case of failure (bad input parameter value), JumpHDPLvl3 returns 0xF5F5F5F5.

JumpHDPLvl3NS

Secure attribute: Secure callable function.

Prototype:

```
uint32_t JumpHDPLvl3NS(uint32_t VectorTableAddr)
```

User code function call example:

```
RSSLIB_PFUNC->S.JumpHDPLvl3NS((uint32_t)NextVectorTableAddr, 1U );
```

Arguments:

- VectorTableAddr:
  - Input parameter, address of the next vector table to apply.
  - The vector table format is the one used by the Cortex-M33 core.

User calls JumpHDPLvl3NS to close user flash HDPL1 and HDPL2 areas by incrementing HDPL up to 3, to move from secure to non-secure domain, then jump to the non-secure reset handler embedded within the vector table, whose address is passed as input parameter.

After closing HDPL1/2, JumpHDPLvl3 jumps to the non-secure reset handler function supported by the vector table. JumpHDPLvl3NS does not set the new vector table.

On successful execution, the function does not return and does not push LR onto the stack.

In case of failure (bad input parameter value), JumpHDPLvl3NS returns 0xF5F5F5F5.

NSSLIB

The user firmware calls only the NSSLIB function when TrustZone is disabled (TZEN bitfield is set to 0xC3).

The user firmware calls NSSLIB functions using NSSLIB_PFUNC C defined macro, which points to a location within system memory.

Table 81. NSS lib interface functions

Library	Function	Attributes
NSSLIB_PFUNC	JumpHDPLvl2	Non-secure function
	JumpHDPLvl3	

JumpHDPLvl2

Prototype:

```
uint32_t JumpHDPLvl2(uint32_t VectorTableAddr, uint32_t MPUIndex)
```

User code function call example:

```
NSSLIB_PFUNC->JumpHDPLvl2((uint32_t)NextVectorTableAddr, 1U );
```

Arguments:

- VectorTableAddr:
  - Input parameter, address of the next vector table to apply.
  - The vector table format is the one used by the Cortex-M33 core.
- MPUIndex:
  - Input parameter, MPU region index. Caller function shall define but keep disable the corresponding MPU region before calling JumpHDPLvl2. The function enables

the MPU region before jumping to the reset handler of the vector table. The vector table reset handler function belongs to the MPU region.

User calls JumpHDPLvl2 to close user Flash HDPL1 area by incrementing HDPL to 2, and to jump to the reset handler embedded within the vector table which address is passed as input parameter

After closing HDPL1, JumpHDPLvl2 enables the MPU region provided as input parameter. Once the MPU is enabled, the function sets the SP to the address provided by the passed vector table and jumps to the reset handler function supported by the vector table too. JumpHDPLvl2 does not set the new vector table.

On successful execution, the function does not return and does not push LR onto the stack.

In case of failure (bad input parameter value), RSSLIB_Sec_JumpHDPLvl2 returns 0xF5F5F5F5.

JumpHDPLvl3

Prototype:

```
uint32_t JumpHDPLvl3(uint32_t VectorTableAddr, uint32_t MPUIndex)
```

User code function call example:

```
NSSLIB_PFUNC->JumpHDPLvl3((uint32_t)NextVectorTableAddr, 1U );
```

Arguments:

- VectorTableAddr:
  - Input parameter, address of the next vector table to apply.
  - The vector table format is the one used by the Cortex-M33 core.
- MPUIndex:
  - Input parameter, MPU region index. Caller function shall define but keep disable the corresponding MPU region before calling JumpHDPLvl3. The function enables the MPU region before jumping to the reset handler of the vector table. The vector table reset handler function belongs to the MPU region.

User calls JumpHDPLvl3 to close user Flash HDPL1 and HDPL2 areas by incrementing HDPL up to 3, and then jump to the reset handler embedded within the vector table, whose address is passed as input parameter

After closing HDPL1/2, JumpHDPLvl3 enables the MPU region provided as input parameter. Once the MPU is enabled, the function sets the SP to the address provided by the passed vector table, and jumps to the reset handler function supported by it. JumpHDPLvl3 does not set the new vector table.

On successful execution, the function does not return and does not push LR onto the stack.

In case of failure (bad input parameter value), JumpHDPLvl3 returns 0xF5F5F5F5.

7.8 FLASH low-power modes

Table 82 summarizes the memory behavior in STM32 low-power modes. Embedded flash memory belongs to the core domain.

Table 82. Effect of low-power modes on the embedded flash memory

Power mode	Core domain voltage range	Allowed if FLASH busy	FLASH power mode
Run	VOS0/1/2/3	Yes	Run
Stop1 (clock stopped)	SVOS3/4/5	No	Clock gated or stopped in case of SVOS5
Standby	Off	No	Off

When the system state changes or within a given system state, the memory can operate with a different voltage supply range (VOS), according to the application. The procedure to switch the memory into power modes (run, clock gated, stopped, off) is described hereafter.

Note: For more information on the microcontroller power states, refer to Section 10: Power control (PWR).

Managing the FLASH domain switching to Stop or Standby

As explained in *Table 82*, if the memory informs the reset and clock controller (RCC) that it is busy (BSY, DBNE, WBNE is set), the microcontroller cannot switch the core domain to Stop or Standby mode.

There are two ways to release the memory:

- Reset the WBNE busy flag in FLASH_NS/SECSR register by any of the following actions:
  - Complete the write buffer with missing data.
  - Force the write operation without filling the missing data by activating the FW bit in FLASH_NS/SECCR register. This forces all missing data “high”.
- Poll BSY busy bits in FLASH_NS/SECSR register until they are cleared. This indicates that all recorded write, erase and option change operations are complete.

The microcontroller can then switch the domain to Stop or Standby mode.

7.9 FLASH error management

7.9.1 Introduction

The memory automatically reports when an error occurs during a read, program or erase operation. A wide range of errors are reported:

- *Non-secure write protection error (WRPERR)*
- *Secure write protection error (WRPERR)*
- *Non secure programming sequence error (PGSERR)*
- *Secure programming sequence error (PGSERR)*
- *Secure strobe error (STRBERR)*
- *Error correction code error (ECCC, ECCD)*
- *Illegal access (ILAFM/ILAP)*
- *Option byte change error (OPTCHANGEERR)*
- OBK non-secure general error (OBKERR)
- OBK secure general error (OBKERR)
- OBK non-secure write error (OBKWERR)
- OBK secure write error (OBKWERR)

The application software can individually enable the interrupt for each error, as detailed in *Section 7.10*.

The flash memory interface uses different interrupt lines to trigger the event. There are two direct lines to NVIC, one for secure interrupts, other for non secure. Third line leads to TZIC, for the ILAFM interrupt.

Note: For all errors, the application software must clear the error flag before attempting a new modify operation.

Since there is just one write buffer and only one operation is allowed at a time, the write control bits and status bits are shared by both banks:

- Write control bits (PG, FW, EOPIE, WRPERRIE, PGSERRIE, STRBERRIE, INCERRIE, CLR_EOP, CLR_WRPERR, CLR_STRBERR, CLR_INCERR, CLR_PGSERR) controls Bank1 and 2 at the same time.
- Write status flag reports (WBNE, DBNE, EOP, WRPERR, PGSERR, STRBERR, INCERR, BSY) error for Bank1 and 2 at the same time.

7.9.2 Non-secure write protection error (WRPERR)

When an illegal non-secure erase/program operation is attempted to the nonvolatile memory, the flash interface sets the write protection error flag WRPERR in FLASH_NSSR register.

An erase operation is rejected and flagged as illegal if it targets one of the following memory areas:

- A sector write-protected with WRPSGn
- An HDP area while the HDPL has made it inaccessible
- Secure (TZ) state not matching the erased memory attribute
- Attempt to erase privilege sector within unprivileged mode

A program operation is ignored and flagged as illegal if it targets one of the following memory areas:

- System flash memory
- A user sector write-protected with WRPSGn
- An OTP block, locked with LOCKBL
- A read-only section
- A reserved area
- Secure area
- Privileged area from within unprivileged mode
- OBK access conditions check error

When WRPERR flag is raised, the operation is rejected and nothing is changed in the corresponding bank. If this error is detected, the write buffer is invalidated.

Note: WRPERR flag must be cleared before any erase/program operation.

WRPERR flag is cleared by setting CLR_WRPERR bit to 1 in FLASH_NSCCR register.

If WRPERRIE bit in FLASH_NSCR register is set to 1, an interrupt is generated when WRPERR flag is raised (see *Section 7.10* for details).

7.9.3 Secure write protection error (WRPERR)

When an illegal secure erase/program operation is attempted to the nonvolatile memory, the flash interface sets the write protection error flag WRPERR in FLASH_SECSR register.

An erase operation is rejected and flagged as illegal if it targets one of the following memory areas:

- A sector write-protected with WRPSGn
- An HDP area while the HDPL has made it inaccessible
- Secure (TZ) state not matching the erased memory attribute
- Attempt to erase privilege sector within unprivileged mode

A program operation is ignored and flagged as illegal if it targets one of the following memory areas:

- System flash memory
- A user sector write-protected with WRPSGn
- An OTP block, locked with LOCKBL
- A read-only section
- A reserved area
- Non-secure area
- Privileged area from within unprivileged mode
- OBK access condition check error

When WRPERR flag is raised, the operation is rejected and nothing is changed in the corresponding bank.

If this error is detected the write buffer is invalidated.

Note: WRPERR flag must be cleared before any erase/program operation.

WRPERR flag is cleared by setting CLR_WRPERR bit to 1 in FLASH_SECCCR register.

If WRPERRIE bit in FLASH_SECCR register is set to 1, an interrupt is generated when WRPERR flag is raised (see *Section 7.10* for details).

7.9.4 Non secure programming sequence error (PGSERR)

When the non-secure programming sequence is incorrect, the flash interface sets the programming sequence error flag PGSERR in FLASH_NSSR register.

More specifically, PGSERR flag is set when one of below conditions is met:

- An error like INCERR, WRPERR, PGSERR, STRBERR, OPTCHANGEERR, OBKERR or OBKWERR have not been cleared before requesting a new write or erase operation, OBK operation or options change.
- For erase, PGSERR is set if:
  - missing erase operation (FLASH_NSCR): STRT = 1 with (MER = 0, SER = 0 and BER = 0)
  - sector and bank erase requested at the same time (NSSTRT and more than one of MER, SER and BER)
  - PG set during erase operation (it ensures that no erase req and write req happen at the same time): (STRT = 1) with (PG = 1)
  - Erase operation is started while write buffer is waiting for next data: (STRT = 1) with WBNE = 1.
  - STRT = 1 set by secure access.
  - Erase operation is started while DBNE = 1.
- For programming, PGSERR is set if:
  - missing write flag: A write operation is requested but the program enable bit PG has not been set in FLASH_NSCR register prior to the request.
  - write operation to flash high-cycle data is requested but PG = 0.
  - AHB write request is received and (SER = 1, BER = 1 or MER = 1)
  - 16 bit data access requested while WBNE = 1
- For options change, PGSERR is set if:
  - option change is started while write buffer is waiting for next data: OPTSTRT = 1 with WBNE = 1
  - OPTSTRT set with non-secure DBNE = 1.
- For options byte key storage access, PGSERR is set if:
  - error flags from previous operation not cleared
  - ALT_SECT_ERASE and SWAP_SECT_REQ in FLASH_NSOBKCFGR are set at the same time
  - SWAP_OFFSET value is wrong: when SWAP_SECT_REQ is set to 1, the SWAP_OFFSET should be equal or greater than OBK_HDPL< N - 1 > _SWAP_OFFSET
  - ALT_SECT_ERASE or OBK_SWAP set while DBNE = 1
  - OBK-HDPL is incorrect and non-secure OBK_SWAP is requested
  - If OBKSWAP or OBK alternate erase is started while write buffer is waiting for new data (WBNE = 1).

When PGSERR flag is raised as consequence of failed write attempt (flash programming), the current program operation is aborted and nothing is changed in the corresponding bank. The write data buffer is also invalidated.

Note: When PGSERR flag is raised, there is a risk that the last write operation performed by the application has been lost because of the above protection mechanism. It is recommended to generate interrupts on PGSERR and verify in the interrupt handler if the last write operation has been successful, by reading back the value in the flash memory.

The PGSERR flag also blocks any new program operation. This means that PGSERR must be cleared before starting a new program operation.

PGSERR flag is cleared by setting CLR_PGSERR bit to 1 in FLASH_NSCCR register.

If PGSERRIE bit in FLASH_NSCR register is set to 1, an interrupt is generated when PGSERR flag is raised. See *Section 7.10* for details.

7.9.5 Secure programming sequence error (PGSERR)

When the secure programming sequence is incorrect, the flash interface sets the programming sequence error flag PGSERR in FLASH_SECSR register.

More specifically, PGSERR flag is set when one of below conditions is met:

- An error like INCERR, WRPERR, PGSERR, STRBERR, OBKERR, or OBKWERR has not been cleared before requesting a new write or erase operation, OBK operation or options change.
- For erase, PGSERR is set if:
  - missing erase operation (FLASH_SECCR): STRT = 1 with (SER = 0 and BER = 0, MER = 0)
  - sector and bank erase requested at the same time (SECSTRT and more than one of MER, SER, and BER)
  - PG set during erase operation (it ensures that no erase request and write request happen at the same time): (STRT = 1) with (PG = 1)
  - erase operation is started while write buffer is waiting for next data: (STRT = 1) with WBNE = 1.
  - operation started while secure DBNE = 1.
- For programming, PGSERR is set if:
  - missing write flag: a write operation is requested but the program enable bit PG has not been set in FLASH_SECCR register prior to the request
  - write operation to flash high-cycle data is requested but PG = 0
  - AHB write request is received and (SER = 1, SBER = 1, or MER = 1)
  - 16 bit data access requested while WBNE = 1
- For options change, PGSERR is set if:
  - Option change is started while write buffer is waiting for next data: OPTSTRT = 1 with WBNE = 1
  - OPTSTRT set with secure DBNE = 1
- For options byte key storage access, SECPGSERR is set if:
  - error flags from previous operation not cleared
  - ALT_SECT_ERASE and SWAP_SECT_REQ are set at the same time
  - SWAP_OFFSET value is wrong: when SWAP_SECT_REQ = 1, SWAP_OFFSET must be equal or greater than OBK_HDPL<N-1>_SWAP_OFFSET.
  - SWAP_SECT_REQ or OBK_SWAP set while DBNE = 1
  - OBK-HDPL is incorrect and secure OBK_SWAP is requested

– If OBKSWAP or OBK alternate erase is started while write buffer is waiting for new data (WBNE = 1)

When PGSERR flag is raised as consequence of failed write attempt (flash programming), the current program operation is aborted and nothing is changed in the corresponding bank. The write data buffer is also invalidated.

Note: When PGSERR flag is raised, there is a risk that the last write operation performed by the application has been lost because of the above protection mechanism. It is recommended to generate interrupts on PGSERR and verify in the interrupt handler if the last write operation has been successful by reading back the value in the flash memory.

The PGSERR flag also blocks any new program operation. This means that PGSERR must be cleared before starting a new program operation.

PGSERR flag is cleared by setting CLR_SECPGSERR bit to 1 in FLASH_SECCCR register.

If PGSERRIE bit in FLASH_SECCR register is set to 1, an interrupt is generated when PGSERR flag is raised. See *Section 7.10* for details.

7.9.6 Non-secure strobe error (STRBERR)

When the non-secure application software writes several times to the same byte in the write buffer, the flash interface sets the strobe error flag STRBERR (FLASH_NSSR) whatever the target bank of the write access.

When STRBERR flag is raised, the current program operation is aborted and the write buffer is invalidated.

STRBERR flag is cleared by setting CLR_STRBERR bit to 1 in FLASH_NSCCR register.

If STRBERRIE bit in FLASH_NSCR register is set to 1, an interrupt is generated when STRBERR flag is raised. See *Section 7.10* for details.

7.9.7 Secure strobe error (STRBERR)

When the secure application software writes several times to the same byte in the write buffer, the memory sets the strobe error flag STRBERR (FLASH_SECSR) whatever the target bank of the write access.

When STRBERR flag is raised, the current program operation is aborted and the write buffer is invalidated.

STRBERR flag is cleared by setting CLR_STRBERR bit to 1 in FLASH_SECCCR register.

If STRBERRIE bit in FLASH_SECCR register is set to 1, an interrupt is generated when STRBERR flag is raised. See *Section 7.10* for details.

7.9.8 Non-secure inconsistency error (INCERR)

When a programming inconsistency in non-secure access is detected, the flash interface sets the inconsistency error flag INCERR in register FLASH_NSSR.

More specifically, INCERR flag is set when one of the following conditions is met:

- A write operation is attempted before completion of the previous write operation, for example:
  - The application software starts a write operation to fill the 128-bit write buffer, but sends a new write burst request to a different flash memory address before the buffer is full.
  - One master starts a write operation, but before the buffer is full, another master starts a new write operation to the same address or to a different address.
  - ALT_SECT in FLASH_NSOBKCFGR changed while filling the write buffer.

Note: INCERR flag must be cleared before starting a new write operation, otherwise a sequence error (PGSERR) is raised.

It is recommended to follow the following sequence to avoid losing data when an inconsistency error occurs:

1. Execute a handler routine when INCERR flag is raised.
2. Stop all write requests to the memory.
3. Clear INCERR bit.
4. Restart the write operations from where they have been interrupted.

INCERR flag is cleared by setting CLR_INCERR bit to 1 in FLASH_NSCCR register.

If INCERRIE bit in FLASH_NSCR register is set to 1, an interrupt is generated when INCERR flag is raised (see *Section 7.10* for details).

7.9.9 Secure inconsistency error (INCERR)

When a programming inconsistency is detected, the flash interface sets the inconsistency error flag INCERR in register FLASH_SECSR.

More specifically, INCERR flag is set when one of the following conditions is met:

- A write operation is attempted before completion of the previous write operation, for example:
  - The application software starts a write operation to fill the 128-bit write buffer, but sends a new write burst request to a different flash memory address before the buffer is full.
  - One master starts a write operation, but before the buffer is full, another master starts a new write operation to the same address or to a different address.

Note: INCERR flag must be cleared before starting a new write operation, otherwise a sequence error (PGSERR) is raised.

It is recommended to follow the sequence below to avoid losing data when an inconsistency error occurs:

1. Execute a handler routine when INCERR flag is raised.
2. Stop all write requests to the flash memory.
3. Clear INCERR bit.
4. Restart the write operations from where they have been interrupted.

INCERR flag is cleared by setting CLR_INCERR bit to 1 in FLASH_SECCCR register.

If INCERRIE bit in FLASH_SECCR register is set to 1, an interrupt is generated when INCERR flag is raised (see *Section 7.10* for details).

7.9.10 Error correction code error (ECCC, ECCD)

When a single-error correction is detected during a read, the flash interface sets the single-error correction flag ECCC in FLASH_ECCCORR register.

When two ECC errors are detected during a read, the flash interface sets the double error detection flag ECCD in FLASH_ECCDETR register.

When the ECCC flag is raised, the corrected read data are returned. The application can ignore the error, and request new read operations. When the ECCD the flag is raised, an NMI is generated, it can be masked in SBS registers (*SBS flift ECC NMI mask register (SBS_ECCNMIR)*) for data access (OTP, data area, RO data). Software must invalidate the instruction cache (CACHEINV = 1) in the NMI interrupt service routine when the ECCD flag is set.

When ECCC or ECCD flag is raised, the address of the flash word that generated the error is saved in the FLASH_ECCCORR (FLASH_ECCDETR) register. If the address corresponds to a read-only area or to an OTP area or flash high-cycle data, the OTP_ECC bit is also set to 1 in the FLASH_ECCCORRR (FLASH_ECCDETR) register. This register is automatically cleared when the associated flag that generated the error is reset.

A BK_ECC flag indicates in which flash bank the error occurred.

A SYSF_ECC flag indicates an error detected in the system flash area.

Table 83. Locating ECC failure

OTP_ECC	SYSF_ECC	BK_ECC	EDATA_ECC	OBK_ECC	Flash area	ADDR_ECC min	ADDR_ECC max
0	0	0/1	0	0	User flash	0x0000	0xFFFF
0	1	0/1	0	0	System flash	0x0000	0x0FFF
1	0	0	0	0	OTP	0x0600	0x07FF
0	0	1	0	1	OBKeys	0x0000	0x03FF
0	0	0/1	1	0	Data area sector 7	0xF000	0xF1FF
0	0	0/1	1	0	Data area sector 6	0xF200	0xF3FF
0	0	0/1	1	0	Data area sector 5	0xF400	0xF5FF
0	0	0/1	1	0	Data area sector 4	0xF600	0xF7FF
0	0	0/1	1	0	Data area sector 3	0xF800	0xF9FF
0	0	0/1	1	0	Data area sector 2	0xFA00	0xFBFF
0	0	0/1	1	0	Data area sector 1	0xFC00	0xFDFF
0	0	0/1	1	0	Data area sector 0	0xFE00	0xFFFF

Note: In case of successive single correction or double detection errors, only the address corresponding to the first error is stored in FLASH_ECCCORR (FLASH_ECCDETR) *register.*

It is mandatory to clear ECCC or ECCD flags before starting a new read operation.

As the ECC interface is shared by the two banks, if the same error is registered simultaneously, a physical Bank1 error is registered. Both errors are registered if one bank reports ECCD and the other bank ECCC.

ECCC (respectively ECCD) flag is cleared by setting to 1 ECCC bit (respectively ECCD bit) in FLASH_ECCCORR (FLASH_ECCDETR) register.

If ECCC bit in FLASH_ECCCORR register is set to 1, an interrupt is generated when ECCC flag is raised. Only NMI is generated for the ECCD. See *Section 7.10* for details.

7.9.11 Illegal access (ILAFM/ILAP)

Illegal access is a signal to the TZIC, triggering a secure interrupt in there. It is complementary to the other interrupts and only generated when the TZ is enabled (TZ_STATE = 0xB4).

It can be masked only on GTZC level.

ILAFM is generated when rules on secure flash memory access are violated:

- Attempt to access secure memory location in non-secure mode.
- Attempt to access non-secure memory location in secure mode.

ILAP is generated on:

- Attempt to access secure register in non-secure mode.

If ILAFM is detected during write, the write buffer is invalidated.

Note: ILAFM and ILAP has no flag within the flash memory controller to clear to resume operation. It is dealt within the TZIC.

7.9.12 Option byte change error (OPTCHANGEERR)

When the flash interface finds an error during an option change operation, it aborts the operation and sets the option byte change error flag OPTCHANGEERR in FLASH_NSSR register.

The error is raised after OPTSTRT bit set if:

- TZEN is not set and transition to TZ-Closed or NS-Regression state is requested. Also any other forbidden PRODUCT_STATE transition (*Table 72*).
- SWAP_BANK is locked out by combination of TZEN and BOOT_LOCK.
- OB modification is attempted in wrong HDPL (selected PRODUCT_STATE transitions)
- Attempt to modify OB with invalid value
  - New EPOCH value not greater than current
  - Value not recognized by the specification (enumerated magic numbers only)
- Attempt to modify OB in product state where it is not allowed (usually a state open for debug is required for change).

Note: Exceptions and details provided in the OB description (see Section 7.4.7 and Table 52).

OPTCHANGEERR flag is cleared by setting CLR_OPTCHANGEERR bit to 1 in FLASH_NSCCR register.

If OPTCHANGEERRIE bit in FLASH_NSCR register is set to 1, an interrupt is generated when OPTCHANGEERR flag is raised (see *Section 7.10* for details).

It is mandatory to clean the OPTCHANGEERR flag before starting a new operation (option change, erase or write).

7.9.13 Miscellaneous HardFault errors

The following events generate a bus error on the corresponding bus interface:

- On main AHB system bus for access targeting code and data with 9 bits ECC.
  - Fetching from secure user flash memory in non-secure mode.
  - Fetching from non-secure user flash memory in secure mode.
  - Access to invalid address (including data addresses forbidden for code use).
  - Fetching from HDP area with incorrect HDPL value.
  - Fetching from privilege area in unprivilege mode.
- On AHB configuration or system bus for accesses targeting OTP/RO (all addresses using 6bits ECC):
  - wrong key input to FLASH_NS/SECKEYR or FLASH_OPTKEYR.
  - 8-bit accesses to system AHB interface.
  - Wrong unlock sequence on a register.

7.9.14 OBK error cases (OBKERR, OBKWERR)

Four error types can be detected while filling the OBK write buffer: STRBERR, INCERR, WRPERR or OBKERR. The OBKERR is specific to the OBK access and cannot be raised by access to any other flash memory asset.

Note: For STRBERR, INCERR and WRPERR, non-secure or secure flags are used depending on the AHB access type (sec or non-secure). If a non-secure access is received, the non-secure error flags are used. If a secure access is received, the secure error flags are used.

For OBKERR and OBKWERR, non-secure or secure flags are used depending on the value of the secure state. If the SOC is in secure state (TZ_STATE active), the secure error flags (OBKERR and OBKWERR) are used. Otherwise the non-secure error flags (OBKERR and OBKWERR) are used.

OBKERR is raised when:

- the OBK-HDPL (input signal from SBS) does not match the HDPL associated to the key during a read or write access, and the TZ and PRIV attributes are correct (See *Section 7.5.2* for more details).

If OBKERR is raised during write access, the write buffer is flushed. If the OBKERR is raised on read access, the write buffer is intact. In both cases the flag must be cleared to enable write/erase access to flash memory.

OBKWERR is detected after filling the write buffer and raised if:

- the address is not virgin during a write access.
- OBK selector in the alternate sector is not virgin during a SWAP operation.

If OBKWERR is raised, the write buffer is flushed. The flag must be cleared to enable write/erase access to flash memory.

7.10 FLASH interrupts

The flash interface can generate a maskable interrupt to signal the following events on a given bank:

- Read and write errors (see *Section 7.9*)
  - Single ECC error correction during read operation
  - Write inconsistency error
  - Bad programming sequence
  - Strobe error during write operations
  - option change operation error
- Security errors (see *Section 7.9*)
  - Write protection error
  - Illegal access error - signal to TZIC
- Miscellaneous events (described below)
  - End of programming

A NMI is raised on double ECC error detection during read operation.

Two interrupt lines gather all the error sources, one for errors in non-secure access and second for errors that happen in secure access.

The user can individually enable or disable flash interface interrupt sources by changing the mask bits in the FLASH_NS/SECCR, FLASH_ECCCORR and FLASH_ECCDETR register. Setting the appropriate mask bit to 1 enables the interrupt.

Note: Prior to writing, FLASH_NS/SECCR register must be unlocked as explained in Section 7.6.7.

Table 84 gives a summary of the available flash interface interrupt features. As mentioned in the table below, some flags need to be cleared before a new operation is triggered.

Table 84. Flash interrupt request

Interrupt event Event flag	Error flag label in register	Enable control bit	Clear flag to resume operation
End of operation event	EOP	EOPIE	N/A
Write protection error	WRPERR	WRPERRIE	Yes
Programming sequence error	PGSERR	PGSERRIE	Yes
Strobe error	STRBERR	STRBERRIE	Yes
OBK write error	OBKWERR	OBKWERRIE	Yes
OBK general error	OBKERR	OBKERRIE	Yes
Inconsistency error	INCERR	INCERRIE	Yes
Illegal access error	ILAFM	TZEN	No
Option byte error	OPTCHANGEERR	-	Yes

Table 84. Flash interrupt request (continued)

Interrupt event Event flag	Error flag label in register	Enable control bit	Clear flag to resume operation
ECC single error correction event	ECCC	ECCCIE	No
ECC double error detection event(1)	ECCD	Disabled	No

1. NMI.

The status of the individual maskable interrupt sources described in *Table 84* (except for option byte error and ECC) can be read from the FLASH_NS/SECSR register. They can be cleared by setting to 1 the adequate bit in FLASH_NS/SECCCR register.

Note: No unlocking mechanism is required to clear an interrupt.

End of operation event

Setting the end of operation interrupt enable bit (EOPIE) in the FLASH_NS/SECCR register enables the generation of an interrupt at the end of an erase operation, a program operation, or an option byte change.

When managing the OBKey storage area, the EOP is associated also with the OBK swap operation and the alternate area erase operation. EOP bit in the FLASH_NS/SECSR register is also set when one of these events occurs.

Setting CLR_EOP bit to 1 in FLASH_NS/SECCCR register clears EOP flag.

7.11 FLASH registers

Each register has an offset address and a reset value. In case of registers representing option bytes, the reset value is determined by the OBL process. In case of success the reset value is loaded from OB, in case of failure, a highly restrictive default value is set.

7.11.1 FLASH access control register (FLASH_ACR)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

For more details, refer to *Section 7.3.4* and *Section 7.3.5*.

Address offset: 0x000

Reset value: 0x0000 0013

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	PRFTEN	Res.	Res.	WRHIGHFREQ[1:0]		LATENCY[3:0]			
							rw			rw	rw	rw	rw	rw	rw

Bits 31:9 Reserved, must be kept at reset value.

Bit 8 **PRFTEN**: Prefetch enable. When bit value is modified, user must read back ACR register to be sure PRFTEN has been taken into account.
Bits used to control the prefetch.
0: prefetch disabled.
1: prefetch enabled when latency is at least one wait state.

Bits 7:6 Reserved, must be kept at reset value.

Bits 5:4 **WRHIGHFREQ[1:0]**: Flash signal delay
These bits are used to control the delay between nonvolatile memory signals during programming operations. Application software has to program them to the correct value depending on the memory interface frequency. Please refer to *Table 45* for details.
Note: No check is performed to verify that the configuration is correct.
Two WRHIGHFREQ values can be selected for some frequencies.

Bits 3:0 **LATENCY[3:0]**: Read latency
These bits are used to control the number of wait states used during read operations on both nonvolatile memory banks. The application software has to program them to the correct value depending on the memory interface frequency and voltage conditions.
0000: no wait states used to read a word from nonvolatile memory
0001: one wait state used to read a word from nonvolatile memory
0010: two wait states used to read a word from nonvolatile memory
...
0111: seven wait states used to read a word from nonvolatile memory
1111: 15 wait states used to read a word from nonvolatile memory
Note: No check is performed by hardware to verify that the configuration is correct.

7.11.2 FLASH non-secure key register (FLASH_NSKEYR)

This register is non-secure. It can written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

FLASH_NSKEYR is a write-only register. The following values must be programmed consecutively to unlock FLASH_NSCR register and allow programming/erasing it.

- First key = 0x4567 0123
- Second key = 0xCDEF 89AB

A wrong sequence locks the FLASH_NSCR register until next system reset.

Address offset: 0x004

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NSKEY[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NSKEY[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **NSKEY[31:0]**: Non-volatile memory non-secure configuration access unlock key

7.11.3 FLASH secure key register (FLASH_SECKEYR)

This register is secure. It can be written only by secure access. A non-secure read/write access is RAZ/WI. This register can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register.

FLASH_SECKEYR is a write-only register. The following values must be programmed consecutively to unlock FLASH_SECCR register and allow programming/erasing it. A wrong sequence locks the FLASH_SECCR register until next system reset.

First key = 0x4567 0123

Second key = 0xCDEF 89AB

Address offset: 0x008

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECKEY[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECKEY[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **SECKEY[31:0]**: Non-volatile memory secure configuration access unlock key

7.11.4 FLASH option key register (FLASH_OPTKEYR)

This write-only register is non-secure. It can be written by both secure and non-secure accesses, and can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

The following values must be programmed consecutively to unlock FLASH_OPTCR register

- First key = 0x0819 2A3B
- Second key = 0x4C5D 6E7F

A wrong sequence locks FLASH_OPTCR register until next system reset.

Address offset: 0x00C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OPTKEY[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OPTKEY[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **OPTKEY[31:0]**: FLASH option bytes control access unlock key

7.11.5 FLASH non-secure OBK key register (FLASH_NSOBKKEYR)

This is a write-only register. The following values must be programmed consecutively to unlock it register (a wrong sequence locks the register until next system reset).

- First key = 0x192A 083B
- Second key = 0x6E7F 4C5D

This register is non-secure. This register can be written only by privileged access.

Address offset: 0x0010

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NSOBKKEY[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NSOBKKEY[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **NSOBKKEY[31:0]**: FLASH non-secure option bytes keys control access unlock key

7.11.6 FLASH secure OBK key register (FLASH_SECOBKKEYR)

This is a write-only register. The following values must be programmed consecutively to unlock it (a wrong sequence locks the register until the next system reset)

- First key = 0x192A 083B
- Second key = 0x6E7F 4C5D

This register is secure, it can be written only by privileged access.

Address offset: 0x0014

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECOBKKEY[31:16]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECOBKKEY[15:0]															
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:0 **SECOBKKEY[31:0]**: FLASH secure option bytes keys control access unlock key

7.11.7 FLASH operation status register (FLASH_OPSR)

This register is non-secure. This register can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x0018

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CODE_OP[2:0]			Res.	Res.	Res.	Res.	OTP_OP	SYSF_OP	BK_OP	DATA_OP	Res.	ADDR_OP[19:16]			
r	r	r					r	r	r	r		r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR_OP[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:29 **CODE_OP[2:0]**: Flash memory operation code
- 000: No flash operation on going during previous reset
- 001: Single write operation interrupted
- 010: OBK alternate sector erase
- 011: Sector erase operation interrupted
- 100: Bank erase operation interrupted
- 101: Mass erase operation interrupted
- 110: Option change operation interrupted
- 111: OBK swap sector request

Bits 28:25 Reserved, must be kept at reset value.

Bit 24 **OTP_OP**: OTP operation interrupted
Indicates that reset interrupted an ongoing operation in OTP area (or OBKeys area).

Bit 23 **SYSF_OP**: Operation in system flash memory interrupted
Indicates that reset interrupted an ongoing operation in system flash.

Bit 22 **BK_OP**: Interrupted operation bank
It indicates which bank was concerned by operation.

Bit 21 **DATA_OP**: Flash high-cycle data area operation interrupted
It indicates if flash high-cycle data area is concerned by operation.

Bit 20 Reserved, must be kept at reset value.

Bits 19:0 **ADDR_OP[19:0]**: Interrupted operation address

7.11.8 FLASH option control register (FLASH_OPTCR)

This register is non-secure. It can be read and written by both secure and non-secure access, and protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Access: No wait states when no memory operations are ongoing. The FLASH_OPTCR register is not accessible in write mode when the BSY bit is set. Any attempt to write to it while the BSY bit set causes the AHB bus to stall until the BSY bit is cleared.

Address offset: 0x01C

Reset value: 0xX000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWAP_BANK	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OPTSTRT	OPTLOCK
														rw	rs

Bit 31 **SWAP_BANK**: Bank swapping option configuration bit
SWAP_BANK controls whether Bank1 and Bank2 are swapped or not. This bit is loaded with the SWAP_BANK bit of FLASH_OPTSR_CUR register only after reset or POR.
0: Bank1 and Bank2 not swapped
1: Bank1 and Bank2 swapped

Bits 30:2 Reserved, must be kept at reset value.

Bit 1 **OPTSTRT**: Option byte start change option configuration bit
OPTSTRT triggers an option byte change operation. The user can set OPTSTRT only when the OPTLOCK bit is cleared to 0. It is set only by Software and cleared when the option byte change is completed or an error occurs (PGSERR or OPTCHANGEERR). It is reset at the same time as BSY bit.
The user application cannot modify any FLASH_XXX_PRG flash interface register until the option change operation has been completed.
Before setting this bit, the user has to write the required values in the FLASH_XXX_PRG registers. The FLASH_XXX_PRG registers are locked until the option byte change operation has been executed in nonvolatile memory.

Bit 0 **OPTLOCK**: FLASH_OPTCR lock option configuration bit

The OPTLOCK bit locks the FLASH_OPTCR register as well as all _PRG registers. The correct write sequence to FLASH_OPTKEYR register unlocks this bit. If a wrong sequence is executed, or the unlock sequence to FLASH_OPTKEYR is performed twice, this bit remains locked until next system reset.

It is possible to set OPTLOCK by programming it to 1. When set to 1, a new unlock sequence is mandatory to unlock it. When OPTLOCK changes from 0 to 1, the others bits of FLASH_OPTCR register do not change.

0: FLASH_OPTCR register unlocked

1: FLASH_OPTCR register locked.

7.11.9 FLASH non-secure status register (FLASH_NSSR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x020

Reset value: 0x0000 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OPT CHANGE ERR	OBKW ERR	OBK ERR	INC ERR	STRB ERR	PGS ERR	WRP ERR	EOP
								r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DBNE	Res.	WBNE	BSY
												r		r	r

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **OPTCHANGEERR**: Option byte change error flag

OPTCHANGEERR flag indicates that an error occurred during an option byte change operation. When OPTCHANGEERR is set to 1, the option byte change operation did not successfully complete. An interrupt is generated when this flag is raised if the OPTCHANGEERRIE bit of FLASH_NSCR register is set to 1.

Writing 1 to CLR_OPTCHANGEERR of register FLASH_NSCCR clears OPTCHANGEERR.

0: no option byte change errors occurred

1: one or more errors occurred during an option byte change operation.

Note: The OPTSTRT bit in FLASH_OPTCR cannot be set while OPTCHANGEERR is set.

Bit 22 **OBKWERR**: OBK write error flag

OBKWERR flag is raised when the address is not virgin on a write access to the OBK storage. Alternatively also when the OBK selector in the alternate sector is not virgin during a swap operation.

0: no OBK write error occurred

1: an OBK write error occurred

Bit 21 **OBKERR**: OBK general error flag

OBKERR flag is raised when the OBK-HDPL signal from the SBS does not match the HDPL value associated with the key slot during access to the key location. Alternatively also when the ALT_SECT is unexpectedly changed while the write buffer is being filled.

0: no OBK general error occurred

1: an OBK general error occurred

Bit 20 **INCERR:** inconsistency error flag

NSINCERR flag is raised when a inconsistency error occurs. An interrupt is generated if INCERRIE is set to 1. Writing 1 to CLR_INCERR bit in the FLASH_NSCCR register clears NSINCERR.

0: no inconsistency error occurs

1: a inconsistency error occurs

Bit 19 **STRBERR:** strobe error flag

STRBERR flag is raised when a strobe error occurs (when the master attempts to write several times the same byte in the write buffer). An interrupt is generated if the STRBERRIE bit is set to 1. Writing 1 to CLR_STRBERR bit in FLASH_NSCCR register clears STRBERR.

0: no strobe error occurred

1: a strobe error occurred

Bit 18 **PGSERR:** programming sequence error flag

PGSERR flag is raised when a sequence error occurs. An interrupt is generated if the PGSERRIE bit is set to 1. Writing 1 to CLR_PGSERR bit in FLASH_NSCCR register clears PGSERR.

0: no sequence error occurred

1: a sequence error occurred

Bit 17 **WRPERR**: write protection error flag

WRPERR flag is raised when a protection error occurs during a program operation. An interrupt is also generated if the WRPERRIE is set to 1. Writing 1 to CLR_WRPERR bit in FLASH_NSCCR register clears WRPERR.

0: no write protection error occurred

1: a write protection error occurred

Bit 16 **EOP**: end of operation flag

EOP flag is set when a operation (program/erase) completes. An interrupt is generated if the EOPIE is set to 1. It is not necessary to reset EOP before starting a new operation. EOP bit is cleared by writing 1 to CLR_EOP bit in FLASH_NSCCR register.

0: no operation completed

1: a operation completed

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **DBNE**: data buffer not empty flag

DBNE flag is set when the flash interface is processing 6-bits ECC data in dedicated buffer. This bit cannot be set to 0 by software. The hardware resets it once the buffer is free.

0: data buffer not used

1: data buffer used, wait

Bit 2 Reserved, must be kept at reset value.

Bit 1 **WBNE**: write buffer not empty flag

WBNE flag is set when the flash interface is waiting for new data to complete the write buffer. In this state, the write buffer is not empty. WBNE is reset by hardware each time the write buffer is complete or the write buffer is emptied following one of the events below:

– the application software forces the write operation using FW bit in FLASH_NSCR
– the memory detects an error that involves data loss

This bit cannot be reset by software writing 0 directly. To reset it, clear the write buffer by performing any of the above listed actions, or send the missing data.

0: write buffer empty or full

1: write buffer waiting data to complete

Bit 0 **BSY**: busy flag

BSY flag indicates that a flash memory is busy by an operation (write, erase, option byte change, OBK operation). It is set at the beginning of a flash memory operation and cleared when the operation finishes, or an error occurs.

0: no programming, erase or option byte change operation being executed

1: programming, erase or option byte change operation being executed

7.11.10 FLASH secure status register (FLASH_SECSR)

This register is secure. It can be read only by secure access. A non-secure read/write access is RAZ/WI. This register can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x024

Reset value: 0x0000 000X

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OBKWERR	OBKERR	INCERR	STRBERR	PGSERR	WRPERR	EOP
									r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DBNE	Res.	WBNE	BSY
												r		r	r

Bits 31:23 Reserved, must be kept at reset value.

Bit 22 **OBKWERR**: OBK write error flag

OBKWERR flag is raised when the address is not virgin on a write access to the OBK storage. Alternatively also when the OBK selector in the alternate sector is not virgin during a swap operation.

0: no OBK write error occurred

1: an OBK write error occurred

Bit 21 **OBKERR**: OBK general error flag

OBKERR flag is raised when the OBK-HDPL signal from the SBS does not match the HDPL value associated with the key slot during access to the key location. Alternatively also when the ALT_SECT is unexpectedly changed while the write buffer is being filled.

0: no OBK general error occurred

1: an OBK general error occurred

Bit 20 **INCERR:** inconsistency error flag

INCERR flag is raised when a inconsistency error occurs. An interrupt is generated if INCERRIE is set to 1. Writing 1 to CLR_INCERR bit in the FLASH_SECCCR register clears INCERR.

0: no inconsistency error occurred

1: a inconsistency error occurred

Bit 19 **STRBERR:** strobe error flag

STRBERR flag is raised when a strobe error occurs (when the master attempts to write several times the same byte in the write buffer). An interrupt is generated if the STRBERRIE bit is set to 1. Writing 1 to CLR_STRBERR bit in FLASH_SECCCR register clears STRBERR.

0: no strobe error occurred

1: a strobe error occurred

Bit 18 **PGSERR:** programming sequence error flag

PGSERR flag is raised when a sequence error occurs. An interrupt is generated if the PGSERRIE bit is set to 1. Writing 1 to CLR_PGSERR bit in FLASH_SECCCR register clears PGSERR.

0: no sequence error occurred

1: a sequence error occurred

Bit 17 **WRPERR**: write protection error flag

WRPERR flag is raised when a protection error occurs during a program operation. An interrupt is also generated if the WRPERRIE is set to 1. Writing 1 to CLR_WRPERR bit in FLASH_SECCCR register clears WRPERR.

0: no write protection error occurred

1: a write protection error occurred

Bit 16 **EOP**: end of operation flag

EOP flag is set when a operation (program/erase) completes. An interrupt is generated if the EOPIE is set to. It is not necessary to reset EOP before starting a new operation. EOP bit is cleared by writing 1 to CLR_EOP bit in FLASH_SECCCR register.

0: no operation completed

1: a operation completed

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **DBNE**: data buffer not empty flag

DBNE flag is set when the memory interface is processing 6-bits ECC data in dedicated buffer. This bit cannot be set to 0 by software. The hardware resets it once the buffer is free.

0: data buffer not used

1: data buffer used, wait

Bit 2 Reserved, must be kept at reset value.

Bit 1 **WBNE**: write buffer not empty flag

WBNE flag is set when the flash interface is waiting for new data to complete the write buffer. In this state, the write buffer is not empty. WBNE is reset by hardware each time the write buffer is complete or the write buffer is emptied following one of the events below:

- the application software forces the write operation using FW bit in FLASH_SECCR
- the flash interface detects an error that involves data loss

This bit cannot be reset by writing 0 directly by software. To reset it, clear the write buffer by performing any of the above listed actions, or send the missing data.

0: write buffer empty or full

1: write buffer waiting data to complete

Bit 0 **BSY**: busy flag

BSY flag indicates that a FLASH memory is busy (write, erase, option byte change, OBK operations). It is set at the beginning of a flash memory operation and cleared when the operation finishes or an error occurs.

0: no programming, erase or option byte change operation being executed

1: programming, erase or option byte change operation being executed

7.11.11 FLASH non-secure control register (FLASH_NSCR)

This register is non-secure. It can be read and written by both secure and non-secure accesses, and can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Access: No wait states when no memory operations are ongoing. The FLASH_NSCR register is not accessible in write mode when the BSY bit is set. Any attempt to write to it with the BSY bit set causes the AHB bus to stall until the BSY bit is cleared.

Address offset: 0x028

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BKSEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OPT CHANGE ERRIE	OBKW ERRIE	OBK ERRIE	INC ERRIE	STRB ERRIE	PGS ERRIE	WRP ERRIE	EOPIE
rw								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MER	Res.	Res.	SNB[6:0]							STRT	FW	BER	SER	PG	LOCK
rw			rw	rw	rw	rw	rw	rw	rw	rs	rw	rw	rw	rw	rs

Bit 31 **BKSEL**: Bank selector bit

Can be programmed only when LOCK is cleared to 0. The bit selects physical bank, SWAP_BANK setting is ignored.

0: Bank1 is selected for Bank erase / sector erase / interrupt enable

1: Bank2 is selected for BER / SER

Bits 30:24 Reserved, must be kept at reset value.

Bit 23 **OPTCHANGEERRIE**: Option byte change error interrupt enable bit

This bit controls if an interrupt must be generated when an error occurs during an option byte change. It can be programmed only when LOCK bit is cleared to 0.

0: no interrupt is generated when an error occurs during an option byte change

1: an interrupt is generated when and error occurs during an option byte change.

Bit 22 **OBKWERRIE**: OBK write error interrupt enable bit

OBKWERRIE enables generation of interrupt in case of OBK specific write error. This bit can be programmed only when LOCK bit is cleared to 0.

0: no interrupt is generated on OBK write error

1: an interrupt is generated on OBK write error

Bit 21 **OBKERRIE**: OBK general error interrupt enable bit

OBKERRIE enables generating an interrupt in case of OBK specific access error. This bit can be programmed only when LOCK bit is cleared to 0.

0: no interrupt is generated on OBK general access error

1: an interrupt is generated on OBK general access error

Bit 20 **INCERRIE**: inconsistency error interrupt enable bit

When INCERRIE bit is set to 1, an interrupt is generated when an inconsistency error occurs during a write operation. INCERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a inconsistency error occurs

1: interrupt generated when a inconsistency error occurs.

Bit 19 **STRBERRIE**: strobe error interrupt enable bit

When STRBERRIE bit is set to 1, an interrupt is generated when a strobe error occurs (the master programs several times the same byte in the write buffer) during a write operation. STRBERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a strobe error occurs

1: interrupt generated when strobe error occurs.

Bit 18 **PGSERRIE**: programming sequence error interrupt enable bit

When this bit is set to 1, an interrupt is generated when a sequence error occurs during a program operation. PGSERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a sequence error occurs

1: interrupt generated when sequence error occurs

Bit 17 **WRPERRIE**: write protection error interrupt enable bit

When this bit is set to 1, an interrupt is generated when a protection error occurs during a program operation. WRPERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a protection error occurs

1: interrupt generated when a protection error occurs

Bit 16 **EOPIE**: end of operation interrupt control bit

Setting EOPIE bit to 1 enables the generation of an interrupt at the end of a program or erase operation. EOPIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated at the end of operation.

1: interrupt enabled when at the end of operation

Bit 15 **MER**: mass erase request

Setting MER bit to 1 requests a mass erase operation (user flash memory only). MER can be programmed only when LOCK is cleared to 0.

If BER or SER are both set, a PGSERR is raised.

0: mass erase not requested

1: mass erase requested

Note: An error is triggered when a mass erase is required and some sectors are protected.

Bits 14:13 Reserved, must be kept at reset value.

Bits 12:6 **SNB[6:0]**: sector erase selection number

These bits are used to select the target sector for an erase operation, otherwise they are not used. They can be programmed only when LOCK is cleared to 0.

0x00: Sector 0 selected

0x01: Sector 1 selected

..

0x7F: Sector 127 selected

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bit 5 **STRT**: erase start control bit

STRT bit is used to start a sector erase or a bank erase operation. STRT can be programmed only when LOCK is cleared to 0.

STRT is reset at the end of the operation or when an error occurs. It cannot be reset by software.

Bit 4 **FW:** write forcing control bit

FW forces a write operation even if the write buffer is not full. In this case all bits not written are set to 1 by hardware. FW can be programmed only when LOCK is cleared to 0.

The memory resets FW when the corresponding operation has been acknowledged.

Note: Using a force-write operation prevents the application from updating later the missing bits with something else than 1, because it is likely that it leads to permanent ECC error.

Write forcing is effective only if the write buffer is not empty and was filled by non-secure access (in particular, FW does not start several write operations when the force-write operations are performed consecutively).

Since there is just one write buffer, FW can force a write in Bank1 or Bank2.

Bit 3 **BER**: erase request

Setting BER bit to 1 requests a bank erase operation (user flash memory only). BER can be programmed only when LOCK is cleared to 0.

If MER and SER are also set, a PGSERR is raised.

0: bank erase not requested

1: bank erase requested

Note: Write protection error is triggered when a bank erase is required and some sectors are protected.

Bit 2 **SER**: sector erase request

Setting SER bit to 1 requests a sector erase. SER can be programmed only when LOCK is cleared to 0.

If MER and SER are also set, a PGSERR is raised.

0: sector erase not requested

1: sector erase requested

Bit 1 **PG**: programming control bit

PG can be programmed only when LOCK is cleared to 0.

PG allows programming in Bank1 and Bank2.

0: programming disabled

1: programming enabled

Bit 0 **LOCK**: configuration lock bit

This bit locks the FLASH_NSCR register. The correct write sequence to FLASH_NSKEYR register unlocks this bit. If a wrong sequence is executed, or if the unlock sequence to FLASH_NSKEYR is performed twice, this bit remains locked until the next system reset.

LOCK can be set by programming it to 1. When set to 1, a new unlock sequence is mandatory to unlock it. When LOCK changes from 0 to 1, the other bits of FLASH_NSCR register do not change.

0: FLASH_NSCR register unlocked

1: FLASH_NSCR register locked

7.11.12 FLASH secure control register (FLASH_SECCR)

This register is secure, can be read and written only by secure access. A non-secure read/write access is RAZ/WI. This register can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register.

Access: No wait states when no memory operations are ongoing. The FLASH_SECCR register is not accessible in write mode when the BSY bit is set. Any attempt to write to it with the BSY bit set causes the AHB bus to stall until the BSY bit is cleared.

Address offset: 0x02C

Reset value: 0x0000 0001

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BKSEL	Res.	INV	Res.	Res.	Res.	Res.	Res.	Res.	OBKW ERRIE	OBK ERRIE	INC ERRIE	STRB ERRIE	PGS ERRIE	WRP ERRIE	EOPIE
rw		rw							rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MER	Res.	Res.	SNB[6:0]							STRT	FW	BER	SER	PG	LOCK
rw			rw	rw	rw	rw	rw	rw	rw	rs	rw	rw	rw	rw	rs

Bit 31 **BKSEL**: Bank selector bit

BKSEL can only be programmed when LOCK is cleared to 0. The bit selects physical bank, SWAP_BANK setting is ignored.

0: Bank1 is selected for Bank erase / sector erase / interrupt enable

1: Bank2 is selected for BER / SER

Bit 30 Reserved, must be kept at reset value.

Bit 29 **INV**: Flash memory security state invert.

This bit inverts the flash memory security state.

Bits 28:23 Reserved, must be kept at reset value.

Bit 22 **OBKWERRIE**: OBK write error interrupt enable bit

OBKWERRIE enables generation of interrupt in case of OBK specific write error. OBKWERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt is generated on OBK write error

1: an interrupt is generated on OBK write error

Bit 21 **OBKERRIE**: OBK general error interrupt enable bit

OBKERRIE enables generating an interrupt in case of OBK specific access error. OBKERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt is generated on OBK general access error

1: an interrupt is generated on OBK general access error

Bit 20 **INCERRIE**: inconsistency error interrupt enable bit

When INCERRIE bit is set to 1, an interrupt is generated when an inconsistency error occurs during a write operation. INCERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a inconsistency error occurs

1: interrupt generated when a inconsistency error occurs.

Bit 19 **STRBERRIE**: strobe error interrupt enable bit

When STRBERRIE bit is set to 1, an interrupt is generated when a strobe error occurs (the master programs several times the same byte in the write buffer) during a write operation. STRBERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a strobe error occurs

1: interrupt generated when strobe error occurs.

Bit 18 **PGSERRIE**: programming sequence error interrupt enable bit

When PGSERRIE is set to 1, an interrupt is generated when a sequence error occurs during a program operation. PGSERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a sequence error occurs

1: interrupt generated when sequence error occurs

Bit 17 **WRPERRIE**: write protection error interrupt enable bit

When WRPERRIE bit is set to 1, an interrupt is generated when a protection error occurs during a program operation. WRPERRIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated when a protection error occurs

1: interrupt generated when a protection error occurs

Bit 16 **EOPIE**: end of operation interrupt control bit

Setting EOPIE bit to 1 enables the generation of an interrupt at the end of a program/erase operation. EOPIE can be programmed only when LOCK is cleared to 0.

0: no interrupt generated at the end of operation.

1: interrupt enabled when at the end of operation

Bit 15 **MER**: mass erase request

Setting MER bit to 1 requests a mass erase operation (user flash memory only). MER can be programmed only when LOCK is cleared to 0.

If BER or SER are also set, a PGSERR is raised.

0: mass erase not requested

1: mass erase requested

Note: An error is triggered when a mass erase is required and some sectors are protected.

Bits 14:13 Reserved, must be kept at reset value.

Bits 12:6 **SNB[6:0]**: sector erase selection number

These bits are used to select the target sector for an erase operation, otherwise they are unused. SNB can be programmed only when LOCK is cleared to 0.

0x00: Sector 0 selected

0x01: Sector 1 selected

..

0x7F: Sector 127 selected

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bit 5 **STRT**: erase start control bit

STRT bit is used to start a sector erase or a bank erase operation. STRT can be programmed only when LOCK is cleared to 0.

STRT is reset at the end of the operation or when an error occurs. It cannot be reset by software.

Bit 4 **FW:** write forcing control bit

FW forces a write operation even if the write buffer is not full. In this case all bits not written are set to 1 by hardware. FW can be programmed only when LOCK is cleared to 0.

The memory resets FW when the corresponding operation has been acknowledged.

Note: Using a force-write operation prevents the application from updating later the missing bits with something else than 1, because it is likely that it leads to permanent ECC error.

Write forcing is effective only if the write buffer is not empty and was filled by secure access (in particular, FW does not start several write operations when the force-write operations are performed consecutively).

Since there is just one write buffer, FW can force a write in Bank1 or Bank2.

Bit 3 **BER**: erase request

Setting BER bit to 1 requests a bank erase operation (user flash memory only). BER can be programmed only when LOCK is cleared to 0.

If MER and SER are also set, a PGSERR is raised.

0: bank erase not requested

1: bank erase requested

Note: A write protection error is triggered when a bank erase is required and some sectors are protected.

Bit 2 **SER**: sector erase request

Setting SER bit to 1 requests a sector erase. SER can be programmed only when LOCK is cleared to 0.

If BER and MER are also set, a PGSERR is raised.

0: sector erase not requested

1: sector erase requested

Bit 1 **PG**: programming control bit

PG can be programmed only when LOCK is cleared to 0.

PG allows programming in Bank1 and Bank2.

0: programming disabled

1: programming enabled

Bit 0 **LOCK**: configuration lock bit

This bit locks the FLASH_SECCR register. The correct write sequence to FLASH_SECKEYR register unlocks this bit. If a wrong sequence is executed, or if the unlock sequence to FLASH_NSKEYR is performed twice, this bit remains locked until the next system reset. LOCK can be set by programming it to 1. When set to 1, a new unlock sequence is mandatory to unlock it. When LOCK changes from 0 to 1, the other bits of FLASH_SECCR register do not change.

0: FLASH_SECCR register unlocked

1: FLASH_SECCR register locked

7.11.13 FLASH non-secure clear control register (FLASH_NSCCR)

This register is non-secure. It can be written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x030

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLR_OPTCHANGEERR	CLR_OBKWERR	CLR_OBKERR	CLR_INCERR	CLR_STRBERR	CLR_PGSERR	CLR_WRPERR	CLR_EOP
								w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **CLR_OPTCHANGEERR:** Clear the flag corresponding flag in FLASH_NSSR by writing this bit.

Bit 22 **CLR_OBKWERR**: OBKWERR flag clear bit.
Setting this bit to 1 resets to 0 OBKWERR flag in FLASH_NSSR register.

Bit 21 **CLR_OBKERR**: OBKERR flag clear bit.
Setting this bit to 1 resets to 0 OBKERR flag in FLASH_NSSR register.

Bit 20 **CLR_INCERR**: INCERR flag clear bit
Setting this bit to 1 resets to 0 INCERR flag in FLASH_NSSR register.

Bit 19 **CLR_STRBERR**: STRBERR flag clear bit
Setting this bit to 1 resets to 0 STRBERR flag in FLASH_NSSR register.

Bit 18 **CLR_PGSERR**: PGSERR flag clear bit
Setting this bit to 1 resets to 0 PGSERR flag in FLASH_NSSR register.

Bit 17 **CLR_WRPERR**: WRPERR flag clear bit
Setting this bit to 1 resets to 0 WRPERR flag in FLASH_NSSR register.

Bit 16 **CLR_EOP**: EOP flag clear bit
Setting this bit to 1 resets to 0 EOP flag in FLASH_NSSR register.

Bits 15:0 Reserved, must be kept at reset value.

7.11.14 FLASH secure clear control register (FLASH_SECCCR)

This register is secure. It can be written only by secure access. A non-secure read/write access is RAZ/WI. This register can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x034

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLR_ OBKW ERR	CLR_ OBK ERR	CLR_ INC ERR	CLR_ STRB ERR	CLR_ PGS ERR	CLR_ WRP ERR	CLR_ EOP
									w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.

Bits 31:23 Reserved, must be kept at reset value.

Bit 22 **CLR_OBKWERR**: OBKWERR flag clear bit
Setting this bit to 1 resets to 0 OBKWERR flag in FLASH_SECSR register.

Bit 21 **CLR_OBKERR**: OBKWERR flag clear bit
Setting this bit to 1 resets to 0 OBKWERR flag in FLASH_SECSR register.

Bit 20 **CLR_INCERR**: INCERR flag clear bit
Setting this bit to 1 resets to 0 INCERR flag in FLASH_SECSR register.

Bit 19 **CLR_STRBERR**: STRBERR flag clear bit
Setting this bit to 1 resets to 0 STRBERR flag in FLASH_SECSR register.

Bit 18 **CLR_PGSERR**: PGSERR flag clear bit
Setting this bit to 1 resets to 0 PGSERR flag in FLASH_SECSR register.

Bit 17 **CLR_WRPERR**: WRPERR flag clear bit
Setting this bit to 1 resets to 0 WRPERR flag in FLASH_SECSR register.

Bit 16 **CLR_EOP**: EOP flag clear bit
Setting this bit to 1 resets to 0 EOP flag in FLASH_SECSR register.

Bits 15:0 Reserved, must be kept at reset value.

7.11.15 FLASH privilege configuration register (FLASH_PRIVCFGR)

Address offset: 0x03C

Reset value: 0x0000 0000

This register can be read by both privileged and unprivileged access. NSPRIV is a non-secure bit. SPRIV is a secure bit. It can be written only by privileged mode.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	NSPRIV	SPRIV
														rw	rw

Bits 31:2 Reserved, must be kept at reset value.

Bit 1 **NSPRIV**: privilege attribute for non secure registers
0: access to non secure registers is always granted
1: access to non secure registers is denied in case of unprivileged access.

Bit 0 **SPRIV**: privilege attribute for secure registers
0: access to secure registers is always granted
1: access to secure registers is denied in case of unprivileged access.

7.11.16 FLASH non-secure OBK configuration register (FLASH_NSOBKCFGR)

Register is only available when TZ_STATE = 0xC3. This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can only be accessed by privilege code. This register is not accessible in write mode when the BSY bit is set. Any attempt to write to it with the BSY bit set causes the AHB bus to stall until the BSY bit is cleared.

Address offset: 0x040

Reset value: 0x01FF 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWAP_OFFSET[8:0]								
							rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ALT_SECT_ERASE	ALT_SECT	SWAP_SECT_REQ	LOCK
												rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 24:16 **SWAP_OFFSET[8:0]**: Key index (offset /16 bits) pointing for next swap.
0x00: means that no OBK is copied from current to alternate OBK sector during SWAP operation.
0x01 means that only the first OBK data (128 bits) are copied from current to alternate OBK sector
0x02 means that the two first OBK data are copied
…
0x1FF: means that all OBK data (511) are copied

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **ALT_SECT_ERASE**: alternate sector erase bit
0: do not touch OBK sector
1: erase the alternate OBK sector
When ALT_SECT bit is set, use this bit to generate an erase command for the OBK alternate sector. It is set only by Software and cleared when the OBK swap operation is completed or an error occurs (PGSERR). It is reset at the same time as BUSY bit.

Bit 2 **ALT_SECT**: alternate sector bit
0: current OBK sector is mapped to OBK address range for access
1: alternate OBK sector is mapped to OBK address range for access
This bit must not change while filling the write buffer, otherwise an error (OBKERR) is generated

Bit 1 **SWAP_SECT_REQ**: OBK swap sector request bit
0: no swap requested
1: launch the sector swap
When set, all the OBKs not updated in the alternate sector are copied from current sector to alternate one.
The SWAP_OFFSET must have a minimum value to launched the swap. The minimum value is 16 for OBK-HDPL = 1, 144 for OBK-HDPL = 2, and 192 for OBK-HDPL = 3.

Bit 0 **LOCK**: OBKCFGR lock option configuration bit
This bit locks the register, the correct write sequence unlocks it. If a wrong sequence is executed, or if the unlock sequence is performed twice, this bit remains locked until the next system reset. LOCK can be set by programming it to 1. When set to 1, a new unlock sequence is mandatory to unlock it. When LOCK changes from 0 to 1, the other bits of FLASH_NSCR register do not change.
0: FLASH_NSOBKCFGR register unlocked
1: FLASH_NSOBKCFGR register locked

7.11.17 FLASH secure OBK configuration register (FLASH_SECOBKCFGR)

This register can be accessed only in secure privileged mode. The FLASH_SECOBKCFGR register is not accessible in write mode when the BSY bit is set. Any attempt to write to it with the BSY bit set causes the AHB bus to stall until the BSY bit is cleared.

Address offset: 0x044

Reset value: 0x01FF 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWAP_OFFSET[8:0]								
							rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ALT_SECT_ERASE	ALT_SECT	SWAP_SECT_REQ	LOCK
												rw	rw	rw	rw

Bits 31:25 Reserved, must be kept at reset value.

Bits 24:16 **SWAP_OFFSET[8:0]**: key index (offset /16 bits) pointing for next swap.
0x0: no OBK is copied from current to alternate OBK sector during SWAP operation.
0x1: only the first OBK data (128 bits) are copied from current to alternate OBK sector
0x2: the two first OBK data are copied
…
0x1FF: the 511 first OBK data are copied

Bits 15:4 Reserved, must be kept at reset value.

Bit 3 **ALT_SECT_ERASE**: alternate sector erase bit
0: do not touch OBK sector
1: erase the alternate OBK sector
When ALT_SECT bit is set, use this bit to generate an erase command for the OBK alternate sector. It is set only by Software and cleared when the OBK swap operation is completed or an error occurs (PGSERR). It is reset at the same time as the BUSY bit.

Bit 2 **ALT_SECT**: alternate sector bit
0: current OBK sector is mapped to OBK address range for access
1: alternate OBK sector is mapped to OBK address range for access
This bit must not change while filling the write buffer, otherwise an error is generated

Bit 1 **SWAP_SECT_REQ**: OBK swap sector request bit
0: no swap requested
1: launch the sector swap
When set, all the OBKs not updated in the alternate sector are copied from current sector to alternate one.
SWAP_OFFSET must have a minimum value to launch the swap. The minimum value is 16 for OBK-HDPL = 1, 144 for OBK-HDPL = 2, and 192 for OBK-HDPL = 3.

Bit 0 **LOCK**: OBKCFGR lock option configuration bit
This bit locks the FLASH_OBKCFGR register. The correct write sequence to FLASH_SECOBKKEYR register unlocks this bit. If a wrong sequence is executed, or if the unlock sequence to FLASH_SECOBKKEYR is performed twice, this bit remains locked until the next system reset. LOCK can be set by programming it to 1. When set to 1, a new unlock sequence is mandatory to unlock it. When LOCK changes from 0 to 1, the other bits of FLASH_NSCR register do not change.
0: FLASH_OBKCFGR register unlocked
1: FLASH_OBKCFGR register locked

7.11.18 FLASH HDP extension register (FLASH_HDPEXTR)

All bits in this register are non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

The register can be modified only in HDPL ≤ 2.

The values in this register cannot be decremented. Attempts to write a value lower than the current one are ignored.

Address offset: 0x048

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_EXT[6:0]						
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_EXT[6:0]						
									rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **HDP2_EXT[6:0]**: HDP area extension in 8 Kbytes sectors in Bank2. Extension is added after the HDP2_END sector (included).

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **HDP1_EXT[6:0]**: HDP area extension in 8 Kbytes sectors in Bank1. Extension is added after the HDP1_END sector (included).

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.19 FLASH option status register (FLASH_OPTSR_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Default value: 0x0030 5CD8 (value in case of double ECC issue during OBL).

This read-only register reflects the current values of the corresponding option bits.

Address offset: 0x050

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWAP_BANK	Res.	BOOT_UBE[7:0]								IWDG_STDBY	IWDG_STOP	Res.	Res.	IO_VDDIO2_HSLV	IO_VDD_HSLV
r		r	r	r	r	r	r	r	r	r	r			r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRODUCT_STATE[7:0]								NRST_STDBY	NRST_STOP	Res.	WWDG_SW	IWDG_SW	BORH_EN	BOR_LEV[1:0]	
r	r	r	r	r	r	r	r	r	r		r	r	r	r	r

Bit 31 **SWAP_BANK**: Bank swapping option status bit
SWAP_BANK reflects whether Bank1 and Bank2 are swapped or not.
SWAP_BANK is loaded to SWAP_BANK of FLASH_OPTCR after a reset.
0: Bank1 and Bank2 not swapped
1: Bank1 and Bank2 swapped

Bit 30 Reserved, must be kept at reset value.

Bits 29:22 **BOOT_UBE[7:0]:** Available only on cryptography enabled devices.
Unique boot entry control, selects either ST or OEM iRoT for secure boot.
0xC3: ST-iRoT (system flash) selected
0xB4: OEM-iRoT (user flash) selected. In Open PRODUCT_STATE this value selects bootloader. Defaut value.

Bit 21 **IWDG_STDBY**: IWDG Standby mode freeze option status bit
When set the independent watchdog IWDG is frozen in system Standby mode.
0: Independent watchdog frozen in Standby mode
1: Independent watchdog keep running in Standby mode.

Bit 20 **IWDG_STOP**: IWDG Stop mode freeze option status bit
When set the independent watchdog IWDG is in system Stop mode.
0: Independent watchdog frozen in system Stop mode
1: Independent watchdog keep running in system Stop mode.

Bits 19:18 Reserved, must be kept at reset value.

Bit 17 **IO_VDDIO2_HSLV**: High-speed IO at low V_{DDIO2} voltage configuration bit.
This bit can be set only with V_{DDIO2} below 2.7 V.
0: High-speed IO at low V_{DDIO2} voltage feature disabled (V_{DDIO2} can exceed 2.7 V)
1: High-speed IO at low V_{DDIO2} voltage feature enabled (V_{DDIO2} remains below 2.7 V)

Bit 16 **IO_VDD_HSLV**: High-speed IO at low V_{DD} voltage configuration bit.
This bit can be set only with V_{DD} below 2.7 V.
0: High-speed IO at low V_{DD} voltage feature disabled (V_{DD} can exceed 2.7 V)
1: High-speed IO at low V_{DD} voltage feature enabled (V_{DD} remains below 2.7 V)

Bits 15:8 **PRODUCT_STATE[7:0]**: Life state code (based on Hamming 8,4). See *Section 7.6.11*.

Bit 7 **NRST_STDBY**: Core domain Standby entry reset option status bit
0: a reset is generated when entering Standby mode on core domain
1: no reset generated when entering Standby mode on core domain.

Bit 6 **NRST_STOP**: Core domain Stop entry reset option status bit
0: a reset is generated when entering Stop mode on core domain
1: no reset generated when entering Stop mode on core domain.

Bit 5 Reserved, must be kept at reset value.

Bit 4 **WWDG_SW**: WWDG control mode option status bit
0: WWDG watchdog is controlled by hardware
1: WWDG watchdog is controlled by software

Bit 3 **IWDG_SW**: IWDG control mode option status bit
0: IWDG watchdog is controlled by hardware
1: IWDG watchdog is controlled by software

Bit 2 **BORH_EN**: Brownout high enable
0: disable
1: enable

Bits 1:0 **BOR_LEV[1:0]**: Brownout level option status bit
These bits reflects the power level that generates a system reset.
00 or 11: BOR Level 1, the threshold level is low (~ 2.1 V)
01: BOR Level 2, the threshold level is medium (~ 2.4 V)
10: BOR Level 3, the threshold level is high (~ 2.7 V)

7.11.20 FLASH option status register (FLASH_OPTSR_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits. Values after reset reflects the current values of the corresponding option bits.

Address offset: 0x054

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWAP_BANK	Res.	BOOT_UBE[7:0]								IWDG_STDBY	IWDG_STOP	Res.	Res.	IO_VDDIO2_HSLV	IO_VDD_HSLV
rw		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw			rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRODUCT_STATE[7:0]								NRST_STDBY	NRST_STOP	Res.	WWDG_SW	IWDG_SW	BORH_EN	BOR_LEV[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

Bit 31 **SWAP_BANK**: Bank swapping option configuration bit
SWAP_BANK option bit is used to configure whether the Bank1 and Bank2 are swapped or not. This bit is loaded with the SWAP_BANK bit of FLASH_OPTSR_CUR register after a reset.
0: Bank1 and Bank2 not swapped
1: Bank1 and Bank2 swapped

Bit 30 Reserved, must be kept at reset value.

Bits 29:22 **BOOT_UBE[7:0]:** Available only on cryptography enabled devices.
Unique boot entry control, selects either ST or OEM iRoT for secure boot.
0xC3: ST-iRoT (system flash) selected
0xB4: OEM-iRoT (user flash) selected.
In Open PRODUCT_STATE this value selects bootloader. Defaut value.

Bit 21 **IWDG_STDBY**: IWDG Standby mode freeze option status bit
When set the independent watchdog IWDG is frozen in system Standby mode.
0: Independent watchdog frozen in Standby mode
1: Independent watchdog keep running in Standby mode.

Bit 20 **IWDG_STOP**: IWDG Stop mode freeze option status bit
When set the independent watchdog IWDG is in system Stop mode.
0: Independent watchdog frozen in system Stop mode
1: Independent watchdog keep running in system Stop mode.

Bits 19:18 Reserved, must be kept at reset value.

Bit 17 **IO_VDDIO2_HSLV**: High-speed IO at low V_{DDIO2} voltage configuration bit.
This bit can be set only with V_{DDIO2} below 2.7 V.
0: High-speed IO at low V_{DDIO2} voltage feature disabled (V_{DDIO2} can exceed 2.7 V)
1: High-speed IO at low V_{DDIO2} voltage feature enabled (V_{DDIO2} remains below 2.7 V)

Bit 16 **IO_VDD_HSLV**: High-speed IO at low VDD voltage configuration bit.
This bit can be set only with V_{DD} below 2.7 V.
0: High-speed IO at low V_{DD} voltage feature disabled (V_{DD} can exceed 2.7 V)
1: High-speed IO at low V_{DD} voltage feature enabled (V_{DD} remains below 2.7 V)

Bits 15:8 **PRODUCT_STATE[7:0]**: Life state code (based on Hamming 8,4). See *Section 7.6.11*.

Bit 7 **NRST_STDBY**: Core domain Standby entry reset option configuration bit
0: a reset is generated when entering Standby mode on core domain
1: no reset generated when entering Standby mode on core domain.

Bit 6 **NRST_STOP**: Core domain Stop entry reset option configuration bit
0: a reset is generated when entering Stop mode on core domain
1: no reset generated when entering Stop mode on core domain.

Bit 5 Reserved, must be kept at reset value.

Bit 4 **WWDG_SW**: WWDG control mode option configuration bit
0: WWDG watchdog is controlled by hardware
1: WWDG watchdog is controlled by software

Bit 3 **IWDG_SW**: IWDG control mode option configuration bit
0: IWDG watchdog is controlled by hardware
1: IWDG watchdog is controlled by software

Bit 2 **BORH_EN**: Brownout high enable configuration bit
0: disabled
1: enabled

Bits 1:0 **BOR_LEV[1:0]**: Brownout level option configuration bit
These bits reflects the power level that generates a system reset.
00 or 11: BOR Level 1, the threshold level is low (~ 2.1 V)
01: BOR Level 2, the threshold level is medium (~ 2.4 V)
10: BOR Level 3, the threshold level is high (~ 2.7 V)

7.11.21 FLASH non-secure EPOCH register (FLASH_NSEPOCHR_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This read-only register reflects the current values of the corresponding option bits.

Address offset: 0x060

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	NS_EPOCH[23:16]							
								r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NS_EPOCH[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:0 **NS_EPOCH[23:0]:** Non-volatile non-secure EPOCH counter

7.11.22 FLASH secure EPOCH register (FLASH_SECEPOCHR_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This read-only register reflects the current values of the corresponding option bits.

Address offset: 0x068

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SEC_EPOCH[23:16]							
								r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SEC_EPOCH[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:24 Reserved, must be kept at reset value.

Bits 23:0 **SEC_EPOCH[23:0]:** Non-volatile secure EPOCH counter

7.11.23 FLASH option status register 2 (FLASH_OPTSR2_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Default value: 0xB400 0170

This read-only register reflects the current values of the corresponding option bits.

Address offset: 0x070

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TZEN[7:0]								Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
r	r	r	r	r	r	r	r								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	USBPD_DIS	Res.	SRAM2_ECC	SRAM3_ECC	BKPRAM_ECC	SRAM2_RST	SRAM13_RST	Res.	Res.
							r		r	r	r	r	r		

Bits 31:24 **TZEN[7:0]**: TrustZone enable configuration bits
This bit enables the device is in TrustZone mode during an option byte change.
0xC3: TrustZone disabled
0xB4: TrustZone enabled.

Bits 23:15 Reserved, must be kept at reset value.

Bits 14:9 Reserved, must be kept at reset value.

Bit 8 **USBPD_DIS**: USB power delivery configuration option bit
0: Enabled
1: Disabled

Bit 7 Reserved, must be kept at reset value.

Bit 6 **SRAM2_ECC**: SRAM2 ECC detection and correction disable
0: SRAM2 ECC check enabled
1: SRAM2 ECC check disabled

Bit 5 **SRAM3_ECC**: SRAM3 ECC detection and correction disable
0: SRAM3 ECC check enabled
1: SRAM3 ECC check disabled

Bit 4 **BKPRAM_ECC**: Backup RAM ECC detection and correction disable
0: BKPRAM ECC check enabled
1: BKPRAM ECC check disabled

Bit 3 **SRAM2_RST**: SRAM2 erase when system reset
0: SRAM2 erased when a system reset occurs
1: SRAM2 not erased when a system reset occurs.

Bit 2 **SRAM13_RST**: SRAM1 and SRAM3 erase upon system reset
0: SRAM1 and SRAM3 erased when a system reset occurs
1: SRAM1 and SRAM3 not erased when a system reset occurs

Bits 1:0 Reserved, must be kept at reset value.

7.11.24 FLASH option status register 2 (FLASH_OPTSR2_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits. Values after reset reflects the current values of the corresponding option bits.

Address offset: 0x074

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TZEN[7:0]								Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
rw	rw	rw	rw	rw	rw	rw	rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	USBPD_DIS	Res.	SRAM2_ECC	SRAM3_ECC	BKPRAM_ECC	SRAM2_RST	SRAM1_3_RST	Res.	Res.
							rw		rw	rw	rw	rw	rw		

Bits 31:24 **TZEN[7:0]**: TrustZone enable configuration bits
This bit enables the device is in TrustZone mode during an option byte change.
0xC3: TrustZone disabled
0xB4: TrustZone enabled

Bits 23:15 Reserved, must be kept at reset value.

Bits 14:9 Reserved, must be kept at reset value.

Bit 8 **USBPD_DIS**: USB power delivery configuration option bit
0: Enabled
1: Disabled

Bit 7 Reserved, must be kept at reset value.

Bit 6 **SRAM2_ECC**: SRAM2 ECC detection and correction disable
0: SRAM2 ECC check enabled
1: SRAM2 ECC check disabled

Bit 5 **SRAM3_ECC**: SRAM3 ECC detection and correction disable
0: SRAM3 ECC check enabled
1: SRAM3 ECC check disabled

Bit 4 **BKPRAM_ECC**: Backup RAM ECC detection and correction disable
0: BKPRAM ECC check enabled
1: BKPRAM ECC check disabled

Bit 3 **SRAM2_RST**: SRAM2 erase when system reset
0: SRAM2 erased when a system reset occurs
1: SRAM2 not erased when a system reset occurs.
Note: SRAM erase is triggered by option byte change operation, when enabling this feature.

Bit 2 **SRAM1_3_RST**: SRAM1 and SRAM3 erase upon system reset
0: SRAM1 and SRAM3 erased when a system reset occurs
1: SRAM1 and SRAM3 not erased when a system reset occurs
Note: SRAM erase is triggered by option byte change operation, when enabling this feature.

Bits 1:0 Reserved, must be kept at reset value.

7.11.25 FLASH non-secure boot register (FLASH_NSBOOTR_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register reflects the current values of the corresponding option bits.

Address offset: 0x080

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NSBOOTADD[23:8]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NSBOOTADD[7:0]								NSBOOT_LOCK[7:0]							
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:8 **NSBOOTADD[23:0]**: Non secure unique boot entry address
These bits reflect the Non secure UBE address

Bits 7:0 **NSBOOT_LOCK[7:0]:** Field locking the values of SWAP_BANK, and NSBOOTADD settings.
0xC3: SWAP_BANK and NSBOOTADD can still be modified following their individual rules.
0xB4: NSBOOTADD is frozen. SWAP_ BANK can be modified only with TZEN set to 0xB4 (enabled).

7.11.26 FLASH non-secure boot register (FLASH_NSBOOTR_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses, and protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits.

Address offset: 0x084

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
NSBOOTADD[23:8]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NSBOOTADD[7:0]								NSBOOT_LOCK[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 **NSBOOTADD[23:0]**: Non secure unique boot entry address
These bits allow configuring the Non secure BOOT address

Bits 7:0 **NSBOOT_LOCK[7:0]:** Field locking the values of SWAP_ BANK, and NSBOOTADD settings.
0xC3: SWAP_ BANK and NSBOOTADD can still be modified following their individual rules.
0xB4: NSBOOTADD is frozen. SWAP_ BANK can be modified only with TZEN set to 0xB4 (enabled).

7.11.27 FLASH secure boot register (FLASH_SECBOOTR_CUR)

This register is secure. It can be read only by secure access. A non-secure read/write access is RAZ/WI. This register can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register.

This register reflects the current values of the corresponding option bits.

Address offset: 0x088

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECBOOTADD[23:8]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECBOOTADD[7:0]								SECBOOT_LOCK[7:0]							
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:8 **SECBOOTADD[23:0]**: Unique boot entry secure address
These bits reflect the Secure UBE address

Bits 7:0 **SECBOOT_LOCK[7:0]:** Field locking the values of UBE, SWAP_BANK, and SECBOOTADD settings.
0xC3: BOOT_UBE, SWAP_BANK and SECBOOTADD can still be modified following their individual rules.
0xB4: BOOT_UBE and SECBOOTADD are frozen. SWAP_ BANK can be modified only with TZEN set to 0xC3 (disabled).

7.11.28 FLASH secure boot register (FLASH_BOOTR_PRG)

This register is secure, can be read and written only by secure access. A non-secure read/write access is RAZ/WI. This register can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits.

Address offset: 0x08C

Reset value: 0xXXXX XXXX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECBOOTADD[23:8]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECBOOTADD[7:0]								SECBOOT_LOCK[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 **SECBOOTADD[23:0]**: Secure unique boot entry address.
These bits allow configuring the secure UBE address.

Bits 7:0 **SECBOOT_LOCK[7:0]:** Field locking the values of UBE, SWAP_ BANK, and SECBOOTADD setting.
0xC3: BOOT_UBE, SWAP_ BANK and SECBOOTADD can still be modified following their individual rules.
0xB4: BOOT_UBE and SECBOOTADD are frozen. SWAP_BANK can be modified only with TZEN set to 0xC3 (disabled).

7.11.29 FLASH non-secure OTP block lock (FLASH_OTPBLR_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register reflects the current values of the corresponding option bits.

Address offset: 0x090

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCKBL[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCKBL[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **LOCKBL[31:0]**: OTP block lock

Block *n* corresponds to OTP 16-bit word 32 x *n* to 32 x *n* + 31.
LOCKBL[n] = 1 indicates that all OTP 16-bit words in OTP Block *n* are locked and attempt to program them results in WRPERR.
LOCKBL[n] = 0 indicates that all OTP 16-bit words in OTP Block *n* are not locked.
When one block is locked, it is not possible to remove the write protection.
Also if not locked, it is not possible to erase OTP words.

7.11.30 FLASH non-secure OTP block lock (FLASH_OTPBLR_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits.

Address offset: 0x094

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOCKBL[31:16]															
rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCKBL[15:0]															
rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs	rs

Bits 31:0 **LOCKBL[31:0]**: OTP block lock

Block *n* corresponds to OTP 16-bit word 32 x *n* to 32 x *n* + 31.
LOCKBL[n] = 1 indicates that all OTP 16-bit words in OTP Block *n* are locked and attempt to program them results in WRPERR.
LOCKBL[n] = 0 indicates that all OTP 16-bit words in OTP Block *n* are not locked.
When one block is locked, it is not possible to remove the write protection.
LOCKBL bits can be set if the corresponding bit in FLASH_OTPBLR_CUR is cleared.

7.11.31 FLASH secure block based register for Bank1 (FLASH_SECBB1Rx)

These registers (only one for STM32H523/33xx devices, as x = 1) are secure. They can be read and written only by secure access. A non-secure read/write access is RAZ/WI.

Address offset: 0x0A0 + 0x004 * (x-1), (x = 1 to 4)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECBB1[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECBB1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **SECBB1[31:0]**: Secure/non-secure 8 Kbytes flash Bank1 sector attributes (y = 0 to 31)

0: sector (32 * (x-1) + y) in Bank1 is non secure.
1: sector (32 * (x-1) + y) in Bank1 is secure.

7.11.32 FLASH privilege block based register for Bank1 (FLASH_PRIVBB1Rx)

These registers (only one for STM32H523/33xx devices, as x = 1) are privileged. They can be read and written only by a privileged access. This register can be protected against non-secure write access.

Address offset: 0x0C0 + 0x004 * (x-1), (x = 1 to 4)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRIVBB1[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRIVBB1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **PRIVBB1[31:0]**: Privileged/unprivileged 8-Kbyte flash Bank1 sector attribute (y = 0 to 31)

0: sectors (32 * (x-1) + y) in Bank1 is unprivileged.
1: sector (32 * (x-1) + y) in Bank1 is privileged.

7.11.33 FLASH security watermark for Bank1 (FLASH_SECWM1R_CUR)

This read-only register is non-secure, it can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register. It reflects the current values of the corresponding option bits.

Address offset: 0x0E0

Reset value: 0x00XX 00XX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_END[6:0]						
									r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_STRT[6:0]						
									r	r	r	r	r	r	r

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **SECWM1_END[6:0]**: Bank1 security WM area 1 end sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **SECWM1_STRT[6:0]**: Bank1 security WM area 1 start sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.34 FLASH security watermark for Bank1 (FLASH_SECWM1R_PRG)

This register is secure, can be read and written only by secure access. A non-secure read/write access is RAZ/WI. This register can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register. This register is used to program values in corresponding option bits.

Address offset: 0x0E4

Reset value: 0x00XX 00XX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_END[6:0]						
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_STRT[6:0]						
									rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **SECWM1_END[6:0]**: Bank1 security WM area 1 end sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **SECWM1_STRT[6:0]**: Bank1 security WM area 1 start sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.35 FLASH write sector group protection for Bank1 (FLASH_WRP1R_CUR)

This register is non-secure. It can be read by both secure and non-secure accesses, and can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This read-only register reflects the current values of the corresponding option bits.

Address offset: 0x0E8

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WRPSG1[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WRPSG1[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **WRPSG1[31:0]**: Bank1 sector group protection option status byte

Each FLASH_WRP1R_CUR bit reflects the write protection status of the corresponding group of four consecutive sectors in Bank1 (0: the group is write-protected; 1: the group is not write-protected)

Bit 0: Group embedding sectors 0 to 3

Bit 1: Group embedding sectors 4 to 7

Bit N: Group embedding sectors 4 x N to 4 x N + 3

Bit 31: Group embedding sectors 124 to 127

Note: In STM32H523/33xx devices only bits [7:0] are used. Consider bits [31:8] as reserved, and keep them at reset value.

7.11.36 FLASH write sector group protection for Bank1 (FLASH_WRP1R_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits.

Address offset: 0x0EC

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WRPSG1[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WRPSG1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **WRPSG1[31:0]**: Bank1 sector group protection option status byte

Setting WRPSG1 bits to 0 write protects the corresponding group of four consecutive sectors in Bank1 (0: the group is write-protected; 1: the group is not write-protected)

Bit 0: Group embedding sectors 0 to 3

Bit 1: Group embedding sectors 4 to 7

Bit N: Group embedding sectors 4 x N to 4 x N + 3

Bit 31: Group embedding sectors 124 to 127

Note: In STM32H523/33xx devices only bits [7:0] are used. Consider bits [31:8] as reserved, and keep them at reset value.

7.11.37 FLASH data sector configuration Bank1 (FLASH_EDATA1R_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. The register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This read-only register reflects the current values of the corresponding options bits.

Address offset: 0x0F0

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EDATA1_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA1_STRT[2:0]		
r													r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bit 15 **EDATA1_EN:** Bank1 flash high-cycle data enable

0: No flash high-cycle data area

1: Flash high-cycle data is used

Bits 14:3 Reserved, must be kept at reset value.

Bits 2:0 **EDATA1_STRT[2:0]:**

EDATA1_STRT contains the start sectors of the flash high-cycle data area in Bank1 There is no hardware effect to those bits. They shall be managed by ST tools in Flasher.

000: The last sector of Bank1 is reserved for flash high-cycle data

001: The two last sectors of Bank1 are reserved for flash high-cycle data

010: The three last sectors of Bank1 are reserved for flash high-cycle data

...

111: The eight last sectors of Bank1 are reserved for flash high-cycle data

7.11.38 FLASH data sector configuration Bank1 (FLASH_EDATA1R_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding options bits.

Address offset: 0xF4

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EDATA1_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA1_STRT[2:0]		
rw													rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bit 15 **EDATA1_EN:** Bank1 flash high-cycle data enable
0: No flash high-cycle data area
1: Flash high-cycle data is used

Bits 14:3 Reserved, must be kept at reset value.

Bits 2:0 **EDATA1_STRT[2:0]:**
EDATA1_STRT contains the start sectors of the flash high-cycle data area in Bank1 There is no hardware effect to those bits. They must be managed by ST tools in Flasher.
000: The last sector of Bank1 is reserved for flash high-cycle data
001: The two last sectors of Bank1 are reserved for flash high-cycle data
010: The three last sectors of Bank1 are reserved for flash high-cycle data
...
111: The eight last sectors of Bank1 are reserved for flash high-cycle data

7.11.39 FLASH HDP Bank1 configuration (FLASH_HDP1R_CUR)

This register is non-secure. It can be read by both secure and non-secure accesses, and protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register. This register is initially loaded with the values of the corresponding option bits.

Address offset: 0x0F8

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_END[6:0]						
									r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_STRT[6:0]						
									r	r	r	r	r	r	r

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **HDP1_END[6:0]:** HDPL barrier end set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **HDP1_STRT[6:0]:** HDPL barrier start set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.40 FLASH HDP Bank1 configuration (FLASH_HDP1R_PRG)

This register is non-secure. It can be read and written by both secure and non-secure access, and can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

The register is accessible only in HDPL1. In HDPL2 or HDPL3 it is WI, RAZ.

This register is used to program values in corresponding option bits.

Address offset: 0x0FC

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_END[6:0]						
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_STRT[6:0]						
									rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **HDP1_END[6:0]:** HDPL barrier end set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **HDP1_STRT[6:0]:** HDPL barrier start set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.41 FLASH ECC correction register (FLASH_ECCCORR)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x100

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	ECCC	Res.	Res.	Res.	Res.	ECCCIE	OTP_ECC	SYSF_ECC	BK_ECC	EDATA_ECC	OBK_ECC	Res.	Res.	Res.	Res.
	rw					rw	r	r	r	r	r				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR_ECC[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bit 31 Reserved, must be kept at reset value.

Bit 30 **ECCC:** ECC correction set by hardware when single ECC error has been detected and corrected.

Cleared by writing 1.

Bits 29:26 Reserved, must be kept at reset value.

Bit 25 **ECCCIE**: ECC single correction error interrupt enable bit

When ECCCIE bit is set to 1, an interrupt is generated when an ECC single correction error occurs during a read operation.

0: no interrupt generated when an ECC single correction error occurs

1: non-secure interrupt generated when an ECC single correction error occurs

Bit 24 **OTP_ECC**: OTP ECC error bit

This bit is set to 1 when one single ECC correction occurred during the last successful read operation from the read-only/ OTP area. The address of the ECC error is available in ADDR_ECC bitfield.

Bit 23 **SYSF_ECC**: ECC fail for corrected ECC error in system flash memory

It indicates if system flash memory is concerned by ECC error.

Bit 22 **BK_ECC**: ECC fail bank for corrected ECC error

It indicates which bank is concerned by ECC error

Bit 21 **EDATA_ECC**: ECC fail for corrected ECC error in flash high-cycle data area

It indicates if flash high-cycle data area is concerned by ECC error.

Bit 20 **OBK_ECC**: Single ECC error corrected in flash OB Keys storage area. It indicates the OBK storage concerned by ECC error.

Bits 19:16 Reserved, must be kept at reset value.

Bits 15:0 **ADDR_ECC[15:0]**: ECC error address

When an ECC error occurs (for single correction) during a read operation, ADDR_ECC contains the address that generated the error. It is reset when the flag error is reset.

The flash interface programs the address in this register only when no ECC error flags are set. This means that only the first address that generated an ECC error is saved.

The address in ADDR_ECC is relative to the flash memory area where the error occurred (user flash memory, system flash memory, data area, read-only/OTP area).

7.11.42 FLASH ECC detection register (FLASH_ECCDETR)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x104

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ECCD	Res.	Res.	Res.	Res.	Res.	Res.	OTP_ECC	SYSF_ECC	BK_ECC	EDATA_ECC	OBK_ECC	Res.	Res.	Res.	Res.
rw							r	r	r	r	r				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR_ECC[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bit 31 **ECCD:** ECC detection
Set by hardware when two ECC error has been detected.
When this bit is set, a NMI is generated.
Cleared by writing 1. Needs to be cleared in order to detect subsequent double ECC errors.

Bits 30:25 Reserved, must be kept at reset value.

Bit 24 **OTP_ECC**: OTP ECC error bit
This bit is set to 1 when double ECC detection occurred during the last read operation from the read-only/ OTP area. The address of the ECC error is available in ADDR_ECC bitfield.

Bit 23 **SYSF_ECC**: ECC fail for double ECC error in system flash memory
It indicates if system flash memory is concerned by ECC error.

Bit 22 **BK_ECC**: ECC fail bank for double ECC error
It indicates which bank is concerned by ECC error

Bit 21 **EDATA_ECC**: ECC fail double ECC error in flash high-cycle data area
It indicates if flash high-cycle data area is concerned by ECC error.

Bit 20 **OBK_ECC**: ECC fail double ECC error in flash OB Keys storage area. It indicates the OBK storage concerned by ECC error.

Bits 19:16 Reserved, must be kept at reset value.

Bits 15:0 **ADDR_ECC[15:0]**: ECC error address
When an ECC error occurs (double detection) during a read operation, the ADDR_ECC contains the address that generated the error. It is reset when the flag error is reset.
The flash interface programs the address in this register only when no ECC error flags are set. This means that only the first address that generated an double ECC error is saved.
The address in ADDR_ECC is relative to the flash memory area where the error occurred (user flash memory, system flash memory, data area, read-only/OTP area).

7.11.43 FLASH ECC data (FLASH_ECCDR)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x108

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA_ECC[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **DATA_ECC[15:0]**: ECC error data

When an double detection ECC error occurs on special areas with 6-bit ECC on 16-bit data (data, read-only/OTP), the failing data is read to this register.

By checking if it is possible to determine whether the failure was on a real data, or due to access to uninitialized memory.

7.11.44 FLASH secure block-based register for Bank2 (FLASH_SECBB2Rx)

These registers (only one for STM32H523/33xx devices, as x = 1) are secure, they can be read and written only by secure access. A non-secure read/write access is RAZ/WI. The registers can be protected against unprivileged access when SPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x1A0 + 0x004 * (x-1), (x = 1 to 4)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SECBB2[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SECBB2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **SECBB2[31:0]**: Secure/non-secure flash Bank2 sector attribute (y = 0 to 31)

0: sectors (32 * (x-1) + y) in Bank2 is non secure

1: sector (32 * (x-1) + y) in Bank2 is secure

7.11.45 FLASH privilege block-based register for Bank2 (FLASH_PRIVBB2Rx)

These registers (only one for STM32H523/33xx devices, as x = 1) are privileged, they can be read and written only by a privileged access, and can be protected against non-secure write access by watermark.

Address offset: 0x1C0 + 0x004 * (x-1), (x = 1 to 4)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PRIVBB2[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRIVBB2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **PRIVBB2[31:0]:** Privileged / non-privileged 8-Kbyte flash Bank2 sector attribute (y = 0 to 31)
0: sectors (32 * (x-1) + y) in Bank2 is unprivileged
1: sector (32 * (x-1) + y) in Bank2 is privileged

7.11.46 FLASH security watermark for Bank2 (FLASH_SECWM2R_CUR)

This register is non-secure, it can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This read-only register reflects the current values of the corresponding option bits.

Address offset: 0x1E0

Reset value: 0x00XX 00XX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_END[6:0]						
									r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_STRT[6:0]						
									r	r	r	r	r	r	r

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **SECWM2_END[6:0]**: Bank2 security WM end sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **SECWM2_STRT[6:0]**: Bank2 security WM area start sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.47 FLASH security watermark for Bank2 (FLASH_SECWM2R_PRG)

This register is secure, can be read and written only by secure access. A non-secure read/write access is RAZ/WI. The register can be protected against unprivileged accesses when SPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits.

Address offset: 0x1E4

Reset value: 0x00XX 00XX

Bits 0 to 31 are loaded with values from the flash memory at OBL.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_END[6:0]						
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_STRT[6:0]						
									rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **SECWM2_END[6:0]**: Bank2 security WM area end sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **SECWM2_STRT[6:0]**: Bank2 security WM area start sector

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.48 FLASH write sector group protection for Bank2 (FLASH_WRP2R_CUR)

This register is non-secure, can be read by both secure and non-secure access, and protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This read-only register reflects the current values of the corresponding option bits.

Address offset: 0x1E8

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WRPSG2[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WRPSG2[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **WRPSG2[31:0]**: Bank2 sector group protection option status byte

Each bit reflects the write protection status of the corresponding group of four consecutive sectors in Bank2 (0: group is write-protected; 1: group is not write-protected)

Bit 0: Group embedding sectors 0 to 3

Bit 1: Group embedding sectors 4 to 7

Bit N: Group embedding sectors 4 x N to 4 x N + 3

Bit 31: Group embedding sectors 124 to 127

Note: In STM32H523/33xx devices only bits [7:0] are used. Consider bits [31:8] as reserved, and keep them at reset value.

7.11.49 FLASH write sector group protection for Bank2 (FLASH_WRP2R_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This register is used to program values in corresponding option bits.

Address offset: 0x1EC

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WRPSG2[31:16]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WRPSG2[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **WRPSG2[31:0]**: Bank2 sector group protection option status byte
Setting WRPSGn2 bits to 0 write protects the corresponding group of four consecutive sectors in Bank2 (0: group is write-protected; 1: group is not write-protected)
Bit 0: Group embedding sectors 0 to 3
Bit 1: Group embedding sectors 4 to 7
Bit N: Group embedding sectors 4 x N to 4 x N + 3
Bit 31: Group embedding sectors 124 to 127

Note: In STM32H523/33xx devices only bits [7:0] are used. Consider bits [31:8] as reserved, and keep them at reset value.

7.11.50 FLASH data sectors configuration Bank2 (FLASH_EDATA2R_CUR)

This register is non-secure, can be read by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

This read-only register reflects the current values of the corresponding options bits.

Address offset: 0x1F0

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EDATA2_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA2_STRT[2:0]		
r													r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bit 15 **EDATA2_EN:** Bank2 flash high-cycle data enable
0: No flash high-cycle data area
1: Flash high-cycle data is used

Bits 14:3 Reserved, must be kept at reset value.

Bits 2:0 **EDATA2_STRT[2:0]:**
EDATA2_STRT contains the start sectors of the flash high-cycle data area in Bank2 There is no hardware effect to those bits. They shall be managed by ST tools in Flasher.
000: The last sector of Bank2 is reserved for flash high-cycle data
001: The two last sectors of Bank2 are reserved for flash high-cycle data
010: The three last sectors of Bank2 are reserved for flash high-cycle data
...
111: The eight last sectors of Bank2 are reserved for flash high-cycle data

7.11.51 FLASH data sector configuration Bank2 (FLASH_EDATA2R_PRG)

This register is non-secure. It can be read and written by both secure and non-secure accesses, and protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register. It is used to program values in corresponding options bits.

Address offset: 0x1F4

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EDATA2_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA2_STRT[2:0]		
rw													rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bit 15 **EDATA2_EN:** Bank2 flash high-cycle data enable
0: No flash high-cycle data area
1: Flash high-cycle data is used

Bits 14:3 Reserved, must be kept at reset value.

Bits 2:0 **EDATA2_STRT[2:0]:**
EDATA2_STRT contains the start sectors of the flash high-cycle data area in Bank2 There is no hardware effect to those bits. They shall be managed by ST tools in Flasher.
000: The last sector of Bank2 is reserved for flash high-cycle data.
001: The two last sectors of Bank2 are reserved for flash high-cycle data.
010: The three last sectors of Bank2 are reserved for flash high-cycle data.
...
111: The eight last sectors of Bank2 are reserved for flash high-cycle data.

7.11.52 FLASH HDP Bank2 configuration (FLASH_HDP2R_CUR)

This register is initially loaded with the values of the corresponding option bits. This register is non-secure. It can be read by both secure and non-secure accesses, and can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Address offset: 0x1F8

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_END[6:0]						
									r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_STRT[6:0]						
									r	r	r	r	r	r	r

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **HDP2_END[6:0]:** HDPL barrier end set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **HDP2_STRT[6:0]:** HDPL barrier start set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.11.53 FLASH HDP Bank2 configuration (FLASH_HDP2R_PRG)

This register is non-secure. It can be read and written only by both secure and non-secure accesses. This register can be protected against unprivileged access when NSPRIV = 1 in the FLASH_PRIVCFGR register.

Register is accessible only in HDPL1. In HDPL2 or HDPL3 it is WI, RAZ.

This register is used to program values in corresponding option bits.

Address offset: 0x1FC

Reset value: 0xXXXX XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_END[6:0]						
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_STRT[6:0]						
									rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:16 **HDP2_END[6:0]:** HDPL barrier end set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

Bits 15:7 Reserved, must be kept at reset value.

Bits 6:0 **HDP2_STRT[6:0]:** HDPL barrier start set in number of 8-Kbyte sectors

Note: For STM32H523/33xx devices the maximum value is 0x1F, consider bits [6:5] as reserved, and keep them at reset value.

7.12 FLASH register map and reset values

Table 85. Register map and reset value table

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x000	FLASH_ACR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PRFTEN	Res.	Res.	WRHIGHFREQ [1:0]		LATENCY [3:0]			
	Reset value																								0			0	1	0	0	1	1
0x004	FLASH_NSKEYR	NSKEY[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0x008	FLASH_ SECKEYR	SECKEY[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x00C	FLASH_OPTKEYR	OPTKEY[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x010	FLASH_ NSOBKKEYR	NSOBKKEY[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x014	FLASH_ SECOBKKEYR	SECOBKKEY[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x018	FLASH_OPSR	CODE_OP [2:0]			Res.	Res.	Res.	Res.	OTP_OP	SYSF_OP	BK_OP	DATA_OP	Res.	ADDR_OP[19:0]																			
	Reset value	X	X	X					X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x01C	FLASH_OPTCR	SWAP_BANK	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OPTSTRT	OPTLOCK
	Reset value	0																														0	1
0x020	FLASH_NSSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OPTCHANGERR	OBKWERR	OBKERR	INCERR	STRBERR	PGSERR	WRPERR	EOP	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SDBNE	Res.	WBNE	BSY
	Reset value									0	0	0	0	0	0	0	0													0		0	0
0x024	FLASH_SECSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OBKWERR	OBKERR	INCERR	STRBERR	PGSERR	WRPERR	EOP	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DBNE	Res.	WBNE	BSY
	Reset value										0	0	0	0	0	0	0													X		X	X
0x028	FLASH_NSCR	BKSEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OPTCHANGEERRIE	OBKWIE	OBKIE	INCERRIE	STRBERRIE	PGSERRIE	WRPERRIE	EOPIE	MER	Res.	Res.	SNB[6:0]							STRT	FW	BER	SER	PG	LOCK
	Reset value	0									0	0	0	0	0	0	0	0			0	0	0	0	0	0	0	0	0	0	0	0	1

Table 85. Register map and reset value table (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x02C	FLASH_SECCR	BKSEL	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	OBKWIE	OBKERRIE	INCERRIE	STRBERRIE	PGSERRIE	WRPERRIE	EOPIE	MER	Res.	Res.	SNB[6:0]							STRT	FW	BER	SER	PG	LOCK
	Reset value	0	0								0	0	0	0	0	0	0	0			0	0	0	0	0	0	0	0	0	0	0	0	1
0x030	FLASH_NSCCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLR_OPTCHANGEERR	CLR_OBKWERR	CLR_OBKERR	CLR_INCERR	CLR_STRBERR	CLR_PGSERR	CLR_WRPERR	CLR_EOP	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value									0	0	0	0	0	0	0	0																
0x034	FLASH_ SECCCR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	CLR_OBKWERR	CLR_OBKERR	CLR_INCERR	CLR_STRBERR	CLR_PGSERR	CLR_WRPERR	CLR_EOP	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	Reset value										0	0	0	0	0	0	0																
0x03C	FLASH_PRIVCFGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	NSPRIV	SPRIV
	Reset value																															0	0
0x040	FLASH_ NSOBKCFGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWAP_OFFSET[8:0]									Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ALT_SECT_ERASE	ALT_SECT	SWAP_SECT_REQ	LOCK
	Reset value								1	1	1	1	1	1	1	1	1													0	0	0	0
0x044	FLASH_ SECOBKCFGR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWAP_OFFSET[8:0]									Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ALT_SECT_ERASE	ALT_SECT	SWAP_SECT_REQ	LOCK
	Reset value								1	1	1	1	1	1	1	1	1													0	0	0	0
0x048	FLASH_HDPEXTR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_EXT[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_EXT[6:0]						
	Reset value										0	0	0	0	0	0	0										0	0	0	0	0	0	0
0x04C	Reserved	Reserved																															
0x050	FLASH_ OPTSR_CUR	SWAP_BANK	Res.	BOOT_UBE[7:0]								IWDG_STDBY	IWDG_STOP	Res.	Res.	IO_VDDIO2_HSLV	IO_VDD_HSLV	PRODUCT_STATE[7:0]								NRST_STDBY	NRST_STOP	Res.	WWDG_SW	IWDG_SW	BORH_EN	BOR_LEV[1:0]	
	Reset value	X		X	X	X	X	X	X	X	X	X	X			X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X

Table 85. Register map and reset value table (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x054	FLASH_ OPTSR_PRG	SWAP_BANK	Res.	BOOT_UBE[7:0]								IWDG_STDBY	IWDG_STOP	Res.	Res.	IO_VDDIO2_HSLV	IO_VDD_HSLV	PRODUCT_STATE[7:0]								NRST_STDBY	NRST_STOP	Res.	WWDG_SW	IWDG_SW	BORH_EN	BOR_LEV[1:0]	
	Reset value	X		X	X	X	X	X	X	X	X	X	X			X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X
0x058	Reserved	Reserved																															
0x05C	Reserved	Reserved																															
0x060	FLASH_ NSEPOCHR_CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	NS_EPOCH[23:0]																							
	Reset value									X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x064	FLASH_ NSEPOCHR_PRG	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	NS_EPOCH[23:0]																							
	Reset value									X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x068	FLASH_ SECEPOCHR_CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SEC_EPOCH[23:0]																							
	Reset value									X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x06C	FLASH_ SECEPOCHR_PRG	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SEC_EPOCH[23:0]																							
	Reset value									X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x070	FLASH_ OPTSR2_CUR	TZEN[7:0]								Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	USBPD_DIS	SRAM2_ECC	SRAM3_ECC	BKPRAM_ECC	SRAM2_RST	SRAM13_RST	Res.	Res.
	Reset value	X	X	X	X	X	X	X	X																	X	X	X	X	X	X		
0x074	FLASH_ OPTSR2_PRG	TZEN[7:0]								Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	USBPD_DIS	SRAM2_ECC	SRAM3_ECC	BKPRAM_ECC	SRAM2_RST	SRAM13_RST	Res.	Res.
	Reset value	X	X	X	X	X	X	X	X																	X	X	X	X	X	X		
0x080	FLASH_ NSBOOTR_CUR	NSBOOTADD[23:0]																								NSBOOT_LOCK[7:0]							
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x084	FLASH_ NSBOOTR_PRG	NSBOOTADD[23:0]																								NSBOOT_LOCK[7:0]							
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x088	FLASH_ SECBOOTR_CUR	SECBOOTADD[23:0]																								SECBOOT_LOCK[7:0]							
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x08C	FLASH_ SECBOOTR_PRG	SECBOOTADD[23:0]																								SECBOOT_LOCK[7:0]							
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x090	FLASH_ OTPBLR_CUR	LOCKBL[31:0]																															
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Table 85. Register map and reset value table (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x094	FLASH_ OTPBLR_PRG	LOCKBL[31:0]																															
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x0A0 + 0x004 * x (x = 1 to 4)	FLASH_ SECBB1R_x	SECBB1[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0C0 + 0x004 * x (x = 1 to 4)	FLASH_ PRIVBB1R_x	PRIVBB1[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0D0 - 0x0DC	Reserved	Reserved																															
0x0E0	FLASH_SECWM1R _CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_END[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_STRT[6:0]							
	Reset value										X	X	X	X	X	X	X									X	X	X	X	X	X	X
0x0E4	FLASH_SECWM1R _PRG	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_END1[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM1_STRT[6:0]							
	Reset value										X	X	X	X	X	X	X									X	X	X	X	X	X	X
0x0E8	FLASH_WRP1R _CUR	WRPSG1[31:0]																															
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x0EC	FLASH_WRP1R _PRG	WRPSG1[31:0]																															
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x0F0	FLASH_EDATA1R _CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA1_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA1_STRT [2:0]		
	Reset value																	X													X	X	X
0x0F4	FLASH_EDATA1R _PRG	Res.	Res	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA1_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA1_STRT [2:0]		
	Reset value																	X													X	X	X
0x0F8	FLASH_HDP1R _CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_END[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_STRT[6:0]							
	Reset value										0	0	0	0	0	0	0									0	0	0	0	0	0	1
0x0FC	FLASH_HDP1R _PRG	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_END[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP1_STRT[6:0]							
	Reset value										X	X	X	X	X	X	0									X	X	X	X	X	X	X
0x100	FLASH_ECCCORR	Res.	ECCC	Res.	Res.	Res.	Res.	ECCIE	OTP_ECC	SYSF_ECC	BK_ECC	EDATA_ECC	OBK_ECC	Res.	Res.	Res.	Res.	ADDR_ECC[15:0]															
	0x0000 0000		0					0	0	0	0		0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 85. Register map and reset value table (continued)

Offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x104	FLASH_ECCDETR	ECCD	Res.	Res.	Res.	Res.	Res.	Res.	OTP_ECC_FAIL	SYSF_ECC	BK_ECC	EDATA_ECC	OBK_ECC	Res.	Res.	Res.	Res.	ADDR_ECC[15:0]															
	Reset value	0							0	0	0	0	0					0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x108	FLASH_ECCDR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DATA_ECC[15:0]															
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1A0 + 0x004 * x (x = 1 to 4)	FLASH_ SECBB2R_x	SECBB2[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1A0 + 0x004 * x (x = 1 to 4)	FLASH_ PRIVBB2R_x	PRIVBB2[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x1D0 - 0x1DC	Reserved	Reserved																															
0x1E0	FLASH_SECWM2R _CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_END[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_STRT[6:0]						
	Reset value										X	X	X	X	X	X	X										X	X	X	X	X	X	X
0x1E4	FLASH_SECWM2R _PRG	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_END1[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SECWM2_STRT[6:0]						
	Reset value										X	X	X	X	X	X	X										X	X	X	X	X	X	X
0x1E8	FLASH_WRP2R _CUR	WRPSG2[31:0]																															
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x1EC	FLASH_WRP2R _PRG	WRPSG2[31:0]																															
	Reset value	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0x1F0	FLASH_EDATA2R _CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA2_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA2_STRT [2:0]		
	Reset value																	X													X	X	X
0x1F4	FLASH_EDATA2R _PRG	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA2_EN	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EDATA2_STRT [2:0]		
	Reset value																	X													X	X	X
0x1F8	FLASH_HDP2R _CUR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_END[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_STRT[6:0]						
	Reset value										X	X	X	X	X	X	X										X	X	X	X	X	X	X
0x1FC	FLASH_HDP2R _PRG	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_END[6:0]							Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HDP2_STRT[6:0]						
	Reset value										X	X	X	X	X	X	X										X	X	X	X	X	X	X

Refer to *Section 2.3: Memory organization* for the register boundary addresses.

8 Instruction cache (ICACHE)

8.1 ICACHE introduction

The instruction cache (ICACHE) is introduced on the C-AHB code bus of the Cortex®-M33 processor, to improve performance when fetching instructions and data from internal and external memories.

Some specific features, like dual master ports, hit-under-miss, and critical-word-first refill policy, result in close to zero-wait-state performance in most use cases.

8.2 ICACHE main features

The main features of ICACHE are listed below:

- Bus interface
  - One 32-bit AHB slave port, the execution port (input from Cortex®-M33 C-AHB code interface)
  - Two AHB master ports: master1 and master2 ports, with outputs to Fast and Slow buses of main AHB bus matrix, respectively
  - One 32-bit AHB slave port for control (input from AHB peripherals interconnect, for ICACHE registers access)
- Cache access
  - 0 wait-state on hits
  - Hit-under-miss capability: ability to serve processor requests (access to cached data) during an ongoing line refill due to a previous cache miss
  - Dual master access: feature used to decouple the traffic according to targeted memory. For example, the ICACHE assigns fast traffic (addressing flash memories and SRAMs) to the AHB master1 port, and slow traffic (addressing external memories) to the AHB master2 port, thus preventing processor stalls on line refills from external memories. This allows ISR (interrupt service routine) fetching on internal flash memory to take place in parallel with a cache line refill from external memories.
  - Minimal impact on interrupt latency, thanks to dual master
  - Optimal cache line refill thanks to WRAPw bursts of the size of the cache line (32-bit word size, **w**, aligned on cache line size)
  - n-way set-associative default configuration with possibility to configure as 1-way, means direct mapped cache, for applications needing very-low-power consumption profile
- Memory address remap
  - Possibility to remap input address falling into up to four memory regions (used to remap aliased code in external memories to the code region, for execution from the C-AHB code interface)
- Replacement and refill
  - pLRU-t (pseudo-least-recently-used, based on binary tree) replacement policy, algorithm with best complexity/performance balance
  - Critical-word-first refill policy, minimizing processor stalls